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An overview and comprehensive comparison of ensembles for concept drift

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ABSTRACT

Online learning is about extracting information from large data streams which may be affected by changes in the distribution of the data, events known as concept drift. Concept drift detectors are small programs that try to detect these changes and make it possible to replace the base classifier, improving the overall accuracy. Ensembles of classifiers are also common in this application area and some of them are configurable with a drift detector. This article summarizes a large-scale comparison of six ensemble algorithms, configured with 10 different drift detectors, for learning from fully labeled data streams, using a large number of artificial datasets and two popular base learners in the area: Naive Bayes and Hoeffding Tree. In addition, the code of one the ensembles (Leveraging Bagging) was modified to permit its configuration with any drift detector: its original implementation only uses ADWIN. The goal is to assess how good the existing ensemble algorithms configurable with detectors really are and also to verify and challenge a common belief in the area. The results of the experiments suggest that, in most datasets, the choice of ensemble algorithm has much more impact on the final accuracy than the choice of drift detector used in its configuration. They also suggest the best auxiliary detectors to configure the ensembles, i.e. those that maximize the accuracy of the ensembles, are only marginally different from the best detectors in the same datasets in terms of their accuracies (recently reported in another article).

1. Introduction

In data stream environments, large amounts of data (possibly infinite) flow rapidly and continuously. Learning from these streams therefore requires online methods and is often restricted in terms of memory and run-time usage. Also, a common constraint is that each data instance can be read only once. Finally, concept drifts [28,30], usually seen as changes in data distribution, are expected to occur.

A very common categorization of concept drift is based on the speed of the changes. Sudden and/or rapid changes between concepts are called *abrupt* and transitions from one concept to another over a larger number of instances are called *gradual* [29,32,45]. Some authors further categorize gradual drifts as slow, moderate, fast and/or incremental [41]. Concept drifts can also be categorized based on the reason for the change as *real* or *virtual* [30,31].

In real-world applications, concept drifts may occur for several reasons, such as equipment failure, intrusion, seasonal changes, etc. and, thus, it is considered important to detect these drifts in many applications. Examples of real-world applications with concept drifts include the detection of changes in the weather or water temperature, the detection of movement in data collected from sensors [39], separating spam from legitimate e-mail messages [37], predicting equipment failure, and many others [28,56].

A number of different directions has been pursued to learn from data streams containing concept drift. One that is very common is based on concept drift detectors [32], lightweight software that focuses on identifying changes in data distribution by monitoring the prediction results of a base classifier.

Several ensembles employing a base learner have also been proposed to deal with concept drift, sometimes using different strategies and/or weighting functions to compute the resulting classification, including Dynamic Weighted Majority (DWM) [38], Diversity for Dealing with Drifts (DDD) [42], Adaptable Diversity-based Online Boosting (ADOB) [54], Fast Adaptive Stacking of Ensembles (FASE) [27], and Boosting-like Online Learning Ensemble (BOLE) [8]. Other methods proposed to reuse previous classifiers on recurring concepts, e.g. Recurring Concept Drifts (RCD) [31]. Recurring concept drift has been also addressed by fuzzy methods [50]. In addition, it is worth pointing out that some of these ensembles also rely on an auxiliary drift detection method [8,31,42,54].

Ensembles of concept drift detection methods sharing the same base classifier is another approach that has been comparatively less explored [22,40]. Sequential Extreme Learning Machine ELM based computation has also been investigated as the means to address concept drift and imbalanced data [43].

Several concept drift detectors have been proposed over the years and the most well-known are Drift Detection Method (DDM) [29], Early Drift Detection Method (EDDM) [3], Adaptive Windowing (AD-

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WIN) [11], Statistical Test of Equal Proportions (STEPD) [45], Page-Hinkley Test (PHT) [47], Paired Learners (PL) [2], and EWMA for Concept Drift Detection (ECDD) [52]. Of these, DDM and STEPD are among the most simple methods and present a good all-round performance [32].

Numerous recent drift detection methods have also been proposed, including Sequential Drift (SeqDrift) [48], SEED Drift Detector (SEED) [34], Drift Detection Methods based on Hoeffding's Bounds (HDDM) [26] with versions (hddm)$_A$ and HDDM$_W$, Fast Hoeffding Drift Detection Method (FHDDM) [49], Equal Means Z-Test Drift Detector (EMZD) [17], Fisher Test Drift Detector (FTDD) [17,18], Reactive Drift Detection Method (RDDM) [4,5], and Wilcoxon Rank Sum Test Drift Detector (WSTD) [4,6].

In our previous work [7], we experimentally challenged a common wisdom in the area that the best concept drift detectors are necessarily those that detect all the existing drifts closer to their correct points, ideally detecting only them. Specifically, we concluded that minimizing the distance of the true positive detections, false negatives, and false positives often does *not* maximize the accuracy of the classifiers. In other words, at least in some scenarios, a small number of false negatives and/or false positives might indeed help to improve the global accuracy of the classifiers.

Accordingly, one of the objectives of this work is to verify and challenge a similar expectation in the area that assumes ensembles which internally use concept drift detectors would deliver their best accuracy results when using the best drift detection methods according to the aforementioned understanding. Again, the motivation is the fact that the real world often does not behave as predicted or expected by currently accepted theoretical models.

This article presents a comprehensive comparison of ensembles configurable with concept drift detection methods, including detailed information on all relevant aspects of the experiments and analyzing their results. More explicitly, *six* configurations of ensembles, that are configurable with an auxiliary drift detector, were paired with each of *10* selected drift detectors and the accuracies of these 60 combinations were compared among themselves as well as against the selected detectors individually.

This work summarizes and extends the experiments reported in [4]. To achieve this, the code of one the ensembles, Leveraging Bagging (LevBag) [13], available in the Massive Online Analysis (MOA) framework [12], was modified to permit its configuration with any drift detector, since its original implementation has ADWIN hard-coded. In addition, more concept drift detectors were included in the experiments. To the best of our knowledge, these are the largest and most comprehensive comparisons of ensembles configurable with concept drift detectors ever reported in the area.

All these comparisons were performed using a considerably large number of artificial datasets, with both abrupt and gradual concept drift versions of several sizes, using two different base classifiers — Naive Bayes (NB) [36] and Hoeffding Tree (HT) [35], and were all run in the MOA framework, release 2014.11. Also, note the configuration of these experiments is, as much as possible, quite similar to those of Barros [4] and Barros and Santos [7].

The results of these large-scale experiments provide indications, for two different base classifiers, of (a) the best concept drift detectors, both individually and as auxiliary methods inside ensembles, (b) the best ensemble algorithms, irrespective of drift detector adopted, and (c) the best ensemble-detector combinations. They also give indications of how much do the type of concept drift, the dataset generators, and the size of the concepts affect these answers.

More specifically, these experiments are used to extend the research questions answered in Barros and Santos [7] (**RQ1** to **RQ5**), investigating the following set, originally proposed in [4]:

- **RQ6:** What are the best ensemble plus drift detector combinations in terms of final accuracy in abrupt and gradual concept drift datasets?
- **RQ7:** What are the best ensembles in terms of accuracy in abrupt and gradual drift datasets irrespective of the auxiliary concept drift detector used?
- **RQ8:** What are the best concept drift detectors as auxiliary methods in ensembles in terms of accuracy of the ensembles in abrupt and gradual concept drift datasets?
- **RQ9:** Do the answers of **RQ6, RQ7**, and **RQ8** vary with the different dataset generators used in the experiments? How much?
- **RQ10:** Do the answers of **RQ6, RQ7**, and **RQ8** depend on the size of the concepts included in the datasets? How much?
- **RQ11:** In the same datasets, are the best ensembles of **RQ6** and **RQ7** the same?
- **RQ12:** In the same datasets, are the best concept drift detectors of **RQ1** (reported in Barros and Santos [7]), **RQ6**, and **RQ8** the same? To what extent?

The rest of this document is organized into four sections: Section 2 reviews the published literature on concept drift detection methods and ensembles; Section 3 shows the configuration of the experiments, also including brief descriptions of the datasets used in the tests; Section 4 analyses the accuracy results of the experiments, evaluating them statistically to answer the research questions, and discusses the time complexity of the tested methods; and, finally, Section 5 draws conclusions.

2. Related work

This section has two subsections, which cover the literature on concept drift detectors and ensembles, respectively.

2.1. Drift detection methods

The use of a concept drift detector together with a base learner to learn from data streams is quite common. Generally, the drift detector analyses the prediction results of the base classifier and applies some chosen decision model to detect possible changes in the data distribution. Several methods follow this approach.

Existing detectors use distinct strategies and/or statistics to observe the performance of the base learner and signal concept drifts. Most methods also use a lower confidence level to indicate warnings, meaning that concept drifts may have occurred. At warning points, the methods usually create a new instance of the base learner that is trained in parallel. If a concept drift is confirmed, the new classifier replaces the original one, and, when the warning is a false alarm, the new instance is discarded. It is also important to say that, in the MOA framework, the detectors only signal the *warning* and *drift* positions. The interface with the base classifiers is implemented in other classes of MOA.

DDM [29] monitors the classification error rate (p_i) and its standard deviation (s_i) to detect drifts: the error rate is seen as the probability of making an incorrect prediction. Based on the Probably Approximately Correct (PAC) learning model [44], the authors argue the error rate should decrease as the number of examples increases, as long as the data distribution is stationary, and, when the error rate increases, DDM implies the data distribution has changed.

EDDM [3] is very similar to DDM except that it analyzes the distance between consecutive errors, rather than the error rate. Thus, in stationary concepts, the distance between errors should increase and, accordingly, warnings and drifts are triggered when the distance decreases. The authors of EDDM claim it is better than DDM for detecting gradual concept drifts whereas DDM is more effective for detecting abrupt concept drifts.

ADWIN [11] maintains a sliding window of instances (W) divided in two dynamically adjusted sub-windows, which represent older and recent data, respectively. The size of W is reduced when drifts are detected and increased within a stationary concept. Drifts are detected when the difference of the means of these sub-windows is higher than a given threshold.

SeqDrift [48] also uses two sub-windows to represent old and new data and was proposed to improve on some deficiencies of ADWIN. The newer version, SeqDrift2, uses a computationally efficient reservoir sampling to manage the old data, a one-pass method to obtain a random sample of fixed size from a data pool whose size is unknown. It uses the Bernstein bound [9] to compare the sample means of the sub-windows, which presents good results compared to other bounds in distributions with low variance.

SEED [34] is also based on ADWIN and compares two sub-windows. When the means of the sub-windows are higher than a chosen threshold, the older sub-window is dropped. It uses the Hoeffding Inequality with Bonferroni correction to calculate its test statistic and it also performs block compression to eliminate cut points and merge homogeneous blocks. The authors of SEED claim it is faster and more memory-efficient than ADWIN.

STEPD [45] adopts a statistical test of equal proportions, with continuity correction, using two different windows of the processed data (*recent* and *older*). The accuracies of the base classifier on these windows should be the same within a stationary concept and, thus, significant differences in the accuracy of the *recent* window are used to signal warnings and drifts.

WSTD [4,6] uses two windows of data and detects drifts based on an efficient implementation of the Wilcoxon rank sum statistical test [55], instead of the test of equal proportions used in STEPD. In addition, it limits the size of the *older* window using an extra parameter. WSTD detects fewer false positives than STEPD and is particularly efficient in detecting abrupt drifts.

FTDD [17,18] is one of three drift detectors based on an efficient implementation of Fisher's Exact test [23]. It draws on STEPD and on the deficiency of its statistical test of equal proportions in small or imbalanced data samples. FTDD detects drifts using Fisher's Exact test in all situations whereas its two sibling methods, Fisher Proportions Drift Detector (FPDD) and Fisher Squared Drift Detector (FSDD), adopt hybrid applications of Fisher's Exact test.

ECDD [52] is an adaptation of Exponentially Weighted Moving Average (EWMA) [51] to be used in data streams with concept drifts. EWMA detects significant changes in the mean of a sequence of random variables provided that the mean and standard deviation of the data are known in advance, an unreasonable constraint in data stream scenarios. However, in the algorithm of ECDD the mean and standard deviation are not needed.

HDDM [26] monitors the performance of the base classifier by applying "some probability inequalities that assume only independent, uni-variate and bounded random variables to obtain theoretical guarantees for the detection of such distributional changes". It provides bounds on both false positive and false negative rates and does *not* assume that measured values are given according to a Bernoulli distribution. The authors proposed two versions and claim [26] the A_Test ($HDDM_A$) "involves moving averages and is more suitable to detect abrupt changes" whereas the W_Test ($HDDM_W$) "follows a widespread intuitive idea to deal with gradual changes using weighted moving averages".

FHDDM [49] uses a sliding window and Hoeffding's inequality [33] to calculate the maximum (best) probability of the correct predictions observed so far, which is compared to the most recent probability of correct predictions to detect drifts. Its authors claim that their algorithm delivers less detection delay, fewer false positives and fewer false negatives than other state-of-the-art detectors.

RDDM [4,5] tackles a performance loss problem of DDM caused by decreased sensitivity when concepts are very large. It periodically discards older instances of very long concepts, recalculating the statistics responsible for signaling warnings and drifts, and forces concept drifts when the warning period is very long. RDDM usually improves the accuracy of DDM, especially in datasets with gradual drifts, by detecting more drifts and detecting them earlier, in spite of moderately increasing false positives and memory consumption.

2.2. Ensemble methods

As previously discussed, ensembles of classifiers to learn from data streams with concept drift are also fairly common in the literature.

DWM [38] is a weighted ensemble that extends the Weighted Majority Algorithm (WMA) [15] and aims to adapt to concept drifts without explicitly detecting them. DWM adds and removes classifiers according to its global performance: a classifier is added when the ensemble commits an error; the weight of each classifier is reduced when it commits an error; and it is removed when its weight is very low, indicating it has low accuracy on many examples.

Bagging [16] and Boosting [25] are well-known methods for improving the accuracy of other algorithms ("weak" learners). They both use a set of learners and combine them in an ensemble by aggregating their responses to deliver better predictions but using different strategies. Bagging uses different randomly generated bootstrap samples for training, by resampling from the training set with repetitions. Boosting, on the other hand, trains several learners using different distributions over the training data and varies the amount of diversity given to each classifier depending on their previous predictions. Notice that many boosting algorithms come with theoretical guarantees about their results.

Oza and Russell's Online Bagging (OzaBag) and Online Boosting (OzaBoost) [46] both use a Poisson distribution to simulate the behavior of their corresponding offline algorithms in online environments. Adwin Online Bagging (Adwin OzaBag) [14] is basically OZABAG using ADWIN to detect concept drifts. Similarly, Adwin Online Boosting (Adwin OzaBoost), available in the MOA framework, is OzaBoost equipped with ADWIN as its drift detector.

LevBag [13] is a modified version of OzaBag also adding ADWIN as a hard-coded concept drift detector. In addition, it introduces two changes: the first is to increase the value of diversity ($\lambda = 6$) in the Poisson distribution which, as a consequence, leads to an increase in the probability that experts train on the same instance. The second is to change the way the experts predict instances in order to increase diversity and reduce the correlation. With these modifications, LevBag usually delivers better accuracies than (Adwin OzaBag).

DDD [42] is also a variation of Online Bagging and uses four ensembles of classifiers with high and low diversity, before and after a concept drift is detected. A preliminary study [41] analyzed how these ensembles behaved in different sets of data, with abrupt and gradual drifts of different speeds and different lengths of concepts. Based on the obtained results, DDD tries to select the best weighted majority of ensembles before and after the concept drifts detected by a configurable auxiliary drift detector (default is EDDM).

FASE [27] is another algorithm based on OzaBag. It proposes to use a meta-classifier to combine the predictions from the set of base adaptive learners that compose the ensemble and to use $HDDM_A$ for the detection of concept drifts, though it can be configured to use other methods. When a drift is detected, the worst classifier is removed from the ensemble and a new one is added. The proposed voting strategy is weighted and the weights are inversely proportional to the error rates of the components. According to its authors, FASE "is able to process the input data in constant time and space computational complexity".

Online Boost-by-majority (Online BBM) [10] is an online version of Freund's BBM algorithm [24]. Online BBM does not require importance weighted online learning and can achieve results similar to other methods with fewer weak learners. Beygelzimer et al. also proposed another version of online boosting (AdaBoost.OL), which weighs the experts taking into account their accuracy (adaptive).

ADOB [54] is a boosting ensemble based on OzaBoost which uses a different strategy to speed up the experts recovery after concept drifts. ADOB sorts the experts by accuracy before processing each instance affecting the way diversity is distributed to the classifiers and slightly improving the accuracy of the ensemble just after the concept drifts, especially when these drifts are abrupt. Note ADOB also uses a configurable auxiliary drift detector.

BOLE [4,8] is based on simple heuristic modifications to ADOB. More precisely, BOLE weakens the requirements to allow the experts to vote, improving the ensemble accuracy in most situations, specially when the concept drifts are frequent and/or abrupt, where the accuracy gains can be very high. In addition, BOLE delivers very strong performance in most datasets, irrespective of the auxiliary drift detection method used.

Finally, the following two methods are ensembles of detectors rather than ensembles of classifiers, i.e. multiple drift detection methods share a single base learner. e-Detector [22] is a selective detector ensemble which aims to detect both abrupt and gradual drifts. It clusters detector candidates grouping by homogeneous methods and uses a Coefficient of Failure (CoF) to form the ensemble with the best component of each cluster. e-Detector signals drifts and warnings when one of its components does so and this strategy was named the early-find-early-report rule. The authors claim it improves recall and false negatives without significantly increasing false positives and also that it has stronger generalization ability than the detectors.

Drift Detection Ensemble (DDE) [40] is a configurable three-component lightweight ensemble of detectors aimed at delivering better detections of drifts, improving the final accuracy with no substantial effect on the execution time. Its detections are based on a sensitivity parameter, which specifies the minimum number of components needed to signal warnings and drifts. With sensitivity 1, its strategy is similar to the early-find-early-report rule of e-Detector. The other values (2 and 3) make DDE more robust to false positives but the true detections may be delayed.

3. Experimental setting

This section provides all the relevant information regarding the experiments reported in this article. In the carried out experiments, *six* ensemble versions that use auxiliary concept drift detection methods were paired with each of *10* selected drift detectors. The accuracies of these 60 combinations are then compared among themselves and against the selected detectors individually.

The ensembles chosen for the experiments are ADOB [54], DDD [42], FASE [27], and LevBag [13], as well as the $BOLE_4$ and $BOLE_5$ configurations [8]. These are nearly all the ensembles configurable with detectors we are aware of. As previously mentioned, to permit including LevBag in these tests, it was necessary to change its implementation, since its auxiliary drift detector (ADWIN) was hard-coded, rather than parametrized. The ensembles were all set up to use 10 experts and all their specific parameters with respective default values, except for the auxiliary drift detector.

The selected detectors are a subset of the ones used in our recently published comparison of detectors [7], namely: ADWIN, DDM, ECDD, FHDDM, FTDD, $HDDM_A$, RDDM, SeqDrift2, STEPD, and WSTD. The rationale to choose these detectors was to cover a broad range of strategies as well as a mix of traditional and newer detectors, avoiding more than one variation or parametrization of the same method. In particular, in the experiments reported in Barros [4], more versions of both DDM and RDDM were tested to allow a fairer comparison of these two methods. The reason why they were not included here was to make room for the inclusion of other detectors. Finally, other exclusions were necessary because the tables of results would only accommodate 10 detectors and these were based on their results in the experiments of [7], with the worst performing methods being discarded.

The experiments were also run in the MOA framework [12], release 2014.11, and used basically the same base learners, datasets, and set ups adopted in the experiments of both [4] and [7]. Accordingly, the tests used both Naive Bayes NB and Hoeffding Tree HT as base learners, the two most frequently used classifiers in experiments in the area, with implementations available in the MOA framework.

Also, six artificial dataset generators were used to build abrupt and gradual concept drift datasets of seven different sizes, i.e., there are seven sizes of each generator, with 10K, 20K, 50K, 100K, 500K, 1 million, and 2 million instances, respectively. However, in the tests using HT, only the four smaller sizes were tested, i.e., a subset of the ones reported in Chapter 7 of [4] and in [7], because the execution time of the ensembles would become prohibitive.

Again, all generated datasets have four concept drifts distributed at regular intervals making the size of the concepts in each dataset version of the same generator different to cover different scenarios. For instance, in the 20K instances datasets, the four concept drifts are in positions 4K, 8K, 12K, and 16K.

The concept drifts were simulated by concatenating different concepts. In the gradual concept drift datasets the changes take 500 instances and were generated by the sigmoid function available in the MOA framework. This means that, in the 250 instances before the specified points and the 250 instances after them, the class labels are progressively more likely to be based on the newer concept and less likely to be based on the older concept. At the change point, the probability is 50% for each concept.

The experiments using the datasets with 10K, 20K, 50K and 100K instances were executed 30 times to calculate the accuracies of the methods and the mean results were computed with 95% confidence intervals. In the datasets with 500K, 1 million, and 2 million instances there were only 10 repetitions.

Finally, the accuracy evaluation adopted the Prequential methodology [20] with a sliding window of size 1,000, the default in MOA. Thus, each instance is used initially for testing and then for training. The accuracy is based on the cumulative sum of the sequential errors over time, only using 1000 instances in the calculation of each instance.

3.1. Datasets

The dataset generators chosen for the experiments of this article (Section 4) are the same used in the experiments of both [4] and [7]. All of them have been previously used in the area and are publicly available, in the MOA framework or at https://sites.google.com/site/moaextensions. The specific generators are Agrawal, LED, Mixed, Random RBF, Sine, and Waveform. In the case of Agrawal, it was used twice: $Agrawal_1$ is based on its first five functions (F1 to F5) whereas $Agrawal_2$ is based on its remaining functions (F6 to F10), creating very different datasets.

Agrawal generator [1,40] datasets store information from people willing to receive a loan. Their attributes are: salary, commission, age, education level, make of their car, zip code, value of the house, number of years house is owned, and the amount of the loan. From this information, they are classified as group A or group B. The classification is performed according to 10 functions, each providing a different evaluation.

LED generator [26,54] datasets are about the prediction of the digit shown by a seven-segment LED display based on 24 categorical attributes (17 of them are irrelevant) and a categorical class, with 10 possible values. Concept drifts are simulated by changing the position of the seven relevant attributes and each attribute has a 10% probability of being inverted (noise).

Mixed generator [3,31] datasets have two boolean (v and w) and two numeric attributes (x and y). Each instance is to be classified as positive or negative. They are positive if at least two of the three following conditions are met: v, w, and $y < 0.5 + 0.3 sin(3\pi x)$. To simulate a concept drift, the labels of the aforementioned conditions are reversed.

RandomRBF generator [40,48] datasets use n centroids, with centers, labels, and weights randomly defined, and a Gaussian distribution to determine the values of m attributes. The centroids also determine the class labels of the examples. This creates a normally distributed hyper-sphere of examples surrounding each central point with varying densities. Note that this is a very hard problem for learning algorithms. Concept drifts are simulated by changing the positions of the centroids. The generated datasets have six classes, 40 attributes, and 50 centroids.

Sine generator [29,52] datasets are based on two numeric attributes (x and y) and two contexts (Sine1 and Sine2). In the former, a given instance is classified as positive if the point (x, y) is below the curve $y = \sin(x)$. In the latter, the condition $y < 0.5 + 0.3\sin(3\pi x)$ must be satisfied. Concept drifts are simulated either by alternating between Sine1 and Sine2 or by reversing the aforementioned conditions, i.e. points below the curves become negative.

Waveform generator [48,53] datasets have three classes and 40 numerical attributes, with the last 19 used to produce noise. The goal of the problem is to detect the waveform generated by combining two of three base waves. To perform changes, the positions of the attributes representing a certain context are reversed.

4. Results and analysis of the experiments

This section shows and carefully analyses the results of the comprehensive experiments carried out to compare ensembles for mining data streams containing concept drifts. These results give indications of the best ensemble-detector combinations, the best ensemble algorithms, and the best drift detectors to configure the ensembles. In particular, these experiments were designed to answer research questions **RQ6** to **RQ12**, originally introduced in Barros [4] and presented in Section 1.

Tables 1–7 show the ensemble accuracy results in the tests using the *abrupt* datasets (separated by size) and NB as base classifier. It is important to point out that, in all these tables, the last column, named *None*, refers to the results of the concept drift detectors without an ensemble. Also, observe the best result in each dataset is written in **bold**.

The first table presents the results referring to the datasets with 10,000 instances. The others cover the other dataset sizes used in the experiments with NB, i.e., 20,000, 50,000, 100,000, 500,000, 1 Million, and 2 Million instances.

Notice the number of tables to exhibit all the remaining accuracy results of these experiments is fairly big. For this reason, the remaining data are omitted from the main text of the article. Even so, the results of the tests in the *gradual* datasets using NB as base learner are included in the Appendix as Tables 8–14.

Similarly, all the results of the experiments with the ensemble configurations using HT as base classifier are presented in the Appendix. Observe Tables 15–18 comprise the results of the tests in the *abrupt* datasets whereas Tables 19–22 refer to the tests in the *gradual* datasets.

After the accuracy results were obtained, they were compared using the F_F statistic [21] and the Nemenyi post-hoc test to compare all the methods against all the others and find out in what method(s) there is statistical difference.

It is worthwhile saying that the number of statistical comparisons carried out using the results of these experiments with ensembles was very large because the results were compared in three different dimensions: the ensemble-detector configurations, the ensembles irrespective of the detectors, and the detectors without regard to the ensembles.

For this reason, not all these statistical evaluations are explicitly presented here. Nevertheless, the best methods are always enumerated and the ranks have been used as subsidy to answer the research questions.

In the evaluation of the ensemble-detector pairs, because of the large number of configurations, no explicit representation of the results is given. In the tests using NB, the BOLE$_5$ + HDDM$_A$ pair was the best in the abrupt datasets, followed by BOLE$_5$ + RDDM, BOLE$_5$ + WSTD, BOLE$_5$ + FTDD, BOLE$_4$ + HDDM$_A$, and FASE + RDDM. In the case of the gradual datasets, the best pairs were BOLE$_5$ + HDDM$_A$, BOLE$_5$ + RDDM, BOLE$_4$ + HDDM$_A$, FASE + RDDM, ADOB + HDDM$_A$, and BOLE$_5$ + FTDD. In all the datasets together, the bests ranks were those of BOLE$_5$ + HDDM$_A$, BOLE$_5$ + RDDM, BOLE$_5$ + FTDD, BOLE$_5$ + WSTD, BOLE$_4$ + HDDM$_A$, and FASE + RDDM. In the three scenarios, the best pairs were statistically indistinguishable from each other as well as from a number of other combinations.

On the other hand, in the tests using HT, LevBag + HDDM$_A$ was the best pair in the abrupt datasets, followed by BOLE$_5$ + HDDM$_A$, FASE + HDDM$_A$, FASE + DDM, BOLE$_5$ + RDDM, FASE + FHDDM, and FASE + RDDM. In the gradual datasets, the best pairs were LevBag + HDDM$_A$, LevBag + ADWIN, FASE + FHDDM, FASE + DDM, LevBag + STEPD, FASE + HDDM$_A$, and FASE + RDDM. In these two subsets, there were comparatively fewer statistical differences, with many indistinguishable pairs, because fewer datasets were tested with HT. In the evaluation with all the datasets, the bests ranks were those of LevBag + HDDM$_A$, FASE + DDM, FASE + HDDM$_A$, FASE + FHDDM, LevBag + STEPD, FASE + RDDM, and BOLE$_5$ + HDDM$_A$. In this last set, the best pairs were again statistically indistinguishable from each other and from several other combinations.

An interesting information that came out of this round of evaluations is that the choice of ensemble algorithm seems to have much more influence than the choice of detector on the final accuracy results, confirming the results of Barros [4]. This probably happens because all these ensembles are variations of either bagging or boosting, which are meant to improve the performance of weak learners. Thus, the accuracy of individual components have limited impact on the ensemble results. In addition, different algorithms were the best in the tests with the two selected base learners. In fact, based on the ranks, BOLE$_5$ dominated the set of best results using NB whereas LevBag and FASE did the same in the experiments using HT. These results seem to indicate that boosting based ensembles are more effective with NB whereas bagging ones are better with HT. Even so, it is important to remember that a large number of combinations were statistically similar, especially in the tests using HT. Therefore, reaching a definitive conclusion regarding the best combinations based on these results would be premature.

The second round of statistical evaluations concentrated on the ensemble algorithms disregarding the influence of the different concept drift detection methods. Figs. 1–3 show the evaluation of the tests with NB considering *abrupt, gradual*, and *all* the datasets, respectively, using a graphical notation, where the critical difference CD is represented by bars and methods connected by a bar are statistically similar.

Notice that, in these three subsets, the order of the ranks was almost the same: BOLE$_5$ and FASE were the best algorithms, statistically similar, and with statistical superiority to all the other ensembles; BOLE$_4$ and ADOB were the next best, also statistically similar, and significantly better than the last three methods; and the fifth best configuration was *None*, statistically better than DDD and LevBag, except in the gradual datasets.

In the tests using HT, the order of the ranks was also nearly the same in the three subsets, but less statistical differences were observed. Figs. 4–6 introduce the results referring to the evaluation of the tests considering *abrupt, gradual*, and *all* the datasets, respectively.

This time FASE achieved the best rank, with statistical superiority to all the other ensembles only when all the datasets were considered. LevBag and BOLE$_5$ were the following two best, with different orders in different subsets but no significant difference between them. The order of the other methods was BOLE$_4$, ADOB, None, and DDD in all three subsets.

Finally, the third dimension refers to the statistical evaluation of the concept drift detectors inside ensembles ignoring the influence of the different ensemble algorithms. Fig. 7 presents the evaluation based on the results of the experiments in the *abrupt* datasets using NB, i.e., those presented in Tables 1–7.

According to the results, RDDM and HDDM$_A$ were the best detectors in this subset, with close ranks, statistically similar, and significantly better than all the other methods except WSTD, which was third. These were followed by FTDD, FHDDM, ADWIN, STEPD, etc.

Similarly, Fig. 8 presents the corresponding evaluation based on the *gradual* datasets using NB as base learner. In these datasets, the best results were again those of RDDM and HDDM$_A$, also statistically similar, but this time they were significantly better than all the other methods without exceptions. It is also worth pointing out that, except for the first three, the order of the ranks among the other detectors was considerably different.

Table 1
Mean accuracies of Ensembles in percentage (%) in 10K instances abrupt datasets, with 95% confidence intervals, using NB.

Dataset	Ensemble	ADOB	$BOLE_4$	$BOLE_5$	DDD	FASE	LevBag	None
$Agraw_1$	ADWIN	61.19 ± 0.30	61.19 ± 0.30	61.17 ± 0.29	61.42 ± 0.33	63.70 ± 0.24	62.50 ± 0.21	61.83 ± 0.40
	DDM	61.17 ± 0.34	61.17 ± 0.34	61.21 ± 0.34	61.06 ± 0.53	63.94 ± 0.22	62.01 ± 0.23	61.56 ± 0.52
	ECDD	61.93 ± 0.24	61.93 ± 0.24	62.14 ± 0.23	61.89 ± 0.26	64.02 ± 0.21	61.34 ± 0.19	61.81 ± 0.31
	FHDDM	63.18 ± 0.24	63.18 ± 0.24	63.23 ± 0.26	62.94 ± 0.33	**64.11 ± 0.25**	63.23 ± 0.22	63.33 ± 0.29
	FTDD	60.77 ± 0.28	60.77 ± 0.28	60.80 ± 0.30	60.14 ± 0.38	63.60 ± 0.23	59.64 ± 0.43	60.85 ± 0.29
	$HDDM_A$	62.18 ± 0.29	62.17 ± 0.29	62.32 ± 0.27	62.45 ± 0.37	63.97 ± 0.25	63.24 ± 0.19	63.17 ± 0.32
	RDDM	62.63 ± 0.24	62.62 ± 0.24	62.79 ± 0.23	62.81 ± 0.34	**64.11 ± 0.23**	62.21 ± 0.21	63.56 ± 0.26
	SeqDr2	60.97 ± 0.28	60.97 ± 0.28	61.08 ± 0.30	60.93 ± 0.36	63.75 ± 0.25	59.08 ± 0.20	61.28 ± 0.40
	STEPD	62.86 ± 0.19	62.86 ± 0.19	62.94 ± 0.21	62.24 ± 0.35	63.97 ± 0.23	63.17 ± 0.22	62.72 ± 0.35
	WSTD	62.26 ± 0.24	62.26 ± 0.24	62.39 ± 0.25	61.74 ± 0.41	63.85 ± 0.25	61.92 ± 0.39	62.07 ± 0.36
$Agraw_2$	ADWIN	81.69 ± 0.21	81.69 ± 0.21	81.67 ± 0.22	78.41 ± 0.35	81.71 ± 0.25	77.90 ± 0.62	79.22 ± 0.28
	DDM	79.79 ± 0.21	79.79 ± 0.21	79.84 ± 0.24	73.08 ± 1.29	82.06 ± 0.26	81.91 ± 0.21	72.68 ± 1.59
	ECDD	82.53 ± 0.21	82.53 ± 0.21	82.62 ± 0.21	81.99 ± 0.32	82.73 ± 0.22	81.25 ± 0.26	80.99 ± 1.15
	FHDDM	82.99 ± 0.18	82.99 ± 0.18	**83.06 ± 0.19**	80.67 ± 0.46	82.16 ± 0.21	77.72 ± 0.55	81.48 ± 0.50
	FTDD	82.04 ± 0.18	82.05 ± 0.18	82.09 ± 0.19	78.18 ± 0.66	81.74 ± 0.25	75.34 ± 0.81	79.15 ± 0.67
	$HDDM_A$	82.43 ± 0.19	82.44 ± 0.19	82.47 ± 0.19	79.12 ± 0.70	81.83 ± 0.28	79.71 ± 0.55	80.11 ± 0.66
	RDDM	81.44 ± 0.32	81.45 ± 0.32	81.58 ± 0.32	79.43 ± 0.93	82.21 ± 0.22	82.19 ± 0.20	79.63 ± 0.96
	SeqDr2	81.10 ± 0.23	81.10 ± 0.23	81.13 ± 0.23	77.34 ± 0.32	81.67 ± 0.24	71.91 ± 0.29	77.70 ± 0.36
	STEPD	82.92 ± 0.19	82.92 ± 0.19	82.93 ± 0.20	81.05 ± 0.44	82.28 ± 0.20	79.98 ± 0.62	81.43 ± 0.47
	WSTD	82.58 ± 0.23	82.58 ± 0.23	82.64 ± 0.23	79.83 ± 0.46	81.97 ± 0.26	78.22 ± 0.56	80.69 ± 0.52
LED	ADWIN	67.68 ± 0.30	67.68 ± 0.30	67.75 ± 0.29	66.33 ± 0.44	68.29 ± 0.27	67.52 ± 0.43	63.65 ± 0.52
	DDM	68.53 ± 0.28	68.53 ± 0.28	68.55 ± 0.28	68.57 ± 0.35	68.55 ± 0.27	51.36 ± 0.37	69.57 ± 0.30
	ECDD	50.10 ± 5.68	50.10 ± 5.68	67.12 ± 1.04	68.73 ± 0.41	69.03 ± 0.28	59.71 ± 0.54	67.48 ± 0.41
	FHDDM	68.89 ± 0.28	68.89 ± 0.28	68.97 ± 0.28	68.97 ± 0.32	68.70 ± 0.24	67.31 ± 0.65	69.32 ± 0.37
	FTDD	66.40 ± 1.59	68.49 ± 0.31	68.09 ± 0.41	65.81 ± 0.94	68.25 ± 0.25	65.44 ± 0.87	67.20 ± 0.75
	$HDDM_A$	68.92 ± 0.27	68.99 ± 0.27	69.03 ± 0.27	68.59 ± 0.33	68.64 ± 0.25	**69.95 ± 0.31**	69.72 ± 0.29
	RDDM	68.76 ± 0.28	68.78 ± 0.27	68.82 ± 0.27	68.59 ± 0.36	68.58 ± 0.26	63.35 ± 0.54	69.80 ± 0.29
	SeqDr2	66.65 ± 0.31	66.65 ± 0.31	66.95 ± 0.25	65.15 ± 0.55	66.99 ± 0.38	61.82 ± 0.58	60.40 ± 0.58
	STEPD	66.14 ± 0.43	66.14 ± 0.43	66.35 ± 0.43	66.74 ± 0.54	66.37 ± 0.48	69.29 ± 0.35	61.03 ± 1.92
	WSTD	67.88 ± 0.33	67.95 ± 0.34	68.13 ± 0.32	68.26 ± 0.45	67.27 ± 0.36	66.81 ± 0.69	67.60 ± 0.80
Mixed	ADWIN	89.55 ± 0.24	89.55 ± 0.24	89.51 ± 0.24	87.46 ± 0.54	90.02 ± 0.18	74.63 ± 1.45	89.50 ± 0.26
	DDM	89.87 ± 0.21	89.87 ± 0.21	89.87 ± 0.22	88.83 ± 0.51	89.92 ± 0.17	88.29 ± 0.96	89.74 ± 0.29
	ECDD	88.74 ± 0.19	88.74 ± 0.19	88.72 ± 0.19	88.96 ± 0.38	90.09 ± 0.21	82.62 ± 1.41	89.06 ± 0.29
	FHDDM	90.15 ± 0.21	90.15 ± 0.21	90.12 ± 0.21	86.79 ± 0.37	89.79 ± 0.17	68.53 ± 1.53	90.20 ± 0.21
	FTDD	90.47 ± 0.21	90.47 ± 0.21	90.47 ± 0.21	88.05 ± 0.92	89.87 ± 0.18	72.25 ± 1.39	90.39 ± 0.22
	$HDDM_A$	90.40 ± 0.21	90.39 ± 0.21	90.41 ± 0.21	87.40 ± 0.74	89.95 ± 0.17	74.67 ± 2.09	90.39 ± 0.21
	RDDM	90.22 ± 0.17	90.22 ± 0.17	90.23 ± 0.18	87.55 ± 0.74	89.89 ± 0.18	87.49 ± 1.26	90.22 ± 0.23
	SeqDr2	87.75 ± 0.21	87.75 ± 0.21	87.57 ± 0.20	82.06 ± 0.32	90.29 ± 0.19	63.00 ± 0.16	83.83 ± 0.21
	STEPD	90.05 ± 0.22	90.05 ± 0.22	90.02 ± 0.21	87.63 ± 0.85	89.96 ± 0.19	75.13 ± 1.70	90.40 ± 0.28
	WSTD	90.44 ± 0.22	90.44 ± 0.22	**90.49 ± 0.22**	88.02 ± 0.93	89.85 ± 0.18	72.26 ± 1.42	90.41 ± 0.22
RBF	ADWIN	19.86 ± 0.84	19.86 ± 0.84	29.77 ± 0.96	30.71 ± 0.50	31.64 ± 0.36	30.94 ± 0.51	30.55 ± 0.48
	DDM	19.89 ± 0.85	19.89 ± 0.85	29.87 ± 1.01	30.81 ± 0.61	31.42 ± 0.34	29.62 ± 0.41	30.87 ± 0.59
	ECDD	19.90 ± 0.85	19.90 ± 0.85	29.64 ± 1.07	30.94 ± 0.66	**31.73 ± 0.37**	30.86 ± 0.63	30.87 ± 0.64
	FHDDM	19.56 ± 0.88	19.56 ± 0.88	29.49 ± 0.89	30.40 ± 0.47	31.39 ± 0.37	30.43 ± 0.47	29.98 ± 0.47
	FTDD	19.69 ± 0.79	24.49 ± 0.71	30.76 ± 0.68	30.98 ± 0.59	31.71 ± 0.35	31.07 ± 0.60	31.08 ± 0.53
	$HDDM_A$	19.87 ± 0.84	24.66 ± 0.71	30.60 ± 0.63	30.89 ± 0.54	31.58 ± 0.36	30.56 ± 0.43	30.56 ± 0.43
	RDDM	19.86 ± 0.84	24.55 ± 0.66	30.57 ± 0.54	30.44 ± 0.40	31.45 ± 0.31	29.75 ± 0.41	30.53 ± 0.43
	SeqDr2	19.59 ± 0.70	19.59 ± 0.70	29.62 ± 1.00	30.90 ± 0.63	31.72 ± 0.34	31.06 ± 0.66	30.92 ± 0.52
	STEPD	20.25 ± 0.91	20.25 ± 0.91	29.66 ± 0.81	30.34 ± 0.51	31.40 ± 0.36	30.88 ± 0.57	30.01 ± 0.51
	WSTD	20.10 ± 1.07	25.01 ± 0.83	30.19 ± 0.66	30.91 ± 0.65	31.59 ± 0.39	31.10 ± 0.63	30.70 ± 0.56
Sine	ADWIN	87.71 ± 0.17	87.71 ± 0.17	87.65 ± 0.17	84.38 ± 0.36	86.23 ± 0.19	76.51 ± 0.72	85.99 ± 0.22
	DDM	88.12 ± 0.17	88.12 ± 0.17	88.06 ± 0.16	83.95 ± 0.88	86.30 ± 0.21	85.68 ± 0.28	85.10 ± 0.69
	ECDD	87.22 ± 0.16	87.22 ± 0.16	87.31 ± 0.17	86.22 ± 0.25	86.77 ± 0.22	80.94 ± 0.74	86.23 ± 0.22
	FHDDM	88.46 ± 0.19	88.46 ± 0.19	88.47 ± 0.19	84.61 ± 0.33	86.29 ± 0.21	71.32 ± 0.76	86.67 ± 0.22
	FTDD	88.64 ± 0.20	88.66 ± 0.20	88.67 ± 0.18	84.62 ± 0.51	86.38 ± 0.20	71.22 ± 1.01	86.75 ± 0.23
	$HDDM_A$	88.66 ± 0.15	88.68 ± 0.15	88.69 ± 0.16	84.83 ± 0.33	86.42 ± 0.20	77.98 ± 1.72	86.62 ± 0.21
	RDDM	88.55 ± 0.15	88.57 ± 0.15	88.60 ± 0.14	85.38 ± 0.51	86.45 ± 0.21	86.22 ± 0.21	86.58 ± 0.24
	SeqDr2	86.72 ± 0.18	86.72 ± 0.18	86.52 ± 0.19	79.98 ± 0.28	86.11 ± 0.21	63.89 ± 0.17	81.36 ± 0.21
	STEPD	88.30 ± 0.17	88.30 ± 0.17	88.34 ± 0.17	85.37 ± 0.44	86.54 ± 0.20	76.87 ± 1.14	86.78 ± 0.21
	WSTD	88.68 ± 0.14	88.70 ± 0.14	**88.72 ± 0.14**	85.07 ± 0.40	86.40 ± 0.21	71.45 ± 0.82	86.76 ± 0.22
Wavef.	ADWIN	79.54 ± 0.44	79.54 ± 0.44	79.07 ± 0.46	78.15 ± 0.47	78.84 ± 0.42	78.26 ± 0.40	78.54 ± 0.42
	DDM	79.31 ± 0.40	79.31 ± 0.40	78.89 ± 0.38	78.26 ± 0.45	79.01 ± 0.41	79.34 ± 0.32	78.49 ± 0.45
	ECDD	**81.08 ± 0.30**	**81.08 ± 0.30**	81.07 ± 0.29	79.08 ± 0.41	79.74 ± 0.35	79.60 ± 0.36	78.33 ± 0.39
	FHDDM	80.57 ± 0.33	80.57 ± 0.33	80.55 ± 0.31	78.64 ± 0.48	79.20 ± 0.41	77.64 ± 0.37	79.02 ± 0.46
	FTDD	79.67 ± 0.47	79.67 ± 0.48	79.29 ± 0.48	77.73 ± 0.58	78.84 ± 0.44	77.20 ± 0.42	78.06 ± 0.61
	$HDDM_A$	80.14 ± 0.40	80.14 ± 0.41	80.00 ± 0.42	78.41 ± 0.52	79.00 ± 0.43	78.67 ± 0.41	78.73 ± 0.48
	RDDM	80.10 ± 0.35	80.10 ± 0.36	79.91 ± 0.41	78.94 ± 0.46	79.38 ± 0.39	79.68 ± 0.36	79.12 ± 0.47
	SeqDr2	79.26 ± 0.44	79.26 ± 0.44	78.96 ± 0.43	78.38 ± 0.42	78.86 ± 0.41	77.02 ± 0.39	78.56 ± 0.44
	STEPD	80.63 ± 0.32	80.63 ± 0.32	80.57 ± 0.33	79.03 ± 0.41	79.56 ± 0.41	79.01 ± 0.40	79.25 ± 0.42
	WSTD	80.27 ± 0.35	80.27 ± 0.35	80.13 ±0.38	78.26 ± 0.52	79.07 ± 0.40	77.75 ± 0.39	78.79 ± 0.51

Table 2
Mean accuracies of Ensembles in percentage (%) in 20K instances abrupt datasets, with 95% confidence intervals, using NB.

Dataset	Ensemble	ADOB	$BOLE_4$	$BOLE_5$	DDD	FASE	LevBag	None
$Agraw_1$	ADWIN	63.96 ± 0.20	63.96 ± 0.20	64.16 ± 0.18	63.87 ± 0.18	65.02 ± 0.13	64.13 ± 0.15	64.54 ± 0.16
	DDM	63.08 ± 0.34	63.08 ± 0.34	63.24 ± 0.34	62.85 ± 0.40	65.15 ± 0.12	62.76 ± 0.21	63.08 ± 0.59
	ECDD	63.04 ± 0.16	63.04 ± 0.16	63.33 ± 0.16	62.62 ± 0.18	64.74 ± 0.15	61.67 ± 0.15	62.37 ± 0.15
	FHDDM	64.33 ± 0.15	64.33 ± 0.15	64.34 ± 0.15	64.34 ± 0.21	**65.25 ± 0.12**	64.43 ± 0.17	64.41 ± 0.17
	FTDD	62.59 ± 0.23	62.59 ± 0.22	62.72 ± 0.21	61.56 ± 0.44	65.05 ± 0.12	59.44 ± 0.41	62.02 ± 0.35
	$HDDM_A$	64.97 ± 0.21	64.97 ± 0.21	65.22 ± 0.21	64.35 ± 0.19	65.21 ± 0.11	64.68 ± 0.13	64.82 ± 0.17
	RDDM	64.81 ± 0.22	64.81 ± 0.22	65.03 ± 0.22	64.22 ± 0.23	65.24 ± 0.12	62.98 ± 0.18	64.89 ± 0.15
	SeqDr2	63.09 ± 0.29	63.09 ± 0.29	63.28 ± 0.28	63.31 ± 0.34	65.08 ± 0.12	60.04 ± 0.25	63.89 ± 0.40
	STEPD	64.40 ± 0.21	64.40 ± 0.21	64.46 ± 0.21	64.12 ± 0.18	65.24 ± 0.11	64.22 ± 0.18	64.38 ± 0.18
	WSTD	64.73 ± 0.19	64.73 ± 0.19	64.86 ± 0.18	63.88 ± 0.31	65.16 ± 0.12	63.22 ± 0.29	64.48 ± 0.27
$Agraw_2$	ADWIN	84.62 ± 0.11	84.62 ± 0.11	84.62 ± 0.11	81.64 ± 0.36	84.08 ± 0.12	81.82 ± 0.44	82.02 ± 0.27
	DDM	83.02 ± 0.20	83.02 ± 0.20	83.10 ± 0.23	79.27 ± 0.92	84.30 ± 0.12	83.61 ± 0.14	79.00 ± 0.87
	ECDD	84.70 ± 0.12	84.70 ± 0.12	84.81 ± 0.13	83.82 ± 0.14	84.59 ± 0.11	83.44 ± 0.17	83.30 ± 0.18
	FHDDM	85.35 ± 0.10	85.35 ± 0.10	**85.41 ± 0.10**	83.15 ± 0.27	84.42 ± 0.10	80.06 ± 0.66	84.05 ± 0.22
	FTDD	84.70 ± 0.12	84.70 ± 0.12	84.73 ± 0.12	81.58 ± 0.44	84.11 ± 0.14	77.75 ± 0.57	81.90 ± 0.41
	$HDDM_A$	84.95 ± 0.10	84.95 ± 0.10	84.98 ± 0.10	82.52 ± 0.56	84.20 ± 0.11	81.61 ± 0.60	83.00 ± 0.50
	RDDM	84.27 ± 0.15	84.27 ± 0.15	84.38 ± 0.14	82.70 ± 0.46	84.41 ± 0.12	84.06 ± 0.11	83.18 ± 0.56
	SeqDr2	84.09 ± 0.13	84.09 ± 0.13	84.08 ± 0.13	80.70 ± 0.32	84.08 ± 0.11	73.91 ± 0.37	80.97 ± 0.32
	STEPD	85.26 ± 0.10	85.26 ± 0.10	85.29 ± 0.11	83.52 ± 0.33	84.48 ± 0.13	81.31 ± 0.67	83.82 ± 0.37
	WSTD	85.19 ± 0.14	85.19 ± 0.14	85.23 ± 0.15	82.75 ± 0.47	84.16 ± 0.14	80.62 ± 0.62	83.41 ± 0.40
LED	ADWIN	69.94 ± 0.20	69.94 ± 0.20	70.00 ± 0.20	69.07 ± 0.30	70.57 ± 0.20	69.72 ± 0.41	65.97 ± 0.67
	DDM	70.88 ± 0.19	70.88 ± 0.19	70.89 ± 0.19	70.47 ± 0.25	71.03 ± 0.17	51.61 ± 0.31	71.32 ± 0.25
	ECDD	70.69 ± 1.26	70.69 ± 1.26	71.37 ± 0.20	70.16 ± 0.30	71.35 ± 0.18	60.19 ± 0.38	68.15 ± 0.42
	FHDDM	71.27 ± 0.22	71.27 ± 0.22	71.32 ± 0.22	70.82 ± 0.21	71.35 ± 0.17	69.23 ± 0.68	71.28 ± 0.31
	FTDD	70.46 ± 1.12	71.02 ± 0.22	70.94 ± 0.23	69.78 ± 0.37	70.90 ± 0.16	67.59 ± 0.94	70.55 ± 0.47
	$HDDM_A$	71.37 ± 0.18	71.39 ± 0.18	71.40 ± 0.19	70.54 ± 0.23	71.21 ± 0.16	**71.83 ± 0.20**	71.52 ± 0.18
	RDDM	71.19 ± 0.19	71.20 ± 0.19	71.22 ± 0.20	70.64 ± 0.22	71.15 ± 0.16	63.90 ± 0.40	71.74 ± 0.16
	SeqDr2	69.16 ± 0.23	69.16 ± 0.23	69.31 ± 0.21	68.53 ± 0.37	69.66 ± 0.29	62.71 ± 0.67	63.13 ± 0.84
	STEPD	68.49 ± 0.34	68.49 ± 0.34	68.63 ± 0.35	69.70 ± 0.35	69.72 ± 0.32	71.12 ± 0.25	65.79 ± 1.63
	WSTD	70.51 ± 0.29	70.53 ± 0.29	70.62 ± 0.28	70.38 ± 0.24	70.44 ± 0.25	68.44 ± 0.77	70.60 ± 0.44
Mixed	ADWIN	90.79 ± 0.13	90.79 ± 0.13	90.84 ± 0.13	88.94 ± 0.18	90.89 ± 0.11	76.93 ± 1.42	90.79 ± 0.13
	DDM	90.64 ± 0.16	90.64 ± 0.16	90.66 ± 0.15	89.58 ± 0.64	90.87 ± 0.10	90.07 ± 0.21	90.26 ± 0.67
	ECDD	89.03 ± 0.13	89.03 ± 0.13	89.06 ± 0.12	89.92 ± 0.21	90.83 ± 0.11	86.66 ± 1.03	89.41 ± 0.20
	FHDDM	90.77 ± 0.17	90.77 ± 0.17	90.82 ± 0.16	88.40 ± 0.33	90.83 ± 0.10	69.87 ± 1.61	91.11 ± 0.13
	FTDD	91.33 ± 0.15	91.33 ± 0.15	**91.37 ± 0.14**	88.66 ± 0.81	90.90 ± 0.11	71.83 ± 1.52	91.18 ± 0.13
	$HDDM_A$	90.99 ± 0.16	90.99 ± 0.16	91.07 ± 0.16	88.65 ± 0.56	90.89 ± 0.10	78.25 ± 2.28	91.10 ± 0.12
	RDDM	90.78 ± 0.15	90.78 ± 0.15	90.92 ± 0.14	89.53 ± 0.49	90.88 ± 0.10	90.07 ± 0.38	91.03 ± 0.15
	SeqDr2	90.09 ± 0.14	90.09 ± 0.14	90.05 ± 0.13	86.27 ± 0.21	91.07 ± 0.11	65.07 ± 0.12	87.90 ± 0.14
	STEPD	90.36 ± 0.17	90.36 ± 0.17	90.48 ± 0.13	89.87 ± 0.54	90.96 ± 0.11	76.69 ± 2.00	90.95 ± 0.19
	WSTD	91.11 ± 0.17	91.11 ± 0.17	91.16 ± 0.15	88.38 ± 0.77	90.90 ± 0.11	73.86 ± 1.70	91.19 ± 0.13
RBF	ADWIN	19.53 ± 0.69	19.53 ± 0.69	30.26 ± 0.67	30.38 ± 0.43	31.89 ± 0.29	31.17 ± 0.47	30.54 ± 0.37
	DDM	19.62 ± 0.78	19.62 ± 0.78	30.25 ± 0.75	30.82 ± 0.55	31.91 ± 0.30	29.70 ± 0.34	30.79 ± 0.47
	ECDD	19.62 ± 0.78	19.62 ± 0.78	30.10 ± 0.74	30.92 ± 0.63	31.96 ± 0.32	31.14 ± 0.58	31.25 ± 0.62
	FHDDM	19.54 ± 0.70	19.54 ± 0.70	29.74 ± 0.70	30.15 ± 0.44	31.69 ± 0.24	30.28 ± 0.36	29.76 ± 0.37
	FTDD	19.49 ± 0.62	23.62 ± 0.67	30.74 ± 0.56	30.88 ± 0.52	**32.02 ± 0.31**	31.35 ± 0.53	31.15 ± 0.46
	$HDDM_A$	19.54 ± 0.73	23.80 ± 0.69	30.78 ± 0.51	30.80 ± 0.43	31.87 ± 0.29	30.60 ± 0.34	30.69 ± 0.41
	RDDM	19.56 ± 0.72	23.69 ± 0.54	30.50 ± 0.52	30.51 ± 0.42	31.85 ± 0.26	29.85 ± 0.34	30.50 ± 0.41
	SeqDr2	19.29 ± 0.64	19.29 ± 0.64	29.93 ± 0.66	31.02 ± 0.51	32.04 ± 0.30	31.35 ± 0.58	31.07 ± 0.46
	STEPD	19.79 ± 0.71	19.79 ± 0.71	29.72 ± 0.70	29.98 ± 0.33	31.63 ± 0.27	31.16 ± 0.52	29.65 ± 0.39
	WSTD	19.67 ± 0.93	24.16 ± 0.76	30.40 ± 0.64	30.73 ± 0.57	31.98 ± 0.31	31.25 ± 0.58	30.70 ± 0.57
Sine	ADWIN	89.03 ± 0.13	89.03 ± 0.13	89.03 ± 0.13	85.68 ± 0.26	86.95 ± 0.16	78.07 ± 0.62	86.88 ± 0.18
	DDM	88.96 ± 0.15	88.96 ± 0.15	89.00 ± 0.14	85.26 ± 0.74	86.96 ± 0.17	86.27 ± 0.21	83.67 ± 1.77
	ECDD	87.65 ± 0.16	87.65 ± 0.16	87.74 ± 0.15	86.75 ± 0.16	87.35 ± 0.13	84.15 ± 0.46	86.42 ± 0.16
	FHDDM	89.22 ± 0.15	89.22 ± 0.15	89.27 ± 0.15	85.50 ± 0.42	86.99 ± 0.17	71.85 ± 0.85	87.17 ± 0.18
	FTDD	89.55 ± 0.14	89.56 ± 0.14	**89.60 ± 0.12**	85.80 ± 0.44	87.05 ± 0.17	72.57 ± 0.99	87.21 ± 0.19
	$HDDM_A$	89.37 ± 0.13	89.38 ± 0.13	89.42 ± 0.13	85.19 ± 0.34	87.04 ± 0.16	79.62 ± 1.49	87.08 ± 0.18
	RDDM	89.24 ± 0.15	89.25 ± 0.15	89.33 ± 0.14	86.02 ± 0.33	87.11 ± 0.16	86.94 ± 0.18	87.02 ± 0.19
	SeqDr2	88.37 ± 0.17	88.37 ± 0.17	88.33 ± 0.16	83.35 ± 0.23	86.89 ± 0.17	65.00 ± 0.18	84.50 ± 0.18
	STEPD	88.82 ± 0.15	88.82 ± 0.15	88.91 ± 0.13	86.23 ± 0.31	87.18 ± 0.16	77.22 ± 1.08	87.18 ± 0.16
	WSTD	89.43 ± 0.14	89.44 ± 0.14	89.46 ± 0.13	85.71 ± 0.33	87.05 ± 0.17	71.85 ± 0.98	87.21 ± 0.18
Wavef.	ADWIN	80.60 ± 0.23	80.60 ± 0.23	80.44 ± 0.24	79.17 ± 0.31	79.67 ± 0.25	78.91 ± 0.25	79.52 ± 0.31
	DDM	80.11 ± 0.25	80.11 ± 0.25	79.79 ± 0.27	78.83 ± 0.31	79.88 ± 0.23	79.66 ± 0.18	78.98 ± 0.29
	ECDD	**81.59 ± 0.18**	**81.59 ± 0.18**	81.58 ± 0.19	79.53 ± 0.23	80.27 ± 0.21	80.07 ± 0.22	78.67 ± 0.25
	FHDDM	81.30 ± 0.21	81.30 ± 0.21	81.27 ± 0.21	79.29 ± 0.28	79.98 ± 0.23	78.12 ± 0.22	79.69 ± 0.28
	FTDD	80.54 ± 0.25	80.54 ± 0.26	80.30 ± 0.28	78.73 ± 0.42	79.74 ± 0.24	77.55 ± 0.25	79.12 ± 0.44
	$HDDM_A$	80.81 ± 0.23	80.81 ± 0.23	80.77 ± 0.25	79.21 ± 0.27	79.88 ± 0.23	79.37 ± 0.25	79.60 ± 0.26
	RDDM	80.79 ± 0.24	80.79 ± 0.24	80.71 ± 0.25	79.61 ± 0.22	80.14 ± 0.23	80.11 ± 0.21	79.78 ± 0.29
	SeqDr2	80.40 ± 0.21	80.40 ± 0.21	80.27 ± 0.22	79.36 ± 0.29	79.68 ± 0.26	77.35 ± 0.22	79.53 ± 0.28
	STEPD	81.36 ± 0.23	81.36 ± 0.23	81.29 ± 0.22	79.65 ± 0.24	80.29 ± 0.21	79.63 ± 0.25	79.74 ± 0.22
	WSTD	81.04 ± 0.23	81.04 ± 0.23	80.98 ± 0.23	79.17 ± 0.33	79.97 ± 0.23	78.00 ± 0.27	79.71 ± 0.28

Table 3
Mean accuracies of Ensembles in percentage (%) in 50K instances abrupt datasets, with 95% confidence intervals, using NB.

Dataset	Ensemble	ADOB	BOLE$_4$	BOLE$_5$	DDD	FASE	LevBag	None
Agraw$_1$	ADWIN	67.15 ± 0.10	67.15 ± 0.10	67.31 ± 0.12	65.45 ± 0.11	65.93 ± 0.11	65.12 ± 0.16	65.72 ± 0.12
	DDM	65.09 ± 0.24	65.09 ± 0.24	65.30 ± 0.25	63.55 ± 0.47	66.00 ± 0.10	63.28 ± 0.16	63.64 ± 0.63
	ECDD	64.12 ± 0.09	64.12 ± 0.09	64.34 ± 0.09	63.14 ± 0.14	65.25 ± 0.11	61.91 ± 0.07	62.80 ± 0.13
	FHDDM	64.81 ± 0.12	64.81 ± 0.12	64.79 ± 0.11	65.01 ± 0.14	66.02 ± 0.12	65.33 ± 0.10	65.03 ± 0.14
	FTDD	65.60 ± 0.27	65.60 ± 0.27	65.83 ± 0.30	62.61 ± 0.51	65.90 ± 0.12	60.33 ± 0.43	63.55 ± 0.51
	HDDM$_A$	67.58 ± 0.12	67.57 ± 0.12	**67.78 ± 0.11**	65.08 ± 0.20	66.01 ± 0.11	65.59 ± 0.10	65.67 ± 0.16
	RDDM	67.28 ± 0.10	67.27 ± 0.10	67.55 ± 0.12	65.17 ± 0.20	66.06 ± 0.11	63.26 ± 0.10	65.73 ± 0.11
	SeqDr2	66.18 ± 0.16	66.18 ± 0.16	66.56 ± 0.19	65.21 ± 0.18	65.93 ± 0.12	60.64 ± 0.27	65.62 ± 0.10
	STEPD	65.46 ± 0.14	65.46 ± 0.14	65.45 ± 0.16	65.11 ± 0.16	65.93 ± 0.12	65.12 ± 0.13	65.12 ± 0.15
	WSTD	66.71 ± 0.15	66.71 ± 0.15	66.76 ± 0.15	65.30 ± 0.18	66.02 ± 0.12	63.98 ± 0.25	65.57 ± 0.14
Agraw$_2$	ADWIN	86.75 ± 0.10	86.75 ± 0.10	86.75 ± 0.10	84.83 ± 0.23	85.55 ± 0.09	83.34 ± 0.59	85.24 ± 0.17
	DDM	85.67 ± 0.11	85.67 ± 0.11	85.76 ± 0.10	82.99 ± 0.85	85.73 ± 0.06	84.75 ± 0.09	82.40 ± 1.16
	ECDD	86.05 ± 0.07	86.05 ± 0.07	86.16 ± 0.07	84.79 ± 0.10	85.58 ± 0.06	84.59 ± 0.08	84.27 ± 0.09
	FHDDM	86.83 ± 0.07	86.83 ± 0.07	86.89 ± 0.07	84.64 ± 0.12	85.71 ± 0.08	81.63 ± 0.36	85.63 ± 0.15
	FTDD	86.74 ± 0.12	86.74 ± 0.12	86.75 ± 0.12	83.45 ± 0.49	85.63 ± 0.09	78.55 ± 0.59	83.86 ± 0.50
	HDDM$_A$	86.80 ± 0.10	86.81 ± 0.10	86.86 ± 0.10	84.86 ± 0.28	85.63 ± 0.10	83.47 ± 0.56	85.34 ± 0.22
	RDDM	86.43 ± 0.13	86.43 ± 0.13	86.48 ± 0.13	84.72 ± 0.35	85.82 ± 0.06	85.22 ± 0.07	85.38 ± 0.20
	SeqDr2	86.55 ± 0.10	86.55 ± 0.10	86.56 ± 0.10	83.28 ± 0.41	85.53 ± 0.10	75.53 ± 0.45	83.36 ± 0.39
	STEPD	86.80 ± 0.06	86.80 ± 0.06	86.84 ± 0.06	85.10 ± 0.21	85.87 ± 0.07	83.12 ± 0.42	85.33 ± 0.19
	WSTD	87.10 ± 0.09	87.10 ± 0.09	**87.12 ± 0.09**	84.73 ± 0.37	85.70 ± 0.09	81.29 ± 0.69	85.30 ± 0.25
LED	ADWIN	71.31 ± 0.20	71.31 ± 0.20	71.37 ± 0.20	71.37 ± 0.22	72.14 ± 0.16	71.28 ± 0.56	67.83 ± 0.68
	DDM	72.44 ± 0.16	72.44 ± 0.16	72.44 ± 0.17	71.38 ± 0.60	72.60 ± 0.14	51.72 ± 0.21	71.66 ± 0.71
	ECDD	72.50 ± 0.17	72.50 ± 0.17	72.54 ± 0.16	71.01 ± 0.26	72.64 ± 0.17	60.46 ± 0.22	68.73 ± 0.31
	FHDDM	72.66 ± 0.16	72.66 ± 0.16	72.69 ± 0.16	72.26 ± 0.21	72.81 ± 0.15	71.49 ± 0.52	72.12 ± 0.29
	FTDD	72.36 ± 0.17	72.37 ± 0.17	72.36 ± 0.16	71.21 ± 0.30	72.57 ± 0.17	67.76 ± 0.93	72.23 ± 0.21
	HDDM$_A$	72.80 ± 0.16	72.81 ± 0.16	72.81 ± 0.16	71.64 ± 0.23	72.74 ± 0.15	**72.95 ± 0.16**	72.81 ± 0.16
	RDDM	72.67 ± 0.16	72.67 ± 0.16	72.68 ± 0.16	72.06 ± 0.24	72.70 ± 0.14	64.22 ± 0.26	72.89 ± 0.15
	SeqDr2	71.58 ± 0.17	71.58 ± 0.17	71.63 ± 0.16	71.26 ± 0.22	71.87 ± 0.16	63.62 ± 0.83	66.85 ± 1.28
	STEPD	70.38 ± 0.28	70.38 ± 0.28	70.48 ± 0.28	71.50 ± 0.32	72.07 ± 0.19	72.20 ± 0.24	68.73 ± 0.79
	WSTD	72.41 ± 0.19	72.41 ± 0.19	72.45 ± 0.19	71.98 ± 0.23	72.48 ± 0.15	69.48 ± 0.94	72.10 ± 0.33
Mixed	ADWIN	90.97 ± 0.12	90.97 ± 0.12	91.19 ± 0.10	90.84 ± 0.24	91.55 ± 0.10	77.89 ± 1.49	91.58 ± 0.11
	DDM	91.19 ± 0.19	91.19 ± 0.19	91.30 ± 0.12	90.49 ± 0.94	91.56 ± 0.10	90.69 ± 0.31	90.85 ± 0.96
	ECDD	89.39 ± 0.10	89.39 ± 0.10	89.40 ± 0.09	90.44 ± 0.14	91.37 ± 0.10	89.24 ± 0.38	89.82 ± 0.14
	FHDDM	90.91 ± 0.17	90.91 ± 0.17	91.09 ± 0.12	89.50 ± 0.42	91.58 ± 0.10	69.47 ± 1.55	91.68 ± 0.10
	FTDD	91.47 ± 0.17	91.47 ± 0.17	91.66 ± 0.13	89.59 ± 0.77	91.62 ± 0.10	70.62 ± 1.49	91.72 ± 0.10
	HDDM$_A$	91.20 ± 0.15	91.20 ± 0.15	91.45 ± 0.12	89.99 ± 0.47	91.57 ± 0.10	76.62 ± 2.32	91.63 ± 0.11
	RDDM	90.82 ± 0.21	90.81 ± 0.21	91.20 ± 0.15	90.69 ± 0.47	91.58 ± 0.10	91.02 ± 0.35	91.57 ± 0.10
	SeqDr2	90.88 ± 0.12	90.88 ± 0.12	91.05 ± 0.11	89.41 ± 0.30	91.62 ± 0.10	66.34 ± 0.12	90.41 ± 0.09
	STEPD	90.41 ± 0.19	90.41 ± 0.19	90.75 ± 0.13	90.79 ± 0.37	91.63 ± 0.10	78.02 ± 1.77	91.43 ± 0.14
	WSTD	91.13 ± 0.17	91.12 ± 0.17	91.37 ± 0.14	89.08 ± 0.65	91.62 ± 0.10	70.78 ± 1.47	**91.73 ± 0.10**
RBF	ADWIN	19.50 ± 0.64	19.50 ± 0.64	30.05 ± 0.65	30.61 ± 0.38	32.27 ± 0.21	31.37 ± 0.35	30.63 ± 0.33
	DDM	19.56 ± 0.69	19.56 ± 0.69	30.39 ± 0.70	31.11 ± 0.48	32.21 ± 0.18	29.71 ± 0.20	31.06 ± 0.50
	ECDD	19.58 ± 0.71	19.58 ± 0.71	30.41 ± 0.63	31.22 ± 0.59	**32.53 ± 0.27**	31.27 ± 0.63	31.32 ± 0.65
	FHDDM	19.29 ± 0.59	19.29 ± 0.59	29.50 ± 0.68	29.80 ± 0.30	31.76 ± 0.19	30.16 ± 0.23	29.58 ± 0.34
	FTDD	19.46 ± 0.60	23.28 ± 0.54	31.09 ± 0.57	31.16 ± 0.52	32.42 ± 0.23	31.55 ± 0.56	31.03 ± 0.49
	HDDM$_A$	19.49 ± 0.69	23.32 ± 0.66	30.93 ± 0.40	30.95 ± 0.43	32.26 ± 0.22	30.63 ± 0.26	30.91 ± 0.40
	RDDM	19.41 ± 0.66	23.10 ± 0.54	30.91 ± 0.29	30.71 ± 0.34	32.21 ± 0.21	29.76 ± 0.23	30.73 ± 0.36
	SeqDr2	19.04 ± 0.53	19.04 ± 0.53	30.32 ± 0.65	31.15 ± 0.50	32.40 ± 0.25	31.39 ± 0.61	31.26 ± 0.51
	STEPD	19.59 ± 0.58	19.59 ± 0.58	29.51 ± 0.53	29.76 ± 0.29	31.61 ± 0.20	30.95 ± 0.43	29.28 ± 0.30
	WSTD	19.10 ± 0.67	23.22 ± 0.49	30.46 ± 0.65	30.63 ± 0.57	32.24 ± 0.25	31.25 ± 0.65	30.39 ± 0.54
Sine	ADWIN	89.22 ± 0.11	89.22 ± 0.11	89.37 ± 0.10	86.56 ± 0.24	87.21 ± 0.11	77.70 ± 1.14	87.25 ± 0.12
	DDM	89.44 ± 0.12	89.44 ± 0.12	89.55 ± 0.11	82.65 ± 1.44	87.24 ± 0.10	86.36 ± 0.16	84.21 ± 1.32
	ECDD	87.91 ± 0.11	87.91 ± 0.11	88.03 ± 0.11	86.97 ± 0.11	87.48 ± 0.09	86.14 ± 0.24	86.44 ± 0.11
	FHDDM	89.52 ± 0.17	89.52 ± 0.17	89.63 ± 0.14	85.76 ± 0.35	87.30 ± 0.11	72.22 ± 0.97	87.39 ± 0.12
	FTDD	89.79 ± 0.13	89.79 ± 0.13	**89.90 ± 0.11**	86.12 ± 0.33	87.32 ± 0.11	71.44 ± 0.81	87.40 ± 0.12
	HDDM$_A$	89.60 ± 0.16	89.61 ± 0.15	89.74 ± 0.14	85.89 ± 0.33	87.26 ± 0.11	79.60 ± 1.87	87.26 ± 0.10
	RDDM	89.56 ± 0.15	89.56 ± 0.15	89.69 ± 0.14	86.38 ± 0.35	87.33 ± 0.11	87.21 ± 0.10	87.22 ± 0.13
	SeqDr2	89.22 ± 0.13	89.22 ± 0.13	89.31 ± 0.12	85.58 ± 0.24	87.16 ± 0.11	65.53 ± 0.19	86.31 ± 0.12
	STEPD	88.93 ± 0.13	88.93 ± 0.13	89.09 ± 0.12	86.99 ± 0.21	87.39 ± 0.11	79.34 ± 1.02	87.27 ± 0.12
	WSTD	89.50 ± 0.13	89.51 ± 0.14	89.69 ± 0.11	86.05 ± 0.34	87.32 ± 0.11	72.25 ± 0.98	87.40 ± 0.11
Wavef.	ADWIN	81.37 ± 0.15	81.37 ± 0.15	81.33 ± 0.16	79.98 ± 0.17	80.38 ± 0.13	79.50 ± 0.21	80.20 ± 0.13
	DDM	80.79 ± 0.16	80.79 ± 0.16	80.74 ± 0.16	79.51 ± 0.20	80.36 ± 0.13	79.92 ± 0.13	79.60 ± 0.18
	ECDD	**81.97 ± 0.13**	**81.97 ± 0.13**	81.96 ± 0.13	79.91 ± 0.14	80.68 ± 0.14	80.36 ± 0.13	79.02 ± 0.16
	FHDDM	81.71 ± 0.12	81.71 ± 0.12	81.68 ± 0.12	79.85 ± 0.15	80.48 ± 0.13	78.64 ± 0.18	80.18 ± 0.13
	FTDD	81.14 ± 0.14	81.14 ± 0.14	81.06 ± 0.16	79.58 ± 0.24	80.37 ± 0.16	77.95 ± 0.20	79.92 ± 0.25
	HDDM$_A$	81.44 ± 0.15	81.44 ± 0.15	81.41 ± 0.15	79.75 ± 0.18	80.42 ± 0.14	79.78 ± 0.17	80.13 ± 0.15
	RDDM	81.44 ± 0.13	81.44 ± 0.13	81.39 ± 0.13	80.01 ± 0.15	80.50 ± 0.13	80.38 ± 0.13	80.16 ± 0.14
	SeqDr2	81.20 ± 0.16	81.20 ± 0.16	81.17 ± 0.16	80.01 ± 0.15	80.36 ± 0.13	77.58 ± 0.16	80.10 ± 0.13
	STEPD	81.77 ± 0.13	81.77 ± 0.13	81.74 ± 0.13	80.13 ± 0.13	80.72 ± 0.13	80.04 ± 0.16	80.06 ± 0.13
	WSTD	81.59 ± 0.12	81.59 ± 0.12	81.55 ± 0.12	79.88 ± 0.18	80.49 ± 0.13	78.35 ± 0.25	80.21 ± 0.13

Table 4
Mean accuracies of Ensembles in percentage (%) in 100K instances abrupt datasets, with 95% confidence intervals, using NB.

Dataset	Ensemble	ADOB	$BOLE_4$	$BOLE_5$	DDD	FASE	LevBag	None
$Agraw_1$	ADWIN	68.22 ± 0.11	68.22 ± 0.11	68.35 ± 0.12	65.91 ± 0.10	66.24 ± 0.07	65.59 ± 0.14	66.14 ± 0.07
	DDM	66.85 ± 0.17	66.85 ± 0.17	67.25 ± 0.18	64.01 ± 0.47	66.29 ± 0.07	63.53 ± 0.13	64.17 ± 0.68
	ECDD	64.27 ± 0.08	64.27 ± 0.08	64.47 ± 0.08	63.28 ± 0.09	65.51 ± 0.07	62.00 ± 0.05	62.89 ± 0.08
	FHDDM	65.03 ± 0.09	65.03 ± 0.09	65.02 ± 0.09	65.42 ± 0.09	66.27 ± 0.08	65.75 ± 0.09	65.27 ± 0.10
	FTDD	67.30 ± 0.22	67.30 ± 0.22	67.81 ± 0.24	64.44 ± 0.40	66.26 ± 0.08	61.94 ± 0.38	65.04 ± 0.47
	$HDDM_A$	68.35 ± 0.10	68.35 ± 0.10	**68.54 ± 0.11**	65.56 ± 0.18	66.31 ± 0.07	66.02 ± 0.07	66.06 ± 0.08
	RDDM	68.13 ± 0.10	68.13 ± 0.10	68.38 ± 0.10	65.74 ± 0.14	66.36 ± 0.07	63.50 ± 0.07	66.08 ± 0.08
	SeqDr2	67.78 ± 0.17	67.78 ± 0.17	68.39 ± 0.16	65.56 ± 0.14	66.23 ± 0.08	61.38 ± 0.30	66.12 ± 0.08
	STEPD	65.85 ± 0.09	65.85 ± 0.09	65.80 ± 0.11	65.50 ± 0.09	66.27 ± 0.08	65.54 ± 0.10	65.40 ± 0.08
	WSTD	67.32 ± 0.12	67.32 ± 0.12	67.37 ± 0.13	65.82 ± 0.12	66.33 ± 0.07	64.72 ± 0.25	65.96 ± 0.11
$Agraw_2$	ADWIN	87.68 ± 0.06	87.68 ± 0.06	87.68 ± 0.07	86.03 ± 0.11	86.23 ± 0.05	83.74 ± 0.69	86.22 ± 0.05
	DDM	86.71 ± 0.09	86.71 ± 0.09	86.77 ± 0.08	82.98 ± 1.29	86.30 ± 0.04	85.04 ± 0.09	84.29 ± 0.62
	ECDD	86.44 ± 0.07	86.44 ± 0.07	86.56 ± 0.06	85.08 ± 0.06	85.81 ± 0.05	84.97 ± 0.04	84.49 ± 0.07
	FHDDM	87.38 ± 0.05	87.38 ± 0.05	87.43 ± 0.05	85.18 ± 0.11	86.24 ± 0.04	82.32 ± 0.38	86.14 ± 0.06
	FTDD	87.62 ± 0.11	87.62 ± 0.11	87.63 ± 0.11	84.22 ± 0.54	86.27 ± 0.06	78.72 ± 0.76	84.60 ± 0.47
	$HDDM_A$	87.76 ± 0.08	87.76 ± 0.08	**87.83 ± 0.07**	85.61 ± 0.20	86.25 ± 0.05	83.66 ± 0.72	86.14 ± 0.09
	RDDM	87.40 ± 0.07	87.40 ± 0.07	87.46 ± 0.08	85.60 ± 0.30	86.32 ± 0.04	85.57 ± 0.04	86.13 ± 0.04
	SeqDr2	87.60 ± 0.09	87.60 ± 0.09	87.61 ± 0.08	84.62 ± 0.46	86.20 ± 0.05	76.96 ± 0.53	84.51 ± 0.45
	STEPD	87.30 ± 0.06	87.30 ± 0.06	87.33 ± 0.06	86.00 ± 0.11	86.29 ± 0.04	84.16 ± 0.38	85.84 ± 0.09
	WSTD	87.75 ± 0.05	87.75 ± 0.05	87.78 ± 0.05	85.58 ± 0.31	86.29 ± 0.05	82.02 ± 0.62	85.84 ± 0.31
LED	ADWIN	72.14 ± 0.16	72.14 ± 0.16	72.18 ± 0.16	72.72 ± 0.12	72.88 ± 0.10	71.84 ± 0.41	69.32 ± 0.57
	DDM	73.06 ± 0.13	73.06 ± 0.13	73.05 ± 0.14	72.12 ± 0.52	73.25 ± 0.11	51.89 ± 0.16	72.54 ± 0.40
	ECDD	73.01 ± 0.11	73.01 ± 0.11	73.02 ± 0.12	71.48 ± 0.19	73.18 ± 0.13	60.76 ± 0.20	69.02 ± 0.22
	FHDDM	73.24 ± 0.13	73.24 ± 0.13	73.25 ± 0.13	73.00 ± 0.15	73.40 ± 0.11	72.68 ± 0.25	72.53 ± 0.23
	FTDD	72.94 ± 0.14	72.94 ± 0.14	72.93 ± 0.14	71.79 ± 0.20	73.23 ± 0.13	67.80 ± 0.75	72.94 ± 0.19
	$HDDM_A$	73.39 ± 0.12	73.40 ± 0.12	73.40 ± 0.12	72.23 ± 0.17	73.34 ± 0.11	**73.43 ± 0.11**	73.37 ± 0.11
	RDDM	73.27 ± 0.13	73.28 ± 0.13	73.28 ± 0.13	72.65 ± 0.21	73.31 ± 0.11	64.52 ± 0.22	73.39 ± 0.12
	SeqDr2	72.74 ± 0.13	72.74 ± 0.13	72.76 ± 0.13	72.23 ± 0.22	72.86 ± 0.12	64.60 ± 0.78	69.91 ± 0.98
	STEPD	71.50 ± 0.17	71.50 ± 0.17	71.56 ± 0.17	72.36 ± 0.25	72.95 ± 0.15	72.69 ± 0.19	69.46 ± 0.60
	WSTD	73.22 ± 0.12	73.22 ± 0.12	73.25 ± 0.12	72.67 ± 0.21	73.24 ± 0.11	70.91 ± 0.70	72.85 ± 0.20
Mixed	ADWIN	90.68 ± 0.15	90.68 ± 0.15	91.11 ± 0.09	91.44 ± 0.09	91.79 ± 0.05	77.78 ± 1.54	91.82 ± 0.06
	DDM	91.08 ± 0.14	91.08 ± 0.14	91.28 ± 0.11	89.98 ± 1.56	91.79 ± 0.06	91.10 ± 0.29	90.70 ± 1.17
	ECDD	89.45 ± 0.05	89.45 ± 0.05	89.46 ± 0.05	90.60 ± 0.10	91.50 ± 0.06	90.63 ± 0.17	89.81 ± 0.09
	FHDDM	90.72 ± 0.13	90.72 ± 0.13	90.98 ± 0.10	89.88 ± 0.31	91.83 ± 0.06	68.76 ± 1.13	91.88 ± 0.06
	FTDD	91.49 ± 0.10	91.49 ± 0.10	91.69 ± 0.08	90.12 ± 0.55	91.85 ± 0.06	69.17 ± 1.28	**91.90 ± 0.06**
	$HDDM_A$	90.87 ± 0.18	90.87 ± 0.18	91.33 ± 0.12	90.35 ± 0.44	91.80 ± 0.06	81.64 ± 2.91	91.81 ± 0.07
	RDDM	90.76 ± 0.17	90.76 ± 0.17	91.27 ± 0.12	91.08 ± 0.40	91.81 ± 0.05	91.60 ± 0.11	91.78 ± 0.06
	SeqDr2	90.92 ± 0.13	90.92 ± 0.13	91.27 ± 0.11	90.74 ± 0.15	91.81 ± 0.06	66.68 ± 0.11	91.25 ± 0.06
	STEPD	90.40 ± 0.11	90.40 ± 0.11	90.76 ± 0.08	91.41 ± 0.17	91.82 ± 0.06	82.08 ± 1.71	91.54 ± 0.08
	WSTD	90.95 ± 0.17	90.95 ± 0.17	91.35 ± 0.11	90.37 ± 0.57	91.85 ± 0.06	69.59 ± 1.18	**91.90 ± 0.06**
RBF	ADWIN	19.30 ± 0.63	19.30 ± 0.63	29.91 ± 0.73	30.55 ± 0.33	32.72 ± 0.19	31.68 ± 0.31	30.45 ± 0.28
	DDM	19.45 ± 0.66	19.45 ± 0.66	30.89 ± 0.64	31.47 ± 0.50	32.67 ± 0.16	29.83 ± 0.19	31.38 ± 0.42
	ECDD	19.51 ± 0.68	19.51 ± 0.68	30.64 ± 0.55	31.53 ± 0.60	**33.06 ± 0.22**	31.56 ± 0.60	31.51 ± 0.58
	FHDDM	19.17 ± 0.50	19.17 ± 0.50	29.54 ± 0.47	30.02 ± 0.23	32.03 ± 0.12	30.12 ± 0.20	29.53 ± 0.25
	FTDD	19.28 ± 0.63	23.05 ± 0.52	31.15 ± 0.45	31.67 ± 0.47	32.93 ± 0.20	31.89 ± 0.47	31.65 ± 0.45
	$HDDM_A$	19.34 ± 0.62	22.84 ± 0.44	30.55 ± 0.33	31.32 ± 0.34	32.73 ± 0.19	30.89 ± 0.22	31.13 ± 0.34
	RDDM	19.35 ± 0.68	22.67 ± 0.47	30.82 ± 0.30	31.12 ± 0.24	32.66 ± 0.15	29.84 ± 0.15	31.16 ± 0.28
	SeqDr2	19.06 ± 0.49	19.06 ± 0.49	30.57 ± 0.52	31.68 ± 0.46	32.96 ± 0.20	31.82 ± 0.50	31.40 ± 0.41
	STEPD	19.15 ± 0.48	19.15 ± 0.48	29.35 ± 0.54	29.38 ± 0.20	31.92 ± 0.12	31.23 ± 0.35	29.12 ± 0.18
	WSTD	18.81 ± 0.49	22.90 ± 0.42	30.37 ± 0.53	30.77 ± 0.40	32.58 ± 0.16	31.63 ± 0.57	30.69 ± 0.44
Sine	ADWIN	89.16 ± 0.11	89.16 ± 0.11	89.37 ± 0.09	86.68 ± 0.22	87.30 ± 0.08	78.12 ± 0.79	87.36 ± 0.09
	DDM	89.34 ± 0.18	89.34 ± 0.18	89.52 ± 0.13	82.65 ± 1.89	87.29 ± 0.09	86.59 ± 0.17	83.77 ± 1.40
	ECDD	88.00 ± 0.07	88.00 ± 0.07	88.13 ± 0.07	87.05 ± 0.08	87.56 ± 0.07	86.83 ± 0.12	86.45 ± 0.10
	FHDDM	89.35 ± 0.13	89.35 ± 0.13	89.54 ± 0.11	86.05 ± 0.36	87.37 ± 0.09	70.20 ± 0.69	87.42 ± 0.09
	FTDD	89.66 ± 0.19	89.66 ± 0.19	**89.89 ± 0.15**	86.28 ± 0.28	87.39 ± 0.09	71.90 ± 1.08	87.43 ± 0.09
	$HDDM_A$	89.51 ± 0.18	89.51 ± 0.18	89.79 ± 0.14	86.28 ± 0.40	87.29 ± 0.09	80.46 ± 1.82	87.27 ± 0.10
	RDDM	89.66 ± 0.09	89.67 ± 0.09	89.84 ± 0.08	86.87 ± 0.27	87.40 ± 0.09	87.39 ± 0.09	87.31 ± 0.10
	SeqDr2	89.31 ± 0.17	89.31 ± 0.17	89.50 ± 0.14	86.30 ± 0.20	87.24 ± 0.09	65.63 ± 0.13	86.88 ± 0.09
	STEPD	89.08 ± 0.11	89.08 ± 0.11	89.24 ± 0.09	87.23 ± 0.13	87.47 ± 0.08	81.69 ± 0.96	87.30 ± 0.08
	WSTD	89.43 ± 0.15	89.43 ± 0.15	89.67 ± 0.12	86.47 ± 0.29	87.39 ± 0.09	71.24 ± 0.91	87.43 ± 0.09
Wavef.	ADWIN	81.62 ± 0.10	81.62 ± 0.10	81.60 ± 0.10	80.27 ± 0.11	80.56 ± 0.11	79.81 ± 0.16	80.31 ± 0.11
	DDM	80.99 ± 0.15	80.99 ± 0.15	80.96 ± 0.15	79.55 ± 0.24	80.57 ± 0.10	80.00 ± 0.10	79.67 ± 0.22
	ECDD	**82.02 ± 0.10**	**82.02 ± 0.10**	**82.02 ± 0.10**	80.01 ± 0.12	80.81 ± 0.11	80.46 ± 0.10	79.13 ± 0.13
	FHDDM	81.80 ± 0.10	81.80 ± 0.10	81.79 ± 0.10	80.08 ± 0.12	80.64 ± 0.10	79.35 ± 0.10	80.31 ± 0.11
	FTDD	81.44 ± 0.12	81.44 ± 0.12	81.42 ± 0.12	79.90 ± 0.17	80.59 ± 0.10	78.12 ± 0.21	80.23 ± 0.18
	$HDDM_A$	81.62 ± 0.10	81.62 ± 0.10	81.60 ± 0.10	80.11 ± 0.12	80.57 ± 0.11	80.03 ± 0.14	80.27 ± 0.11
	RDDM	81.58 ± 0.09	81.58 ± 0.09	81.56 ± 0.09	80.14 ± 0.12	80.66 ± 0.10	80.46 ± 0.10	80.25 ± 0.11
	SeqDr2	81.55 ± 0.10	81.55 ± 0.10	81.53 ± 0.10	80.18 ± 0.11	80.56 ± 0.10	77.72 ± 0.12	80.29 ± 0.10
	STEPD	81.84 ± 0.10	81.84 ± 0.10	81.82 ± 0.10	80.30 ± 0.10	80.84 ± 0.10	80.29 ± 0.10	80.21 ± 0.10
	WSTD	81.70 ± 0.10	81.70 ± 0.10	81.69 ± 0.10	80.12 ± 0.15	80.66 ± 0.11	78.83 ± 0.23	80.33 ± 0.10

Table 5
Mean accuracies of Ensembles in percentage (%) in 500K instances abrupt datasets, with 95% confidence intervals, using NB.

Dataset	Ensemble	ADOB	$BOLE_4$	$BOLE_5$	DDD	FASE	LevBag	None
$Agraw_1$	ADWIN	68.99 ± 0.09	68.99 ± 0.09	69.02 ± 0.09	66.39 ± 0.07	66.52 ± 0.05	66.05 ± 0.26	66.45 ± 0.04
	DDM	68.21 ± 0.32	68.21 ± 0.32	68.59 ± 0.31	63.71 ± 0.85	66.52 ± 0.04	63.85 ± 0.26	64.72 ± 0.74
	ECDD	64.54 ± 0.04	64.54 ± 0.04	64.72 ± 0.04	63.44 ± 0.05	65.63 ± 0.05	62.06 ± 0.05	62.96 ± 0.06
	FHDDM	65.21 ± 0.06	65.21 ± 0.06	65.14 ± 0.07	65.64 ± 0.03	66.47 ± 0.05	66.08 ± 0.05	65.35 ± 0.06
	FTDD	69.02 ± 0.14	69.02 ± 0.14	**69.31 ± 0.18**	65.96 ± 0.22	66.51 ± 0.04	63.24 ± 0.34	66.32 ± 0.07
	$HDDM_A$	68.91 ± 0.13	68.91 ± 0.13	69.12 ± 0.14	66.11 ± 0.15	66.52 ± 0.04	66.22 ± 0.33	66.40 ± 0.05
	RDDM	68.58 ± 0.21	68.58 ± 0.21	68.86 ± 0.18	66.21 ± 0.13	66.59 ± 0.06	63.56 ± 0.07	66.39 ± 0.06
	SeqDr2	69.10 ± 0.13	69.10 ± 0.13	69.49 ± 0.12	66.10 ± 0.09	66.52 ± 0.05	62.98 ± 0.36	66.46 ± 0.04
	STEPD	66.21 ± 0.08	66.21 ± 0.08	66.10 ± 0.08	65.81 ± 0.07	66.49 ± 0.04	65.82 ± 0.08	65.58 ± 0.08
	WSTD	67.86 ± 0.11	67.86 ± 0.11	67.78 ± 0.13	66.28 ± 0.07	66.53 ± 0.05	65.76 ± 0.23	66.23 ± 0.06
$Agraw_2$	ADWIN	88.50 ± 0.06	88.50 ± 0.06	88.54 ± 0.03	86.76 ± 0.20	86.89 ± 0.05	82.31 ± 1.44	86.87 ± 0.03
	DDM	88.22 ± 0.14	88.22 ± 0.14	88.26 ± 0.15	85.56 ± 0.65	86.90 ± 0.05	85.49 ± 0.10	85.89 ± 0.90
	ECDD	86.94 ± 0.05	86.94 ± 0.05	87.01 ± 0.05	85.36 ± 0.03	86.08 ± 0.04	85.30 ± 0.03	84.75 ± 0.05
	FHDDM	88.04 ± 0.04	88.04 ± 0.04	88.05 ± 0.03	85.73 ± 0.03	86.72 ± 0.04	85.33 ± 0.34	86.64 ± 0.03
	FTDD	88.73 ± 0.12	88.73 ± 0.12	88.72 ± 0.12	85.45 ± 0.81	86.90 ± 0.06	80.75 ± 1.37	86.17 ± 0.64
	$HDDM_A$	88.77 ± 0.04	88.77 ± 0.04	**88.78 ± 0.03**	86.51 ± 0.30	86.91 ± 0.04	84.52 ± 0.86	86.83 ± 0.06
	RDDM	88.61 ± 0.05	88.61 ± 0.05	88.66 ± 0.04	86.54 ± 0.20	86.90 ± 0.04	85.91 ± 0.04	86.78 ± 0.06
	SeqDr2	88.77 ± 0.04	88.77 ± 0.04	88.77 ± 0.05	86.53 ± 0.47	86.89 ± 0.05	78.43 ± 1.08	86.55 ± 0.47
	STEPD	87.84 ± 0.06	87.84 ± 0.06	87.82 ± 0.06	86.41 ± 0.15	86.70 ± 0.05	86.29 ± 0.08	86.28 ± 0.09
	WSTD	88.55 ± 0.07	88.55 ± 0.07	88.56 ± 0.06	86.15 ± 0.37	86.89 ± 0.04	82.12 ± 0.44	86.74 ± 0.11
LED	ADWIN	73.20 ± 0.16	73.20 ± 0.16	73.21 ± 0.16	73.59 ± 0.13	73.65 ± 0.10	72.90 ± 0.55	72.82 ± 0.29
	DDM	73.45 ± 0.09	73.45 ± 0.09	73.38 ± 0.10	72.67 ± 0.64	73.74 ± 0.09	52.01 ± 0.13	72.63 ± 0.60
	ECDD	73.28 ± 0.10	73.28 ± 0.10	73.28 ± 0.10	71.64 ± 0.18	73.53 ± 0.09	60.90 ± 0.21	69.18 ± 0.14
	FHDDM	73.62 ± 0.11	73.62 ± 0.11	73.62 ± 0.11	73.72 ± 0.11	**73.80 ± 0.09**	73.70 ± 0.09	72.93 ± 0.13
	FTDD	73.64 ± 0.09	73.64 ± 0.09	73.64 ± 0.09	72.54 ± 0.29	73.79 ± 0.09	69.34 ± 1.16	73.49 ± 0.28
	$HDDM_A$	**73.80 ± 0.10**	**73.80 ± 0.10**	**73.80 ± 0.10**	72.99 ± 0.28	73.78 ± 0.09	73.77 ± 0.10	73.77 ± 0.11
	RDDM	73.78 ± 0.10	73.78 ± 0.10	73.78 ± 0.10	73.66 ± 0.10	73.78 ± 0.09	64.66 ± 0.16	73.75 ± 0.08
	SeqDr2	73.68 ± 0.11	73.68 ± 0.11	73.68 ± 0.11	73.45 ± 0.24	73.67 ± 0.10	67.38 ± 1.94	72.71 ± 0.81
	STEPD	72.59 ± 0.12	72.59 ± 0.12	72.60 ± 0.12	72.85 ± 0.23	73.64 ± 0.11	72.99 ± 0.16	70.03 ± 0.29
	WSTD	73.76 ± 0.11	73.76 ± 0.11	73.76 ± 0.11	73.57 ± 0.12	73.78 ± 0.08	73.44 ± 0.21	73.45 ± 0.10
Mixed	ADWIN	90.16 ± 0.28	90.16 ± 0.28	91.22 ± 0.16	91.96 ± 0.05	92.03 ± 0.03	79.18 ± 2.98	92.05 ± 0.03
	DDM	91.35 ± 0.29	91.35 ± 0.29	91.57 ± 0.22	90.34 ± 2.38	92.01 ± 0.04	91.59 ± 0.14	91.21 ± 1.20
	ECDD	89.46 ± 0.06	89.46 ± 0.06	89.46 ± 0.06	90.76 ± 0.06	91.64 ± 0.03	91.38 ± 0.03	89.94 ± 0.07
	FHDDM	90.46 ± 0.23	90.46 ± 0.23	90.84 ± 0.14	91.08 ± 0.52	92.05 ± 0.03	68.53 ± 2.07	92.06 ± 0.03
	FTDD	91.07 ± 0.24	91.07 ± 0.24	91.59 ± 0.15	91.35 ± 0.79	92.05 ± 0.03	68.30 ± 1.55	**92.07 ± 0.03**
	$HDDM_A$	90.15 ± 0.33	90.15 ± 0.33	91.13 ± 0.19	91.41 ± 0.51	92.01 ± 0.04	85.09 ± 3.95	92.02 ± 0.05
	RDDM	90.40 ± 0.29	90.40 ± 0.29	91.31 ± 0.21	91.89 ± 0.21	92.03 ± 0.03	91.99 ± 0.04	92.01 ± 0.03
	SeqDr2	90.49 ± 0.32	90.49 ± 0.32	91.36 ± 0.19	91.85 ± 0.04	92.01 ± 0.04	68.02 ± 1.44	91.94 ± 0.03
	STEPD	90.25 ± 0.17	90.25 ± 0.17	90.74 ± 0.10	91.76 ± 0.10	91.99 ± 0.02	89.66 ± 0.80	91.64 ± 0.05
	WSTD	90.29 ± 0.37	90.29 ± 0.37	91.16 ± 0.32	90.72 ± 0.85	92.05 ± 0.03	68.09 ± 1.40	**92.07 ± 0.03**
RBF	ADWIN	18.74 ± 1.01	18.74 ± 1.01	30.37 ± 1.00	31.01 ± 0.41	33.46 ± 0.19	32.95 ± 0.33	31.21 ± 0.38
	DDM	18.38 ± 0.60	18.38 ± 0.60	31.35 ± 1.01	33.12 ± 0.44	33.82 ± 0.16	30.79 ± 0.48	33.42 ± 0.35
	ECDD	18.38 ± 0.60	18.38 ± 0.60	31.77 ± 0.75	33.20 ± 0.61	**34.40 ± 0.13**	33.26 ± 0.66	33.26 ± 0.66
	FHDDM	18.06 ± 0.30	18.06 ± 0.30	29.92 ± 0.78	30.16 ± 0.20	32.55 ± 0.07	30.35 ± 0.15	29.93 ± 0.18
	FTDD	18.88 ± 1.17	21.50 ± 1.23	32.43 ± 0.57	33.49 ± 0.39	34.01 ± 0.17	33.72 ± 0.33	33.12 ± 0.31
	$HDDM_A$	19.10 ± 0.72	22.37 ± 0.76	31.58 ± 0.48	32.77 ± 0.37	33.73 ± 0.14	31.58 ± 0.24	32.54 ± 0.29
	RDDM	18.93 ± 0.80	21.11 ± 1.11	31.52 ± 0.35	32.01 ± 0.24	33.65 ± 0.15	29.86 ± 0.09	32.13 ± 0.26
	SeqDr2	18.38 ± 0.52	18.38 ± 0.52	31.57 ± 0.79	33.39 ± 0.45	34.14 ± 0.18	33.62 ± 0.47	33.16 ± 0.43
	STEPD	18.45 ± 0.58	18.45 ± 0.58	29.14 ± 0.69	29.44 ± 0.13	32.48 ± 0.07	31.90 ± 0.30	29.07 ± 0.12
	WSTD	18.90 ± 0.85	22.07 ± 0.91	31.26 ± 0.38	31.08 ± 0.37	33.18 ± 0.08	33.55 ± 0.34	31.00 ± 0.29
Sine	ADWIN	88.79 ± 0.12	88.79 ± 0.12	89.08 ± 0.10	87.07 ± 0.37	87.38 ± 0.06	79.17 ± 1.79	87.38 ± 0.06
	DDM	89.27 ± 0.32	89.27 ± 0.32	89.62 ± 0.16	83.18 ± 2.50	87.35 ± 0.06	86.77 ± 0.15	79.19 ± 4.28
	ECDD	88.05 ± 0.07	88.05 ± 0.07	88.16 ± 0.07	87.10 ± 0.07	87.60 ± 0.04	87.32 ± 0.05	86.45 ± 0.05
	FHDDM	89.31 ± 0.18	89.31 ± 0.18	89.50 ± 0.15	86.32 ± 0.37	87.39 ± 0.06	66.01 ± 1.09	87.40 ± 0.06
	FTDD	89.31 ± 0.33	89.31 ± 0.33	89.68 ± 0.25	86.28 ± 0.71	87.40 ± 0.06	71.04 ± 1.45	87.41 ± 0.06
	$HDDM_A$	89.37 ± 0.34	89.37 ± 0.34	89.64 ± 0.26	86.77 ± 0.41	87.33 ± 0.07	84.41 ± 2.27	87.33 ± 0.07
	RDDM	89.63 ± 0.23	89.63 ± 0.23	**89.82 ± 0.16**	87.24 ± 0.14	87.44 ± 0.06	87.54 ± 0.05	87.40 ± 0.06
	SeqDr2	88.85 ± 0.23	88.85 ± 0.23	89.26 ± 0.20	86.76 ± 0.40	87.34 ± 0.06	67.12 ± 0.90	87.30 ± 0.06
	STEPD	89.17 ± 0.15	89.17 ± 0.15	89.33 ± 0.13	87.42 ± 0.04	87.52 ± 0.05	86.58 ± 0.21	87.35 ± 0.05
	WSTD	89.25 ± 0.26	89.25 ± 0.26	89.54 ± 0.20	86.89 ± 0.39	87.40 ± 0.06	71.30 ± 1.19	87.40 ± 0.06
Wavef.	ADWIN	81.61 ± 0.11	81.61 ± 0.11	81.61 ± 0.11	80.39 ± 0.12	80.52 ± 0.12	79.80 ± 0.37	80.40 ± 0.12
	DDM	81.08 ± 0.15	81.08 ± 0.15	81.07 ± 0.15	79.76 ± 0.42	80.64 ± 0.14	80.07 ± 0.10	79.80 ± 0.30
	ECDD	**82.00 ± 0.09**	**82.00 ± 0.09**	**82.00 ± 0.09**	80.09 ± 0.09	80.90 ± 0.10	80.53 ± 0.10	79.19 ± 0.10
	FHDDM	81.82 ± 0.11	81.82 ± 0.11	81.82 ± 0.11	80.34 ± 0.11	80.61 ± 0.12	80.05 ± 0.15	80.39 ± 0.11
	FTDD	81.64 ± 0.11	81.64 ± 0.11	81.64 ± 0.11	80.08 ± 0.23	80.58 ± 0.14	78.60 ± 0.37	80.39 ± 0.11
	$HDDM_A$	81.63 ± 0.11	81.63 ± 0.11	81.63 ± 0.11	80.26 ± 0.15	80.54 ± 0.12	80.03 ± 0.22	80.38 ± 0.12
	RDDM	81.66 ± 0.11	81.66 ± 0.11	81.65 ± 0.11	80.33 ± 0.13	80.64 ± 0.12	80.52 ± 0.09	80.37 ± 0.11
	SeqDr2	81.63 ± 0.11	81.63 ± 0.11	81.63 ± 0.11	80.29 ± 0.16	80.52 ± 0.13	77.74 ± 0.18	80.39 ± 0.12
	STEPD	81.83 ± 0.10	81.83 ± 0.10	81.83 ± 0.10	80.38 ± 0.11	80.89 ± 0.11	80.42 ± 0.12	80.26 ± 0.11
	WSTD	81.71 ± 0.11	81.71 ± 0.11	81.71 ± 0.11	80.32 ± 0.12	80.62 ± 0.12	79.55 ± 0.28	80.38 ± 0.11

Table 6
Mean accuracies of Ensembles in percentage (%) in 1 Million instances abrupt datasets, with 95% confidence intervals, using NB.

Dataset	Ensemble	ADOB	BOLE$_4$	BOLE$_5$	DDD	FASE	LevBag	None
Agraw$_1$	ADWIN	69.02 ± 0.14	69.02 ± 0.14	69.04 ± 0.13	66.37 ± 0.12	66.55 ± 0.04	66.14 ± 0.23	66.51 ± 0.04
	DDM	67.85 ± 0.34	67.85 ± 0.34	68.14 ± 0.45	63.53 ± 0.81	66.55 ± 0.04	63.95 ± 0.22	64.28 ± 1.19
	ECDD	64.62 ± 0.04	64.62 ± 0.04	64.79 ± 0.04	63.47 ± 0.06	65.71 ± 0.04	62.07 ± 0.04	62.98 ± 0.05
	FHDDM	65.28 ± 0.07	65.28 ± 0.07	65.20 ± 0.06	65.70 ± 0.05	66.49 ± 0.04	66.15 ± 0.04	65.39 ± 0.05
	FTDD	68.77 ± 0.31	68.77 ± 0.31	69.08 ± 0.15	66.25 ± 0.16	66.54 ± 0.04	63.84 ± 0.47	66.45 ± 0.07
	HDDM$_A$	69.00 ± 0.07	69.00 ± 0.07	69.18 ± 0.10	66.28 ± 0.11	66.55 ± 0.04	66.46 ± 0.04	66.46 ± 0.05
	RDDM	68.65 ± 0.14	68.65 ± 0.14	68.95 ± 0.18	66.48 ± 0.06	66.65 ± 0.04	63.60 ± 0.06	66.49 ± 0.05
	SeqDr2	69.27 ± 0.21	69.27 ± 0.21	**69.57 ± 0.17**	66.08 ± 0.18	66.55 ± 0.04	63.16 ± 0.46	66.52 ± 0.04
	STEPD	66.25 ± 0.05	66.25 ± 0.05	66.10 ± 0.04	65.86 ± 0.06	66.53 ± 0.04	65.89 ± 0.05	65.67 ± 0.05
	WSTD	68.00 ± 0.11	68.00 ± 0.11	67.95 ± 0.10	66.40 ± 0.05	66.59 ± 0.03	66.16 ± 0.11	66.30 ± 0.05
Agraw$_2$	ADWIN	88.70 ± 0.04	88.70 ± 0.04	88.72 ± 0.04	86.87 ± 0.20	86.98 ± 0.03	84.72 ± 1.42	86.97 ± 0.02
	DDM	88.66 ± 0.07	88.66 ± 0.07	88.69 ± 0.04	85.05 ± 1.14	86.97 ± 0.03	85.63 ± 0.10	85.99 ± 1.15
	ECDD	87.00 ± 0.03	87.00 ± 0.03	87.06 ± 0.03	85.39 ± 0.03	86.00 ± 0.05	85.35 ± 0.01	84.77 ± 0.04
	FHDDM	88.13 ± 0.06	88.13 ± 0.06	88.13 ± 0.04	85.77 ± 0.03	86.76 ± 0.02	85.94 ± 0.21	86.70 ± 0.02
	FTDD	88.83 ± 0.08	88.83 ± 0.08	88.87 ± 0.06	86.19 ± 0.33	86.98 ± 0.03	81.38 ± 1.12	86.70 ± 0.27
	HDDM$_A$	88.91 ± 0.06	88.91 ± 0.06	**88.93 ± 0.06**	86.43 ± 0.33	86.97 ± 0.03	83.84 ± 0.88	86.91 ± 0.03
	RDDM	88.84 ± 0.04	88.84 ± 0.04	88.87 ± 0.03	86.86 ± 0.04	86.97 ± 0.02	85.97 ± 0.02	86.86 ± 0.02
	SeqDr2	88.90 ± 0.06	88.90 ± 0.06	**88.93 ± 0.05**	86.71 ± 0.19	86.98 ± 0.03	78.63 ± 0.37	86.90 ± 0.06
	STEPD	87.88 ± 0.02	87.88 ± 0.02	87.86 ± 0.02	86.50 ± 0.04	86.75 ± 0.02	86.40 ± 0.06	86.34 ± 0.05
	WSTD	88.68 ± 0.06	88.68 ± 0.06	88.67 ± 0.05	86.24 ± 0.29	86.96 ± 0.03	83.01 ± 0.43	86.89 ± 0.05
LED	ADWIN	73.43 ± 0.15	73.43 ± 0.15	73.45 ± 0.15	73.81 ± 0.05	73.80 ± 0.05	72.90 ± 0.44	73.40 ± 0.16
	DDM	73.34 ± 0.14	73.34 ± 0.14	73.23 ± 0.17	71.68 ± 1.55	73.83 ± 0.05	51.97 ± 0.11	72.95 ± 0.35
	ECDD	73.35 ± 0.07	73.35 ± 0.07	73.35 ± 0.07	71.65 ± 0.12	73.60 ± 0.05	60.91 ± 0.13	69.25 ± 0.12
	FHDDM	73.70 ± 0.07	73.70 ± 0.07	73.71 ± 0.07	73.82 ± 0.07	73.87 ± 0.06	73.83 ± 0.07	72.99 ± 0.11
	FTDD	73.81 ± 0.06	73.81 ± 0.06	73.81 ± 0.06	72.95 ± 0.32	73.83 ± 0.09	68.80 ± 1.06	73.49 ± 0.27
	HDDM$_A$	73.87 ± 0.06	73.87 ± 0.06	73.87 ± 0.06	73.48 ± 0.20	73.86 ± 0.05	**73.88 ± 0.06**	73.84 ± 0.06
	RDDM	73.87 ± 0.07	73.87 ± 0.07	73.87 ± 0.07	73.80 ± 0.06	**73.88 ± 0.05**	64.67 ± 0.08	73.82 ± 0.06
	SeqDr2	73.83 ± 0.07	73.83 ± 0.07	73.83 ± 0.07	73.74 ± 0.05	73.81 ± 0.05	66.92 ± 1.84	73.34 ± 0.40
	STEPD	72.64 ± 0.10	72.64 ± 0.10	72.65 ± 0.10	73.02 ± 0.15	73.73 ± 0.07	73.10 ± 0.09	70.16 ± 0.30
	WSTD	73.85 ± 0.07	73.85 ± 0.07	73.86 ± 0.07	73.74 ± 0.08	73.87 ± 0.05	73.48 ± 0.24	73.52 ± 0.10
Mixed	ADWIN	90.08 ± 0.30	90.08 ± 0.30	91.26 ± 0.16	92.05 ± 0.03	92.09 ± 0.03	76.83 ± 1.78	92.09 ± 0.03
	DDM	91.27 ± 0.25	91.27 ± 0.25	91.47 ± 0.23	90.17 ± 2.43	92.07 ± 0.03	91.75 ± 0.13	90.11 ± 3.29
	ECDD	89.43 ± 0.03	89.43 ± 0.03	89.44 ± 0.03	90.79 ± 0.03	91.67 ± 0.03	91.49 ± 0.04	89.97 ± 0.05
	FHDDM	90.35 ± 0.17	90.35 ± 0.17	90.79 ± 0.14	91.60 ± 0.14	92.09 ± 0.03	67.70 ± 1.11	**92.10 ± 0.03**
	FTDD	90.91 ± 0.43	90.91 ± 0.43	91.53 ± 0.27	90.78 ± 1.23	**92.10 ± 0.03**	67.99 ± 1.44	**92.10 ± 0.03**
	HDDM$_A$	90.12 ± 0.28	90.12 ± 0.28	91.20 ± 0.19	91.86 ± 0.14	92.08 ± 0.03	81.04 ± 4.25	92.08 ± 0.03
	RDDM	90.28 ± 0.30	90.28 ± 0.30	91.32 ± 0.16	92.02 ± 0.04	92.08 ± 0.03	92.04 ± 0.03	92.04 ± 0.04
	SeqDr2	90.01 ± 0.31	90.01 ± 0.31	91.27 ± 0.17	91.85 ± 0.31	92.08 ± 0.03	68.24 ± 0.93	92.04 ± 0.03
	STEPD	90.27 ± 0.07	90.27 ± 0.07	90.81 ± 0.05	91.89 ± 0.05	92.03 ± 0.03	90.64 ± 0.40	91.67 ± 0.03
	WSTD	90.23 ± 0.28	90.23 ± 0.28	91.21 ± 0.20	90.41 ± 1.09	**92.10 ± 0.03**	67.25 ± 1.21	**92.10 ± 0.03**
RBF	ADWIN	18.90 ± 1.09	18.90 ± 1.09	30.37 ± 0.98	31.00 ± 0.22	33.60 ± 0.08	33.24 ± 0.23	31.31 ± 0.29
	DDM	18.10 ± 0.46	18.10 ± 0.46	31.55 ± 0.72	33.23 ± 0.48	34.00 ± 0.08	31.34 ± 0.43	33.40 ± 0.23
	ECDD	18.22 ± 0.49	18.22 ± 0.49	31.68 ± 0.76	33.07 ± 0.49	**34.49 ± 0.11**	33.13 ± 0.47	33.16 ± 0.44
	FHDDM	17.81 ± 0.16	17.81 ± 0.16	29.47 ± 0.78	30.26 ± 0.13	32.57 ± 0.03	30.25 ± 0.10	29.87 ± 0.18
	FTDD	18.93 ± 1.14	20.90 ± 1.24	32.78 ± 0.51	33.29 ± 0.26	34.01 ± 0.14	33.78 ± 0.28	33.27 ± 0.21
	HDDM$_A$	19.26 ± 0.93	22.00 ± 1.02	31.76 ± 0.38	33.01 ± 0.27	33.93 ± 0.04	31.47 ± 0.17	32.93 ± 0.21
	RDDM	18.84 ± 0.87	20.89 ± 0.71	32.05 ± 0.26	32.00 ± 0.11	33.69 ± 0.04	29.86 ± 0.08	32.16 ± 0.13
	SeqDr2	18.28 ± 0.68	18.28 ± 0.68	31.63 ± 0.40	33.37 ± 0.34	34.27 ± 0.05	33.62 ± 0.37	33.50 ± 0.29
	STEPD	18.00 ± 0.31	18.00 ± 0.31	29.25 ± 0.63	29.32 ± 0.07	32.51 ± 0.05	32.01 ± 0.15	29.03 ± 0.08
	WSTD	18.70 ± 0.64	21.35 ± 0.93	30.96 ± 0.35	31.22 ± 0.09	33.14 ± 0.08	33.58 ± 0.24	31.07 ± 0.23
Sine	ADWIN	88.54 ± 0.17	88.54 ± 0.17	88.89 ± 0.17	87.14 ± 0.36	87.42 ± 0.05	78.28 ± 0.87	87.43 ± 0.05
	DDM	89.30 ± 0.30	89.30 ± 0.30	**89.79 ± 0.13**	82.14 ± 3.44	87.40 ± 0.05	86.57 ± 0.26	81.76 ± 4.29
	ECDD	88.15 ± 0.04	88.15 ± 0.04	88.25 ± 0.04	87.14 ± 0.03	87.61 ± 0.03	87.38 ± 0.04	86.48 ± 0.03
	FHDDM	89.41 ± 0.16	89.41 ± 0.16	89.57 ± 0.15	86.15 ± 0.35	87.44 ± 0.05	65.76 ± 1.33	87.44 ± 0.05
	FTDD	89.34 ± 0.33	89.35 ± 0.33	89.69 ± 0.27	86.36 ± 0.36	87.44 ± 0.05	71.61 ± 1.35	87.45 ± 0.05
	HDDM$_A$	89.15 ± 0.18	89.15 ± 0.18	89.48 ± 0.15	86.60 ± 0.44	87.39 ± 0.06	82.50 ± 1.25	87.38 ± 0.07
	RDDM	89.33 ± 0.15	89.33 ± 0.15	89.65 ± 0.12	87.36 ± 0.13	87.48 ± 0.04	87.58 ± 0.03	87.44 ± 0.04
	SeqDr2	88.64 ± 0.22	88.64 ± 0.22	89.14 ± 0.16	86.97 ± 0.36	87.41 ± 0.05	67.61 ± 1.10	87.39 ± 0.05
	STEPD	89.21 ± 0.09	89.21 ± 0.09	89.36 ± 0.09	87.49 ± 0.04	87.56 ± 0.04	86.95 ± 0.13	87.36 ± 0.04
	WSTD	89.44 ± 0.32	89.44 ± 0.32	89.71 ± 0.25	86.82 ± 0.42	87.44 ± 0.05	70.93 ± 0.75	87.44 ± 0.05
Wavef.	ADWIN	81.64 ± 0.06	81.64 ± 0.06	81.64 ± 0.06	80.42 ± 0.08	80.53 ± 0.07	79.69 ± 0.40	80.43 ± 0.07
	DDM	81.06 ± 0.21	81.06 ± 0.21	81.05 ± 0.21	79.54 ± 0.44	80.62 ± 0.11	80.07 ± 0.06	79.85 ± 0.36
	ECDD	**82.00 ± 0.05**	**82.00 ± 0.05**	**82.00 ± 0.05**	80.12 ± 0.08	80.91 ± 0.06	80.52 ± 0.06	79.20 ± 0.08
	FHDDM	81.84 ± 0.06	81.84 ± 0.06	81.84 ± 0.06	80.41 ± 0.08	80.62 ± 0.08	80.31 ± 0.09	80.42 ± 0.08
	FTDD	81.69 ± 0.06	81.69 ± 0.06	81.69 ± 0.06	80.34 ± 0.12	80.56 ± 0.07	78.87 ± 0.37	80.40 ± 0.10
	HDDM$_A$	81.65 ± 0.08	81.65 ± 0.08	81.65 ± 0.08	80.35 ± 0.12	80.54 ± 0.08	80.22 ± 0.20	80.41 ± 0.09
	RDDM	81.69 ± 0.05	81.69 ± 0.06	81.69 ± 0.06	80.39 ± 0.07	80.66 ± 0.07	80.51 ± 0.06	80.41 ± 0.07
	SeqDr2	81.65 ± 0.07	81.65 ± 0.07	81.65 ± 0.07	80.32 ± 0.12	80.52 ± 0.07	78.11 ± 0.26	80.44 ± 0.07
	STEPD	81.85 ± 0.06	81.85 ± 0.06	81.85 ± 0.06	80.41 ± 0.07	80.91 ± 0.07	80.46 ± 0.08	80.27 ± 0.07
	WSTD	81.73 ± 0.06	81.73 ± 0.06	81.73 ± 0.06	80.38 ± 0.09	80.62 ± 0.07	79.99 ± 0.18	80.40 ± 0.06

Table 7
Mean accuracies of Ensembles in percentage (%) in 2 Million instances abrupt datasets, with 95% confidence intervals, using NB.

Dataset	Ensemble	ADOB	$BOLE_4$	$BOLE_5$	DDD	FASE	LevBag	None
$Agraw_1$	ADWIN	68.97 ± 0.05	68.97 ± 0.05	69.03 ± 0.05	66.47 ± 0.08	66.58 ± 0.03	66.32 ± 0.12	66.56 ± 0.04
	DDM	67.83 ± 0.43	67.83 ± 0.43	68.20 ± 0.44	63.67 ± 0.93	66.60 ± 0.02	64.05 ± 0.28	64.12 ± 0.99
	ECDD	64.67 ± 0.02	64.67 ± 0.02	64.82 ± 0.02	63.47 ± 0.04	65.72 ± 0.01	62.07 ± 0.03	62.98 ± 0.03
	FHDDM	65.30 ± 0.03	65.30 ± 0.03	65.23 ± 0.02	65.71 ± 0.03	66.54 ± 0.02	66.18 ± 0.02	65.40 ± 0.02
	FTDD	68.97 ± 0.14	68.97 ± 0.14	69.31 ± 0.16	66.35 ± 0.12	66.58 ± 0.02	64.11 ± 0.33	66.53 ± 0.04
	$HDDM_A$	69.07 ± 0.19	69.07 ± 0.19	69.29 ± 0.20	66.32 ± 0.15	66.58 ± 0.02	66.52 ± 0.02	66.49 ± 0.05
	RDDM	68.78 ± 0.15	68.78 ± 0.15	69.05 ± 0.14	66.44 ± 0.05	66.68 ± 0.02	63.61 ± 0.02	66.52 ± 0.03
	SeqDr2	68.95 ± 0.37	68.95 ± 0.37	**69.35 ± 0.26**	66.31 ± 0.17	66.58 ± 0.03	63.51 ± 0.55	66.56 ± 0.04
	STEPD	66.31 ± 0.04	66.31 ± 0.04	66.17 ± 0.05	65.87 ± 0.04	66.57 ± 0.02	65.92 ± 0.02	65.66 ± 0.03
	WSTD	68.12 ± 0.08	68.12 ± 0.08	68.07 ± 0.07	66.41 ± 0.02	66.61 ± 0.02	66.37 ± 0.04	66.31 ± 0.02
$Agraw_2$	ADWIN	88.74 ± 0.05	88.74 ± 0.05	88.76 ± 0.05	86.91 ± 0.20	87.03 ± 0.02	84.30 ± 0.90	87.01 ± 0.02
	DDM	88.75 ± 0.08	88.75 ± 0.08	88.79 ± 0.06	85.59 ± 0.58	87.02 ± 0.02	85.70 ± 0.07	84.94 ± 1.40
	ECDD	87.03 ± 0.02	87.03 ± 0.02	87.09 ± 0.02	85.40 ± 0.02	86.00 ± 0.07	85.36 ± 0.02	84.78 ± 0.03
	FHDDM	88.23 ± 0.07	88.23 ± 0.07	88.22 ± 0.06	85.81 ± 0.02	86.78 ± 0.01	86.32 ± 0.11	86.72 ± 0.02
	FTDD	88.98 ± 0.06	88.98 ± 0.06	88.98 ± 0.05	86.29 ± 0.29	87.03 ± 0.02	81.98 ± 1.04	86.97 ± 0.04
	$HDDM_A$	89.03 ± 0.05	89.03 ± 0.05	**89.06 ± 0.04**	86.54 ± 0.31	87.02 ± 0.02	84.28 ± 1.04	86.97 ± 0.02
	RDDM	88.95 ± 0.03	88.95 ± 0.03	88.97 ± 0.03	86.91 ± 0.02	87.00 ± 0.02	85.97 ± 0.02	86.90 ± 0.02
	SeqDr2	88.97 ± 0.05	88.97 ± 0.05	88.99 ± 0.05	86.91 ± 0.07	87.03 ± 0.02	79.73 ± 0.63	86.97 ± 0.03
	STEPD	87.92 ± 0.02	87.92 ± 0.02	87.91 ± 0.02	86.54 ± 0.03	86.77 ± 0.01	86.52 ± 0.04	86.34 ± 0.03
	WSTD	88.74 ± 0.04	88.74 ± 0.04	88.72 ± 0.04	86.49 ± 0.31	86.99 ± 0.02	84.43 ± 0.43	86.94 ± 0.03
LED	ADWIN	73.63 ± 0.09	73.63 ± 0.09	73.66 ± 0.09	73.90 ± 0.04	73.90 ± 0.04	73.22 ± 0.49	73.68 ± 0.10
	DDM	73.38 ± 0.20	73.38 ± 0.20	73.06 ± 0.35	72.43 ± 0.69	73.91 ± 0.04	52.07 ± 0.07	72.78 ± 0.72
	ECDD	73.40 ± 0.03	73.40 ± 0.03	73.40 ± 0.03	71.74 ± 0.07	73.65 ± 0.03	60.99 ± 0.06	69.31 ± 0.08
	FHDDM	73.77 ± 0.03	73.77 ± 0.03	73.77 ± 0.03	73.88 ± 0.04	73.94 ± 0.04	73.92 ± 0.05	73.07 ± 0.09
	FTDD	73.92 ± 0.04	73.92 ± 0.04	73.92 ± 0.04	73.32 ± 0.25	73.93 ± 0.04	70.32 ± 1.38	73.77 ± 0.19
	$HDDM_A$	73.93 ± 0.04	73.93 ± 0.04	73.93 ± 0.04	73.72 ± 0.13	73.92 ± 0.04	73.93 ± 0.04	73.89 ± 0.05
	RDDM	73.93 ± 0.04	73.93 ± 0.04	73.93 ± 0.04	73.87 ± 0.04	**73.94 ± 0.03**	64.75 ± 0.06	73.87 ± 0.04
	SeqDr2	73.92 ± 0.04	73.92 ± 0.04	73.92 ± 0.04	73.85 ± 0.07	73.91 ± 0.04	68.03 ± 2.05	73.65 ± 0.17
	STEPD	72.82 ± 0.06	72.82 ± 0.06	72.82 ± 0.06	73.15 ± 0.07	73.80 ± 0.04	73.16 ± 0.04	70.24 ± 0.23
	WSTD	73.92 ± 0.04	73.92 ± 0.04	73.92 ± 0.04	73.82 ± 0.03	73.93 ± 0.03	73.75 ± 0.07	73.64 ± 0.06
Mixed	ADWIN	90.32 ± 0.17	90.32 ± 0.17	91.41 ± 0.09	92.04 ± 0.03	92.06 ± 0.02	78.05 ± 2.91	92.06 ± 0.02
	DDM	91.29 ± 0.22	91.29 ± 0.22	91.50 ± 0.18	87.76 ± 3.22	92.04 ± 0.02	91.73 ± 0.10	89.91 ± 2.38
	ECDD	89.40 ± 0.03	89.40 ± 0.03	89.39 ± 0.03	90.77 ± 0.03	91.64 ± 0.03	91.52 ± 0.04	89.95 ± 0.04
	FHDDM	90.03 ± 0.18	90.03 ± 0.18	90.51 ± 0.15	91.73 ± 0.11	92.06 ± 0.02	68.69 ± 2.58	**92.07 ± 0.02**
	FTDD	90.58 ± 0.30	90.58 ± 0.30	91.57 ± 0.17	91.72 ± 0.23	92.06 ± 0.02	69.16 ± 2.28	**92.07 ± 0.02**
	$HDDM_A$	90.22 ± 0.16	90.22 ± 0.16	91.27 ± 0.10	91.90 ± 0.13	92.04 ± 0.02	84.53 ± 3.98	92.03 ± 0.02
	RDDM	90.22 ± 0.24	90.22 ± 0.24	91.36 ± 0.13	92.00 ± 0.04	92.05 ± 0.02	92.02 ± 0.03	92.01 ± 0.03
	SeqDr2	89.87 ± 0.31	89.87 ± 0.31	91.36 ± 0.19	92.02 ± 0.03	92.04 ± 0.02	66.92 ± 0.43	92.03 ± 0.02
	STEPD	90.18 ± 0.06	90.18 ± 0.06	90.76 ± 0.04	91.87 ± 0.03	92.00 ± 0.02	91.61 ± 0.14	91.64 ± 0.03
	WSTD	90.10 ± 0.24	90.10 ± 0.24	91.15 ± 0.16	91.83 ± 0.22	92.06 ± 0.02	69.63 ± 2.03	**92.07 ± 0.02**
RBF	ADWIN	18.77 ± 0.99	18.77 ± 0.99	30.62 ± 0.75	31.03 ± 0.25	33.77 ± 0.06	33.58 ± 0.14	31.48 ± 0.31
	DDM	17.86 ± 0.23	17.86 ± 0.23	32.37 ± 0.61	33.42 ± 0.30	34.11 ± 0.07	31.62 ± 0.51	33.58 ± 0.28
	ECDD	18.10 ± 0.49	18.10 ± 0.49	31.96 ± 0.57	33.05 ± 0.18	**34.46 ± 0.13**	33.06 ± 0.20	33.07 ± 0.19
	FHDDM	17.71 ± 0.07	17.71 ± 0.07	29.87 ± 0.74	30.25 ± 0.09	32.66 ± 0.05	30.23 ± 0.07	29.84 ± 0.10
	FTDD	18.59 ± 0.76	20.22 ± 1.13	32.69 ± 0.30	33.31 ± 0.17	34.14 ± 0.09	33.72 ± 0.15	33.23 ± 0.14
	$HDDM_A$	18.97 ± 1.02	21.43 ± 1.10	32.36 ± 0.26	33.21 ± 0.19	34.05 ± 0.09	31.54 ± 0.16	33.02 ± 0.20
	RDDM	18.99 ± 0.95	20.60 ± 0.86	32.03 ± 0.25	32.14 ± 0.12	33.71 ± 0.07	29.84 ± 0.06	32.13 ± 0.13
	SeqDr2	17.97 ± 0.38	17.97 ± 0.38	32.16 ± 0.66	33.37 ± 0.26	34.30 ± 0.09	33.97 ± 0.25	33.49 ± 0.27
	STEPD	17.80 ± 0.15	17.80 ± 0.15	29.29 ± 0.60	29.31 ± 0.05	32.60 ± 0.06	32.04 ± 0.12	29.00 ± 0.09
	WSTD	18.26 ± 0.38	20.42 ± 0.84	31.26 ± 0.18	31.22 ± 0.15	33.23 ± 0.08	33.58 ± 0.13	31.16 ± 0.16
Sine	ADWIN	88.72 ± 0.07	88.72 ± 0.07	89.07 ± 0.07	87.19 ± 0.30	87.43 ± 0.02	78.56 ± 1.41	87.44 ± 0.02
	DDM	88.65 ± 0.22	88.65 ± 0.22	89.45 ± 0.21	80.67 ± 4.29	87.41 ± 0.02	86.75 ± 0.20	77.48 ± 6.00
	ECDD	88.13 ± 0.05	88.13 ± 0.05	88.24 ± 0.05	87.14 ± 0.03	87.60 ± 0.02	87.40 ± 0.02	86.47 ± 0.02
	FHDDM	89.46 ± 0.32	89.46 ± 0.32	89.65 ± 0.25	86.67 ± 0.43	87.44 ± 0.02	64.95 ± 1.51	87.44 ± 0.03
	FTDD	89.21 ± 0.42	89.21 ± 0.42	89.53 ± 0.35	86.69 ± 0.38	87.44 ± 0.02	72.47 ± 1.40	87.44 ± 0.03
	$HDDM_A$	89.20 ± 0.17	89.20 ± 0.17	89.47 ± 0.16	87.08 ± 0.33	87.41 ± 0.02	83.60 ± 2.31	87.41 ± 0.02
	RDDM	89.33 ± 0.18	89.33 ± 0.18	89.58 ± 0.15	87.45 ± 0.03	87.50 ± 0.02	87.59 ± 0.02	87.47 ± 0.03
	SeqDr2	88.65 ± 0.23	88.65 ± 0.23	89.11 ± 0.18	87.40 ± 0.02	87.41 ± 0.02	68.57 ± 1.03	87.42 ± 0.02
	STEPD	89.21 ± 0.13	89.21 ± 0.13	89.37 ± 0.12	87.48 ± 0.02	87.55 ± 0.02	87.30 ± 0.05	87.36 ± 0.03
	WSTD	89.46 ± 0.28	89.46 ± 0.28	**89.70 ± 0.22**	86.57 ± 0.36	87.44 ± 0.02	70.28 ± 1.40	87.44 ± 0.02
Wavef.	ADWIN	81.66 ± 0.02	81.66 ± 0.02	81.66 ± 0.02	80.46 ± 0.03	80.58 ± 0.03	80.01 ± 0.21	80.46 ± 0.03
	DDM	81.12 ± 0.13	81.12 ± 0.13	81.10 ± 0.14	79.74 ± 0.31	80.64 ± 0.06	80.03 ± 0.03	79.55 ± 0.29
	ECDD	**82.02 ± 0.03**	**82.02 ± 0.03**	**82.02 ± 0.03**	80.17 ± 0.05	80.94 ± 0.03	80.55 ± 0.04	79.21 ± 0.04
	FHDDM	81.87 ± 0.04	81.87 ± 0.04	81.87 ± 0.04	80.45 ± 0.04	80.62 ± 0.04	80.40 ± 0.04	80.45 ± 0.04
	FTDD	81.72 ± 0.03	81.72 ± 0.03	81.72 ± 0.03	80.47 ± 0.04	80.59 ± 0.03	79.06 ± 0.28	80.47 ± 0.04
	$HDDM_A$	81.67 ± 0.03	81.67 ± 0.03	81.67 ± 0.03	80.41 ± 0.08	80.59 ± 0.04	80.42 ± 0.16	80.47 ± 0.04
	RDDM	81.71 ± 0.03	81.71 ± 0.03	81.71 ± 0.03	80.46 ± 0.04	80.69 ± 0.04	80.54 ± 0.03	80.46 ± 0.03
	SeqDr2	81.68 ± 0.03	81.68 ± 0.03	81.68 ± 0.03	80.41 ± 0.08	80.58 ± 0.04	78.35 ± 0.30	80.49 ± 0.03
	STEPD	81.87 ± 0.04	81.87 ± 0.04	81.87 ± 0.04	80.45 ± 0.04	80.95 ± 0.04	80.50 ± 0.04	80.31 ± 0.04
	WSTD	81.76 ± 0.04	81.76 ± 0.04	81.76 ± 0.04	80.45 ± 0.04	80.63 ± 0.04	80.10 ± 0.11	80.46 ± 0.04

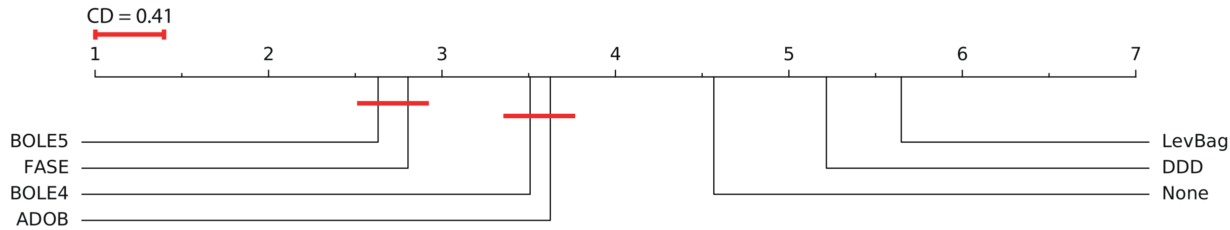

Fig. 1. Comparison results using the Nemenyi test of Ensembles, irrespective of Detectors, with NB in abrupt datasets with 95% confidence.

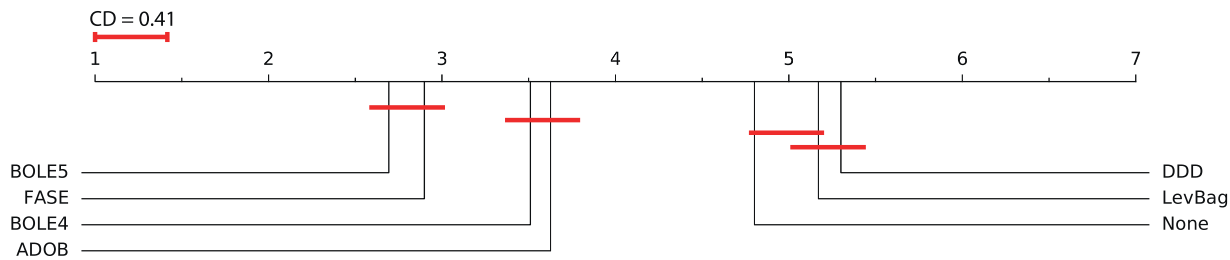

Fig. 2. Comparison results using the Nemenyi test of Ensembles, irrespective of Detectors, with NB in gradual datasets with 95% confidence.

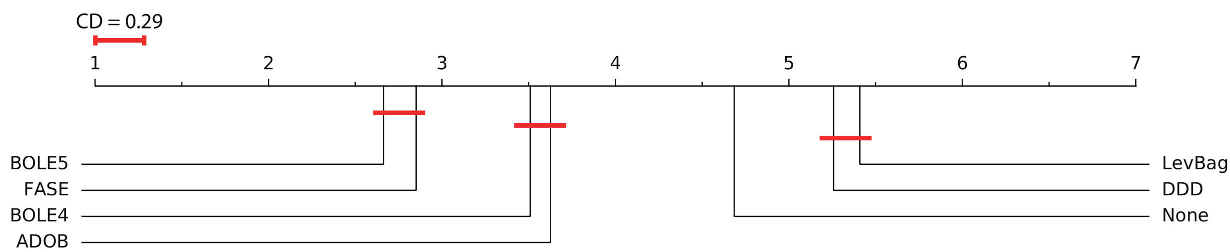

Fig. 3. Comparison results using the Nemenyi test of Ensembles, irrespective of Detectors, with NB in all artificial datasets with 95% confidence.

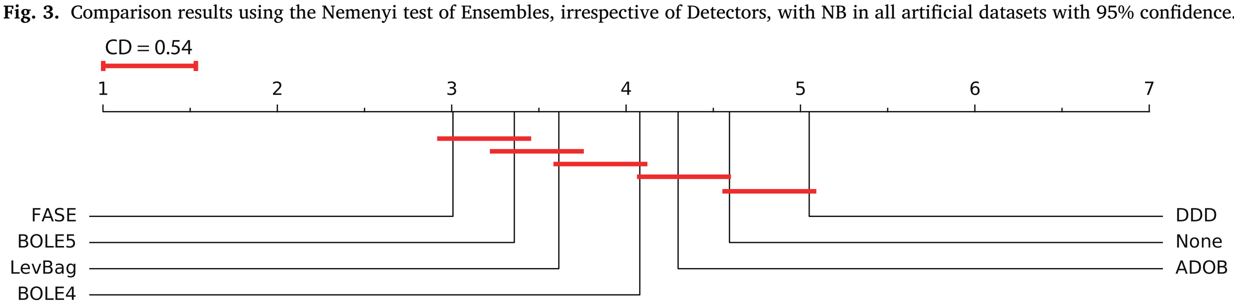

Fig. 4. Comparison results using the Nemenyi test of Ensembles, irrespective of Detectors, with HT in abrupt datasets with 95% confidence.

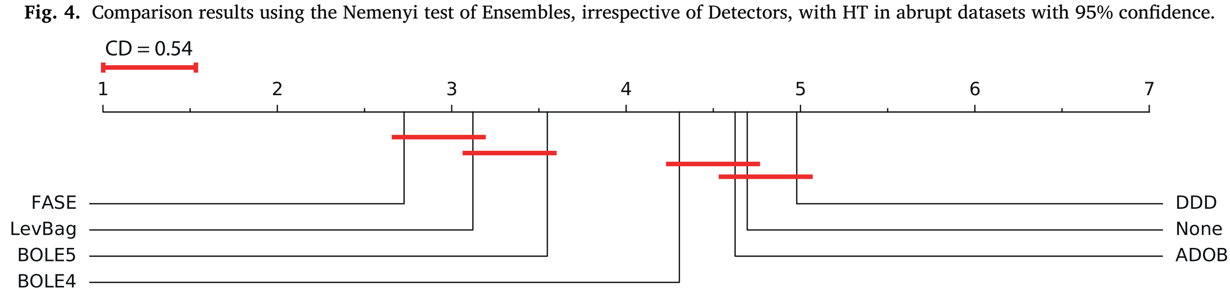

Fig. 5. Comparison results using the Nemenyi test of Ensembles, irrespective of Detectors, with HT in gradual datasets with 95% confidence.

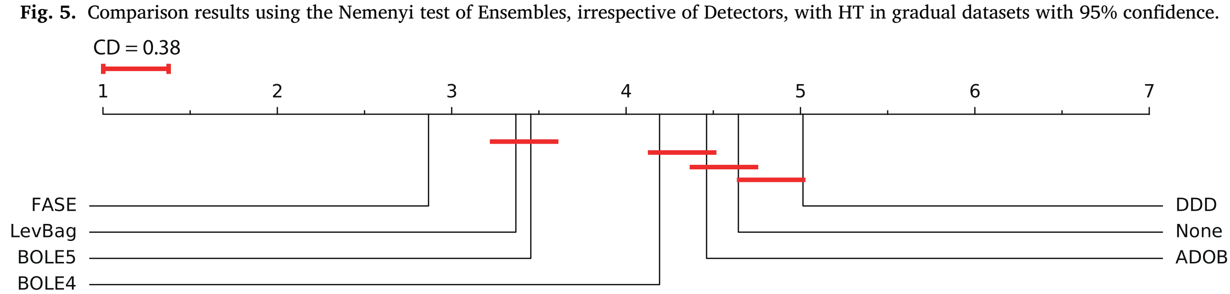

Fig. 6. Comparison results using the Nemenyi test of Ensembles, irrespective of Detectors, with HT in all artificial datasets with 95% confidence.

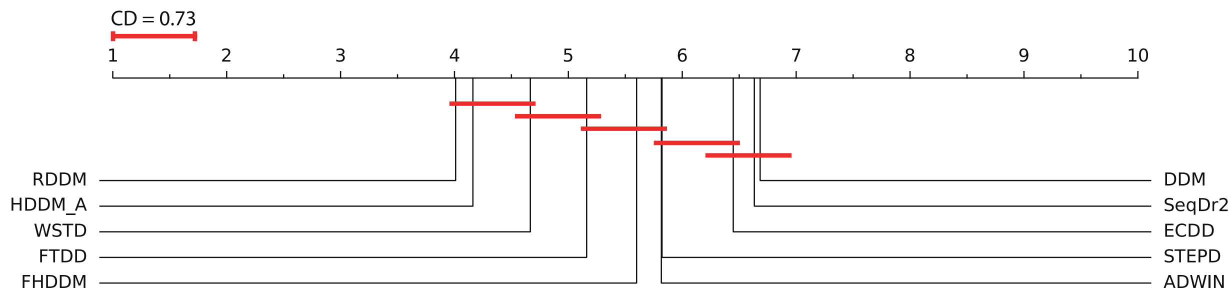

Fig. 7. Comparison results using the Nemenyi test of Detectors inside Ensembles with NB in the abrupt datasets with 95% confidence.

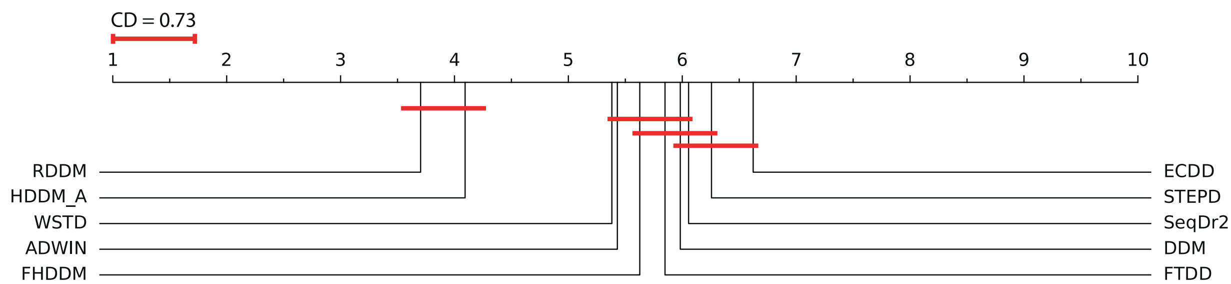

Fig. 8. Comparison results using the Nemenyi test of Detectors inside Ensembles with NB in the gradual datasets with 95% confidence.

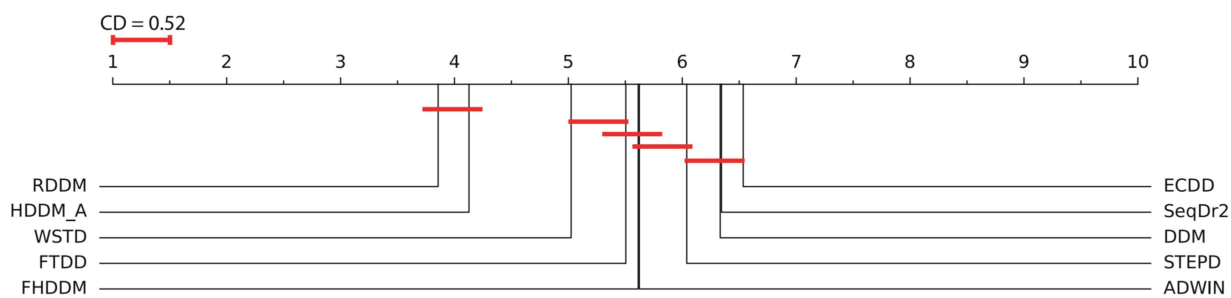

Fig. 9. Comparison results using the Nemenyi test of Detectors inside Ensembles with NB in all artificial datasets with 95% confidence.

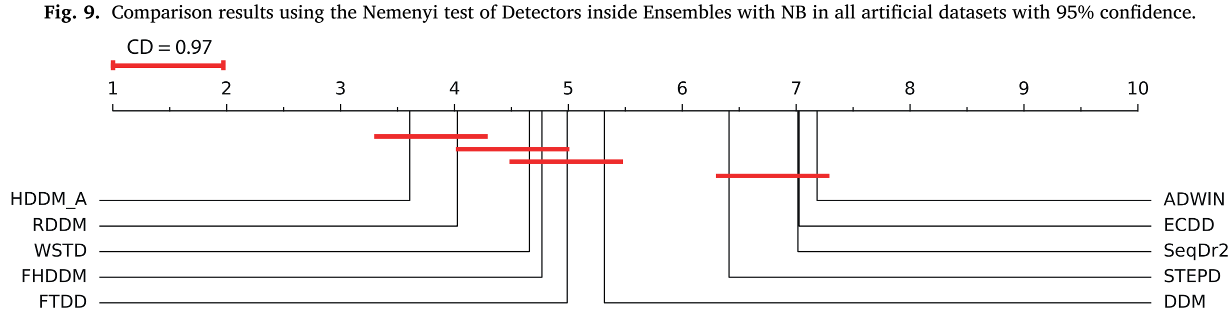

Fig. 10. Comparison results using the Nemenyi test of Detectors inside Ensembles with HT in the abrupt datasets with 95% confidence.

Fig. 9 evaluates the results of the methods aggregating *all* the tests using NB. In this larger view of the data, the best methods are obviously RDDM and HDDM_A, once again statistically similar. Also, observe this time the statistical differences in the ranks of these two detectors to the others are much bigger, well beyond the critical difference.

Figs. 10–12 represent the evaluations based on views similar to those of Figs. 7–9, respectively, but based on the tests using HT as base learner. Fig. 10 refers to the results of the experiments in the *abrupt* datasets. In this subset of the tests, HDDM_A and RDDM have the best ranks and are statistically similar, but only HDDM_A achieved significant superiority to the following three methods, WSTD, FHDDM, and FTDD.

Accordingly, Fig. 11 presents the evaluation in the *gradual* datasets using HT as base classifier. In these datasets, RDDM and HDDM_A were again the best methods with no statistical differences between them, but in this subset of the data, the order of their ranks was the opposite. In addition, this time only RDDM is statistically superior to FHDDM, the third best-ranked detector.

Finally, Fig. 12 captures the evaluation of the accuracy results aggregating *all* the tests using HT as base learner. Considering this subset of the data, the very best configurations were once again RDDM and HDDM_A, with very close ranks and no statistical difference. In addition, these two detectors were significantly superior to the remaining eight methods. Also note there was substantial difference from the following four methods, FHDDM, WSTD, DDM, and FTDD, to the last four methods.

4.1. Answer to research questions

This section examines and answers the seven research questions this work proposed to investigate, i.e., **RQ6** to **RQ12**. The description of **RQ6** was: *What are the best ensemble plus drift detector combinations in terms of final accuracy in abrupt and gradual concept drift datasets?*

Based on the experiments of this chapter, the answer to **RQ6** is: considerable variations happened in the results using the two base learners

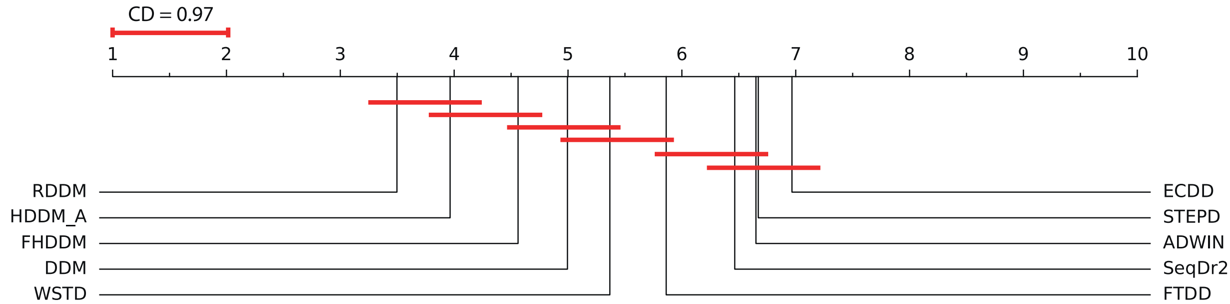

Fig. 11. Comparison results using the Nemenyi test of Detectors inside Ensembles with HT in the gradual datasets with 95% confidence.

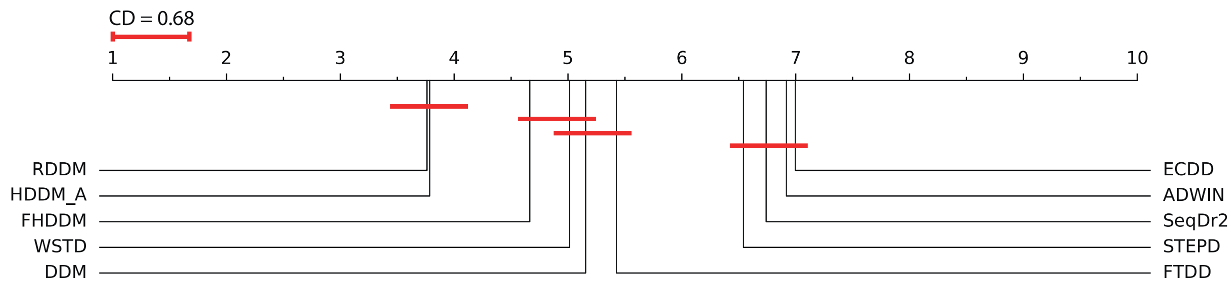

Fig. 12. Comparison results using the Nemenyi test of Detectors inside Ensembles with HT in all artificial datasets with 95% confidence.

(NB and HT), with a large number of statistical similarities among different configurations. Even so, using NB, the chosen ensemble should be BOLE$_5$ combined with either HDDM$_A$ or RDDM, irrespective of the type of concept drift. These two combinations were consistently among the best in most datasets.

Using HT, the chosen ensemble algorithm should be either FASE, LevBag, or BOLE$_5$, but there is no clear choice of detector. In datasets with *abrupt* concept drifts the safer choice is HDDM$_A$, combined with any of these ensembles, though FASE and BOLE$_5$ are probably safer choices. In the ones with *gradual* changes, there were several combinations of LevBag and FASE with distinct detectors that delivered close results. Should the type of concept drift be unknown, FASE is probably the safer choice with DDM, FHDDM, HDDM$_A$ or RDDM as concept drift detector.

The description of **RQ7** was: *What are the best ensembles in terms of accuracy in abrupt and gradual drift datasets irrespective of the auxiliary concept drift detector used?*

The answer to **RQ7** is simple: in all the tested scenarios using NB, BOLE$_5$ achieved the best rank in the vast majority of the datasets and was statistically superior to all the other algorithms, except for FASE. Thus, it is an easy choice. Similarly, using HT, in all the three aggregations, FASE delivered the best ranks in most datasets and is, therefore, declared the best choice, though LevBag and BOLE$_5$ were often statistically similar.

The description of **RQ8** was: *What are the best concept drift detectors as auxiliary methods in ensembles in terms of accuracy of the ensembles in abrupt and gradual concept drift datasets?*

The answer to **RQ8** is: using NB, RDDM is the safer choice since it was the best detector in all three scenarios, especially in the *gradual* datasets. HDDM$_A$ was a consistent second and could be safely considered in the case of *abrupt* datasets, despite being statistically indistinguishable from RDDM in all three scenarios.

Using HT, there were no statistical differences between RDDM and HDDM$_A$ in the three scenarios. However, based on their ranks, the best choices are probably HDDM$_A$, for *abrupt* datasets, and RDDM, for *gradual* datasets. In the tests with *all* the datasets, their ranks were really close: the difference is negligible. Accordingly, when the type of concept drift is unknown, they are both excellent choices.

The description of **RQ9** was: *Do the answers of **RQ6, RQ7**, and **RQ8** vary with the different dataset generators used in the experiments? How much?*

The answer to **RQ9** regarding the combinations ensemble-detector (**RQ6**) is *yes*, there were very considerable differences in the results of the ensembles in different dataset generators. In fact, in this scenario, combinations with BOLE$_5$ only dominated the best results in Sine, also delivering the best rank in another two of the seven dataset generators using NB, and in one generator using HT. Similarly, the dominance of FASE using HT was restricted to Agrawal$_1$ and Agrawal$_2$, the former also in the tests using NB. Notice that LevBag was quite dominant in three generators using HT and, surprisingly, ECDD was part of the best result in several versions of both Random RBF and Waveform datasets. In another surprising result, the *None* configurations (detectors without ensemble) were very competitive in the Mixed datasets using NB and, to some extent, also in Agrawal$_1$ and Agrawal$_2$ datasets using HT.

The strict answer to **RQ9** regarding the ensemble algorithms (**RQ7**) is *yes*, there were some variations in the results considering the different dataset generators. However, BOLE$_5$ was consistently among the best algorithms using NB in all generators, with the exception of Random RBF, as well as in three of the seven generators using HT. On the other hand, FASE was often *not* the best method, even using HT, but delivered reasonably good results in the majority of the dataset generators with both base learners. The stronger performances of LevBag were concentrated in the tests using HT, especially in the gradual datasets.

The answer to **RQ9** regarding the drift detectors inside ensembles (**RQ8**) is again *yes*, there were noticeable differences when the results of different dataset generators were separated. Even though RDDM and HDDM$_A$ consistently delivered good results, they did *not* dominate the ranks in the tests using any of the base learners. Noticeable dominant performances in specific dataset generators were only those of FTDD in the Random RBF datasets with both base learners and RDDM in the Sine datasets using NB.

The description of **RQ10** was: *Do the answers of **RQ6, RQ7**, and **RQ8** depend on the size of the concepts included in the datasets? How much?*

The answer to **RQ10** regarding the combinations ensemble-detector (**RQ6**) is *yes*, there are variations, but they are comparatively small. Using NB, the variations are limited to the 10K instances datasets. With HT, FASE combinations were the best only in the gradual 10K datasets; in most of the others, LevBag combinations achieved the best ranks. Nevertheless, it is worth emphasizing that virtually no statistically significant difference happened in most sizes.

Strictly speaking, the answer to **RQ10** regarding the ensemble algorithms (**RQ7**) would also be *yes*, but the variations were restricted to the 10K *abrupt* and *gradual* datasets using NB, where FASE was ranked in front of BOLE_5, and in the 100K *gradual* datasets using HT, where LevBag was ranked in front of FASE. Interestingly, in all the tests using NB, the orders of the ranks were all very similar irrespective of the size of the datasets.

The answer to **RQ10** regarding the drift detectors inside ensembles (**RQ8**) is again *yes*, there were substantial variations in the results of some detectors when the datasets were separated by size. FTDD and SeqDrift2 are the most affected ones, though ADWIN, ECDD, and FHDDM are also affected. When the size of the datasets increased, FTDD and SeqDrift2 improved their results dramatically with both NB and HT whereas ADWIN also improved but not as much as the others. On the other direction, ECDD and FHDDM became worse with the increase in the size of the datasets with both base learners.

The description of **RQ11** was: *In the same datasets, are the best ensembles of **RQ6** and **RQ7** the same?*

The answer to **RQ11** is definitely *yes*, in both **RQ6** and **RQ7** BOLE_5 was the best choice using NB and FASE was better using HT, despite not existing statistical differences in many scenarios and the strong performance of LevBag in the gradual datasets, especially using HT.

Finally, the description of **RQ12** was: *In the same datasets, are the best concept drift detectors of **RQ1, RQ6**, and **RQ8** the same? To what extent?*

Before answering **RQ12**, to enlighten the reader, lets show the description of **RQ1** answered in Barros and Santos [7]: *What are the best drift detectors in terms of accuracy in abrupt and gradual concept drift datasets?* Its answer basically said that RDDM and HDDM_A were the best detectors, with RDDM slightly better when using NB whereas HDDM_A was marginally superior with HT, and that these detectors were statistically indistinguishable.

The strict answer to **RQ12** is *no*, but note they were *not* very different either, since all scenarios included RDDM and HDDM_A, and in most of the evaluated scenarios they were the only recommended detectors.

4.2. Time complexity analysis

This subsection reviews the asymptotic performance of the ensembles and detectors used in the experiments of this article, ignoring the stream size. For this analysis, it is assumed that variable T corresponds to the size of ensemble.

Initially, regarding the ADOB and BOLE ensembles, both perform a sort operation on ensemble classifiers taking into account the current accuracy. This is the most costly operation of the methods and, for this reason, the time complexity of both is $\Theta(T^2)$ in the worst case scenario, which corresponds to the worst case of insertion sort [19], the sorting algorithm they use. However, as described in [8,54], in most iterations the ensemble members will already be ordered, which implies a time complexity $\Theta(T)$ for both ADOB and BOLE.

In addition to the iterations in T for the training of the T classifiers, DDD uses four ensembles simultaneously: two with low diversity and two with high diversity [42]. This means its time complexity is $O(4 \times T)$ or just $O(T)$. On the other hand, FASE and LevBag perform only one iteration in the ensemble for the training, in addition to their fixed-time operations. Thus, their time complexities are $O(T)$.

Finally, in the case of the detectors, most of them have constant time complexity. The exceptions are ADWIN and SeqDrift, which have logarithmic complexity [11,48], and RDDM, which has constant complexity in most cases [5].

5. Conclusion

This article reported on the extensive experiments designed to evaluate ensembles for data stream mining that are configurable with concept drift detectors. Moreover, it proposed to verify/challenge a common belief in the area, namely that the best ensembles which use auxiliary drift detectors deliver their best accuracy results when using the best concept drift detectors. As such, this work can be seen as a companion to our extensive empirical comparison and evaluation of concept drift detection methods previously published [7], which examined similar issues.

In addition, to analyze this matter, this article introduced and answered *seven* research questions. To answer these questions, a large-scale experiment was carried out to evaluate and compare *six* ensembles configurable with auxiliary concept drift detectors aimed at mining data streams containing concept drift. Moreover, to make this possible, the code of Leveraging Bagging ((LevBag)) [13], which is available in MOA and has ADWIN [11] hard-coded, was changed to permit its configuration with other drift detectors.

More specifically, these *six* ensembles were parametrized with *ten* selected concept drift detection methods, chosen from the ones that delivered the best results in the experiments of Barros and Santos [7], and the accuracies of these 60 combinations were compared among themselves and against the detectors individually. The results were the basis for answering the *seven* proposed research questions. Notice that the experiments reported here were based on and extended the ones reported by Barros [4].

It is worth mentioning that these large-scale experiments were run in the MOA framework [12], release 2014.11, using a very large number of artificial datasets, with *abrupt* and *gradual* concept drift versions of several sizes. Furthermore, these experiments were executed using two different base classifiers, namely Naive Bayes (NB) and Hoeffding Tree (HT). To the best of our knowledge, this is the largest comparison evaluation of its kind ever reported in the area of data stream mining.

The results of these large-scale experiments provide explicit indications of the best concept drift detectors as auxiliary methods inside ensembles; of the best ensemble algorithms, irrespective of drift detector adopted; as well as of the best ensemble-detector combinations. They also laid the basis for making it possible to analyze the influence of the *type of concept drift*, the *dataset generators*, and the *size of the concepts* on the performance of the methods.

As was to be expected, no single ensemble or drift detector is better than all the others in all situations. However, it should be emphasized that BOLE_5 and FASE were the best ensembles using NB and HT as base learners, respectively, whereas HDDM_A and RDDM were consistently among the very best concept drift detectors in terms of *accuracy* with both base learners.

To conclude, it is also important to stress that the answers to the research questions addressed in this article indicated that the aforementioned common belief partially matches the reality. Even though configurations of ECDD achieved the top accuracies in several datasets, especially when using HT as base classifier, overall, the best detectors of our previous study [7] (HDDM_A and RDDM) were the best choices to configure the ensembles.

Acknowledgments

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Appendix

This appendix includes Tables 8–14 which contain the detailed results of the experiments with the ensembles configurations in the gradual datasets using Naive Bayes (NB) as base classifier. It also shows the detailed results of the experiments with the ensembles configurations using Hoeffding Tree (HT) as base learner. Tables 15–18 contain the results in the abrupt datasets and Tables 19–22 the ones in the gradual datasets. All these results, omitted from the main body of the article, are separated by type of concept drift as well as size of the datasets.

Table 8
Mean accuracies of Ensembles in percentage (%) in 10K instances gradual datasets, with 95% confidence intervals, using NB.

Dataset	Ensemble	ADOB	$BOLE_4$	$BOLE_5$	DDD	FASE	LevBag	None
$Agraw_1$	ADWIN	60.54 ± 0.26	60.54 ± 0.26	60.60 ± 0.23	60.70 ± 0.22	62.37 ± 0.21	61.77 ± 0.18	60.69 ± 0.20
	DDM	60.57 ± 0.32	60.57 ± 0.32	60.72 ± 0.29	60.23 ± 0.50	62.36 ± 0.21	60.99 ± 0.22	60.56 ± 0.38
	ECDD	60.87 ± 0.23	60.87 ± 0.23	61.13 ± 0.22	61.04 ± 0.28	**62.67 ± 0.22**	60.28 ± 0.21	60.84 ± 0.26
	FHDDM	61.76 ± 0.23	61.76 ± 0.23	61.85 ± 0.22	61.32 ± 0.30	62.60 ± 0.18	62.13 ± 0.16	61.53 ± 0.30
	FTDD	60.33 ± 0.25	60.33 ± 0.25	60.31 ± 0.27	59.06 ± 0.47	62.10 ± 0.24	59.12 ± 0.31	59.27 ± 0.52
	$HDDM_A$	61.15 ± 0.27	61.14 ± 0.27	61.23 ± 0.27	60.83 ± 0.30	62.26 ± 0.23	62.22 ± 0.18	61.25 ± 0.34
	RDDM	61.55 ± 0.19	61.54 ± 0.19	61.76 ± 0.22	61.40 ± 0.36	62.64 ± 0.18	61.06 ± 0.20	62.05 ± 0.27
	SeqDr2	60.33 ± 0.26	60.33 ± 0.26	60.37 ± 0.26	60.08 ± 0.28	62.23 ± 0.20	58.78 ± 0.33	60.37 ± 0.30
	STEPD	61.45 ± 0.25	61.45 ± 0.25	61.55 ± 0.24	60.92 ± 0.30	62.43 ± 0.18	61.98 ± 0.17	61.28 ± 0.30
	WSTD	61.16 ± 0.30	61.16 ± 0.30	61.24 ± 0.29	60.68 ± 0.29	62.23 ± 0.21	60.92 ± 0.30	60.80 ± 0.30
$Agraw_2$	ADWIN	78.72 ± 0.26	78.72 ± 0.26	78.72 ± 0.26	75.09 ± 0.43	78.71 ± 0.25	77.28 ± 0.42	74.57 ± 0.85
	DDM	76.99 ± 0.31	76.99 ± 0.31	76.88 ± 0.34	68.92 ± 0.89	79.09 ± 0.25	78.94 ± 0.25	69.38 ± 1.23
	ECDD	78.90 ± 0.25	78.90 ± 0.25	78.93 ± 0.27	78.02 ± 0.31	79.42 ± 0.21	78.77 ± 0.24	77.03 ± 1.08
	FHDDM	79.61 ± 0.24	79.61 ± 0.24	**79.65 ± 0.23**	76.73 ± 0.76	79.14 ± 0.22	75.65 ± 0.39	77.33 ± 0.83
	FTDD	78.39 ± 0.30	78.39 ± 0.30	78.51 ± 0.26	71.42 ± 1.07	78.69 ± 0.22	70.52 ± 0.98	71.82 ± 1.30
	$HDDM_A$	79.02 ± 0.21	79.02 ± 0.22	78.96 ± 0.27	75.39 ± 0.55	78.86 ± 0.28	77.83 ± 0.31	76.12 ± 0.85
	RDDM	78.48 ± 0.27	78.48 ± 0.27	78.61 ± 0.26	73.52 ± 1.42	79.14 ± 0.25	78.98 ± 0.23	74.29 ± 1.44
	SeqDr2	78.36 ± 0.33	78.36 ± 0.33	78.24 ± 0.36	74.12 ± 0.63	78.73 ± 0.24	71.84 ± 0.45	73.44 ± 0.77
	STEPD	79.32 ± 0.21	79.32 ± 0.21	79.30 ± 0.22	77.02 ± 0.45	79.13 ± 0.24	77.73 ± 0.31	76.90 ± 0.89
	WSTD	79.21 ± 0.23	79.21 ± 0.23	79.16 ± 0.23	75.21 ± 0.88	78.95 ± 0.23	75.11 ± 0.73	75.61 ± 1.08
LED	ADWIN	66.09 ± 0.32	66.09 ± 0.32	66.16 ± 0.31	64.36 ± 0.56	66.59 ± 0.26	68.34 ± 0.28	62.58 ± 0.47
	DDM	67.18 ± 0.34	67.18 ± 0.34	67.22 ± 0.32	66.56 ± 0.59	67.04 ± 0.25	49.33 ± 0.41	67.78 ± 0.40
	ECDD	54.16 ± 2.55	54.16 ± 2.55	60.83 ± 0.82	66.56 ± 0.42	67.32 ± 0.26	57.50 ± 0.57	65.16 ± 0.40
	FHDDM	67.55 ± 0.28	67.55 ± 0.28	67.58 ± 0.28	67.00 ± 0.36	67.08 ± 0.27	67.57 ± 0.34	66.91 ± 0.35
	FTDD	59.43 ± 2.17	65.82 ± 0.36	65.03 ± 0.46	62.59 ± 0.93	66.31 ± 0.28	63.21 ± 0.81	63.11 ± 0.90
	$HDDM_A$	67.38 ± 0.27	67.42 ± 0.27	67.36 ± 0.27	66.99 ± 0.30	67.10 ± 0.27	**68.38 ± 0.28**	67.65 ± 0.30
	RDDM	67.50 ± 0.29	67.52 ± 0.28	67.52 ± 0.28	67.24 ± 0.28	67.02 ± 0.27	61.07 ± 0.51	67.85 ± 0.29
	SeqDr2	65.45 ± 0.29	65.45 ± 0.29	65.73 ± 0.30	64.27 ± 0.45	65.85 ± 0.32	63.31 ± 0.58	59.93 ± 0.69
	STEPD	65.08 ± 0.31	65.08 ± 0.31	65.29 ± 0.31	65.13 ± 0.55	64.94 ± 0.41	67.73 ± 0.35	59.51 ± 1.55
	WSTD	64.64 ± 1.29	66.24 ± 0.30	66.23 ± 0.30	65.50 ± 0.54	65.64 ± 0.30	66.40 ± 0.60	64.40 ± 0.72
Mixed	ADWIN	83.99 ± 0.21	83.99 ± 0.21	83.97 ± 0.20	83.57 ± 0.26	83.74 ± 0.22	82.71 ± 0.20	83.19 ± 0.27
	DDM	84.41 ± 0.19	84.41 ± 0.19	84.41 ± 0.20	81.96 ± 0.59	83.77 ± 0.27	84.17 ± 0.23	83.65 ± 0.28
	ECDD	82.15 ± 0.20	82.15 ± 0.20	82.13 ± 0.19	82.15 ± 0.50	83.52 ± 0.23	82.30 ± 0.54	83.02 ± 0.31
	FHDDM	83.59 ± 0.19	83.59 ± 0.19	83.56 ± 0.19	81.56 ± 0.44	83.82 ± 0.25	72.09 ± 1.29	84.13 ± 0.27
	FTDD	**84.45 ± 0.17**	**84.45 ± 0.17**	84.39 ± 0.19	80.82 ± 0.47	83.73 ± 0.21	75.20 ± 1.43	83.74 ± 0.24
	$HDDM_A$	83.88 ± 0.20	83.87 ± 0.20	83.90 ± 0.20	81.71 ± 0.48	83.69 ± 0.24	83.13 ± 0.31	83.61 ± 0.27
	RDDM	83.69 ± 0.20	83.69 ± 0.20	83.72 ± 0.20	83.41 ± 0.38	83.61 ± 0.26	83.97 ± 0.25	83.89 ± 0.29
	SeqDr2	83.65 ± 0.20	83.65 ± 0.20	83.65 ± 0.20	82.34 ± 0.44	83.61 ± 0.23	74.01 ± 0.39	83.48 ± 0.25
	STEPD	82.34 ± 0.22	82.34 ± 0.22	82.43 ± 0.20	83.20 ± 0.33	83.19 ± 0.23	80.33 ± 1.03	83.28 ± 0.32
	WSTD	83.40 ± 0.21	83.39 ± 0.21	83.41 ± 0.20	82.08 ± 0.51	83.61 ± 0.24	75.64 ± 1.21	83.42 ± 0.27
RBF	ADWIN	19.91 ± 0.86	19.91 ± 0.86	29.76 ± 0.88	30.36 ± 0.48	31.70 ± 0.38	30.88 ± 0.49	30.44 ± 0.51
	DDM	19.84 ± 0.87	19.84 ± 0.87	29.82 ± 0.95	30.83 ± 0.61	31.57 ± 0.35	29.69 ± 0.43	30.81 ± 0.61
	ECDD	19.84 ± 0.87	19.84 ± 0.87	29.65 ± 0.96	30.82 ± 0.63	31.78 ± 0.41	30.83 ± 0.62	30.91 ± 0.65
	FHDDM	19.71 ± 0.90	19.71 ± 0.90	29.92 ± 0.85	30.51 ± 0.60	31.43 ± 0.36	30.35 ± 0.48	29.95 ± 0.45
	FTDD	19.60 ± 0.79	24.86 ± 0.89	30.68 ± 0.65	30.76 ± 0.56	**31.79 ± 0.40**	31.05 ± 0.56	30.90 ± 0.56
	$HDDM_A$	19.85 ± 0.87	24.80 ± 0.85	30.52 ± 0.63	30.75 ± 0.52	31.61 ± 0.34	30.57 ± 0.47	30.55 ± 0.47
	RDDM	19.82 ± 0.86	24.93 ± 0.79	30.72 ± 0.53	30.63 ± 0.44	31.51 ± 0.30	29.79 ± 0.42	30.39 ± 0.44
	SeqDr2	19.66 ± 0.74	19.66 ± 0.74	29.61 ± 0.85	30.78 ± 0.58	31.66 ± 0.38	31.08 ± 0.59	30.91 ± 0.53
	STEPD	19.95 ± 0.82	19.95 ± 0.82	29.85 ± 0.76	30.27 ± 0.54	31.31 ± 0.40	30.98 ± 0.52	29.78 ± 0.51
	WSTD	19.93 ± 0.99	24.95 ± 0.85	30.36 ± 0.62	30.79 ± 0.62	31.66 ± 0.39	30.98 ± 0.61	30.73 ± 0.61
Sine	ADWIN	81.92 ± 0.20	81.92 ± 0.20	81.96 ± 0.20	81.22 ± 0.20	81.31 ± 0.21	80.50 ± 0.21	80.85 ± 0.19
	DDM	82.96 ± 0.17	82.96 ± 0.17	**82.98 ± 0.16**	79.48 ± 0.53	81.80 ± 0.20	81.79 ± 0.22	81.32 ± 0.27
	ECDD	80.89 ± 0.24	80.89 ± 0.24	80.94 ± 0.24	81.23 ± 0.26	81.67 ± 0.22	80.67 ± 0.40	81.20 ± 0.19
	FHDDM	82.25 ± 0.19	82.25 ± 0.19	82.28 ± 0.20	79.51 ± 0.39	81.95 ± 0.18	72.16 ± 0.80	82.01 ± 0.18
	FTDD	82.48 ± 0.15	82.51 ± 0.15	82.65 ± 0.15	79.50 ± 0.49	81.68 ± 0.20	68.59 ± 0.91	81.26 ± 0.20
	$HDDM_A$	82.46 ± 0.19	82.48 ± 0.19	82.48 ± 0.19	79.98 ± 0.51	81.71 ± 0.20	80.45 ± 0.35	81.51 ± 0.20
	RDDM	82.62 ± 0.18	82.64 ± 0.19	82.65 ± 0.18	81.40 ± 0.39	81.77 ± 0.21	81.89 ± 0.18	81.85 ± 0.18
	SeqDr2	81.89 ± 0.22	81.89 ± 0.22	81.96 ± 0.20	80.01 ± 0.36	81.22 ± 0.21	70.29 ± 0.36	81.15 ± 0.20
	STEPD	81.11 ± 0.22	81.11 ± 0.22	81.27 ± 0.21	80.67 ± 0.29	81.09 ± 0.20	77.24 ± 0.66	80.73 ± 0.24
	WSTD	81.87 ± 0.16	81.89 ± 0.16	81.97 ± 0.16	79.63 ± 0.35	81.60 ± 0.19	70.66 ± 1.09	81.32 ± 0.21
Wavef.	ADWIN	78.67 ± 0.38	78.67 ± 0.38	78.45 ± 0.38	77.61 ± 0.40	78.15 ± 0.37	77.88 ± 0.43	77.87 ± 0.40
	DDM	78.85 ± 0.37	78.85 ± 0.37	78.60 ± 0.38	77.86 ± 0.47	78.26 ± 0.36	78.94 ± 0.33	77.99 ± 0.43
	ECDD	**80.85 ± 0.29**	**80.85 ± 0.29**	80.84 ± 0.29	78.82 ± 0.38	79.37 ± 0.34	79.37 ± 0.36	78.06 ± 0.37
	FHDDM	79.96 ± 0.34	79.96 ± 0.34	79.85 ± 0.35	77.52 ± 0.40	78.28 ± 0.36	77.05 ± 0.36	77.78 ± 0.40
	FTDD	78.49 ± 0.41	78.49 ± 0.42	78.31 ± 0.41	76.54 ± 0.45	78.08 ± 0.34	76.64 ± 0.39	76.65 ± 0.46
	$HDDM_A$	79.10 ± 0.39	79.10 ± 0.40	78.84 ± 0.38	77.72 ± 0.48	78.19 ± 0.38	78.23 ± 0.37	77.82 ± 0.49
	RDDM	79.38 ± 0.37	79.38 ± 0.37	79.15 ± 0.38	78.36 ± 0.42	78.60 ± 0.34	79.39 ± 0.36	78.46 ± 0.37
	SeqDr2	78.77 ± 0.44	78.77 ± 0.44	78.58 ± 0.42	77.71 ± 0.43	78.20 ± 0.38	76.80 ± 0.39	77.74 ± 0.42
	STEPD	80.02 ± 0.38	80.02 ± 0.38	79.88 ± 0.38	78.24 ± 0.40	78.71 ± 0.40	78.60 ± 0.36	78.33 ± 0.38
	WSTD	79.18 ± 0.39	79.18 ± 0.39	78.94 ± 0.38	77.31 ± 0.46	78.17 ± 0.34	76.96 ± 0.43	77.54 ± 0.54

Table 9
Mean accuracies of Ensembles in percentage (%) in 20K instances gradual datasets, with 95% confidence intervals, using NB.

Dataset	Ensemble	ADOB	BOLE$_4$	BOLE$_5$	DDD	FASE	LevBag	None
Agraw$_1$	ADWIN	63.46 ± 0.19	63.46 ± 0.19	63.70 ± 0.18	63.09 ± 0.19	64.37 ± 0.13	63.97 ± 0.13	63.62 ± 0.20
	DDM	62.83 ± 0.29	62.83 ± 0.29	62.97 ± 0.29	62.11 ± 0.57	64.38 ± 0.12	62.21 ± 0.21	62.62 ± 0.51
	ECDD	62.50 ± 0.18	62.50 ± 0.18	62.78 ± 0.17	62.12 ± 0.18	64.10 ± 0.13	60.99 ± 0.14	61.85 ± 0.13
	FHDDM	63.53 ± 0.14	63.53 ± 0.14	63.58 ± 0.13	63.33 ± 0.15	64.42 ± 0.10	63.85 ± 0.13	63.40 ± 0.14
	FTDD	61.96 ± 0.20	61.96 ± 0.21	62.01 ± 0.20	60.77 ± 0.42	64.37 ± 0.12	59.40 ± 0.35	61.14 ± 0.35
	HDDM$_A$	63.98 ± 0.22	63.98 ± 0.22	64.24 ± 0.22	63.49 ± 0.18	64.43 ± 0.11	64.04 ± 0.12	63.92 ± 0.14
	RDDM	64.29 ± 0.21	64.29 ± 0.21	**64.58 ± 0.21**	63.55 ± 0.12	64.49 ± 0.12	62.34 ± 0.15	63.98 ± 0.13
	SeqDr2	62.37 ± 0.30	62.37 ± 0.30	62.42 ± 0.29	62.09 ± 0.32	64.37 ± 0.12	59.89 ± 0.25	62.70 ± 0.41
	STEPD	63.72 ± 0.15	63.72 ± 0.15	63.83 ± 0.16	63.30 ± 0.20	64.39 ± 0.12	63.72 ± 0.16	63.31 ± 0.23
	WSTD	63.84 ± 0.18	63.84 ± 0.18	63.95 ± 0.17	62.71 ± 0.27	64.39 ± 0.11	62.52 ± 0.33	63.15 ± 0.41
Agraw$_2$	ADWIN	83.00 ± 0.14	83.00 ± 0.14	82.93 ± 0.15	79.81 ± 0.27	82.62 ± 0.17	81.70 ± 0.22	79.46 ± 0.60
	DDM	81.44 ± 0.15	81.44 ± 0.15	81.46 ± 0.15	76.02 ± 1.02	82.75 ± 0.16	82.37 ± 0.11	75.89 ± 1.02
	ECDD	82.99 ± 0.12	82.99 ± 0.12	83.05 ± 0.12	81.97 ± 0.17	82.90 ± 0.12	82.17 ± 0.13	81.02 ± 0.92
	FHDDM	83.72 ± 0.11	83.72 ± 0.11	**83.75 ± 0.12**	80.74 ± 0.35	82.90 ± 0.11	79.09 ± 0.40	81.07 ± 0.97
	FTDD	83.00 ± 0.12	83.00 ± 0.13	82.99 ± 0.13	78.72 ± 0.41	82.57 ± 0.19	75.79 ± 0.83	78.79 ± 0.60
	HDDM$_A$	83.27 ± 0.14	83.27 ± 0.14	83.25 ± 0.18	79.44 ± 0.53	82.64 ± 0.17	81.49 ± 0.26	80.47 ± 0.51
	RDDM	82.73 ± 0.17	82.73 ± 0.17	82.90 ± 0.18	80.12 ± 0.81	82.87 ± 0.15	82.46 ± 0.12	80.79 ± 0.91
	SeqDr2	82.86 ± 0.19	82.86 ± 0.19	82.77 ± 0.23	79.27 ± 0.21	82.59 ± 0.17	74.85 ± 0.56	79.26 ± 0.33
	STEPD	83.36 ± 0.11	83.36 ± 0.11	83.36 ± 0.13	81.23 ± 0.39	82.81 ± 0.14	81.25 ± 0.38	81.64 ± 0.38
	WSTD	83.44 ± 0.12	83.45 ± 0.12	83.44 ± 0.13	79.93 ± 0.92	82.70 ± 0.12	78.51 ± 0.63	79.76 ± 1.35
LED	ADWIN	69.10 ± 0.21	69.10 ± 0.21	69.20 ± 0.22	68.02 ± 0.34	69.65 ± 0.22	70.99 ± 0.16	64.84 ± 0.58
	DDM	70.36 ± 0.17	70.36 ± 0.17	70.36 ± 0.17	69.62 ± 0.25	70.19 ± 0.17	50.51 ± 0.31	70.54 ± 0.19
	ECDD	69.72 ± 1.36	69.72 ± 1.36	70.44 ± 0.22	69.15 ± 0.29	70.29 ± 0.18	59.12 ± 0.39	67.21 ± 0.43
	FHDDM	70.43 ± 0.21	70.43 ± 0.21	70.47 ± 0.21	70.01 ± 0.23	70.32 ± 0.17	70.53 ± 0.23	69.81 ± 0.34
	FTDD	68.10 ± 1.32	69.53 ± 0.24	69.20 ± 0.29	67.02 ± 0.73	69.81 ± 0.15	67.26 ± 0.62	67.66 ± 0.87
	HDDM$_A$	70.51 ± 0.18	70.52 ± 0.18	70.53 ± 0.18	69.61 ± 0.16	70.24 ± 0.15	**71.00 ± 0.18**	70.43 ± 0.18
	RDDM	70.50 ± 0.17	70.51 ± 0.17	70.53 ± 0.18	70.17 ± 0.26	70.29 ± 0.17	62.82 ± 0.42	70.66 ± 0.18
	SeqDr2	68.65 ± 0.24	68.65 ± 0.24	68.86 ± 0.22	68.82 ± 0.27	69.05 ± 0.25	65.51 ± 0.57	62.53 ± 0.63
	STEPD	67.72 ± 0.29	67.72 ± 0.29	67.91 ± 0.31	69.05 ± 0.30	68.69 ± 0.24	70.29 ± 0.27	64.11 ± 1.41
	WSTD	69.53 ± 0.27	69.63 ± 0.24	69.69 ± 0.22	69.55 ± 0.27	69.19 ± 0.18	69.71 ± 0.38	68.68 ± 0.52
Mixed	ADWIN	87.81 ± 0.15	87.81 ± 0.15	87.79 ± 0.16	87.63 ± 0.16	87.82 ± 0.13	87.23 ± 0.17	87.46 ± 0.15
	DDM	88.46 ± 0.12	88.46 ± 0.12	**88.47 ± 0.13**	85.94 ± 0.51	87.87 ± 0.16	88.05 ± 0.15	87.85 ± 0.17
	ECDD	86.08 ± 0.16	86.08 ± 0.16	86.00 ± 0.17	87.28 ± 0.18	87.72 ± 0.15	86.82 ± 0.38	86.84 ± 0.19
	FHDDM	87.65 ± 0.15	87.65 ± 0.15	87.64 ± 0.16	85.39 ± 0.51	87.96 ± 0.16	75.51 ± 1.38	88.19 ± 0.18
	FTDD	88.45 ± 0.12	88.45 ± 0.12	88.44 ± 0.12	85.17 ± 0.37	87.84 ± 0.14	76.57 ± 1.68	87.63 ± 0.16
	HDDM$_A$	88.00 ± 0.10	88.00 ± 0.10	88.00 ± 0.11	85.59 ± 0.46	87.89 ± 0.15	87.10 ± 0.20	87.80 ± 0.18
	RDDM	87.92 ± 0.13	87.92 ± 0.13	87.94 ± 0.14	87.29 ± 0.41	87.81 ± 0.16	87.98 ± 0.16	88.01 ± 0.18
	SeqDr2	88.02 ± 0.13	88.02 ± 0.13	88.03 ± 0.13	86.72 ± 0.34	87.86 ± 0.13	79.78 ± 0.70	87.69 ± 0.17
	STEPD	86.75 ± 0.11	86.75 ± 0.11	86.77 ± 0.12	87.54 ± 0.18	87.64 ± 0.13	84.07 ± 0.94	87.50 ± 0.16
	WSTD	87.68 ± 0.13	87.67 ± 0.13	87.68 ± 0.13	85.99 ± 0.37	87.82 ± 0.14	75.97 ± 1.46	87.71 ± 0.16
RBF	ADWIN	19.60 ± 0.69	19.60 ± 0.69	30.33 ± 0.73	30.46 ± 0.41	31.99 ± 0.29	31.20 ± 0.47	30.65 ± 0.44
	DDM	19.60 ± 0.73	19.60 ± 0.73	30.25 ± 0.71	30.82 ± 0.56	31.81 ± 0.29	29.55 ± 0.32	30.89 ± 0.51
	ECDD	19.61 ± 0.74	19.61 ± 0.74	30.11 ± 0.75	31.14 ± 0.62	32.02 ± 0.32	31.23 ± 0.59	31.27 ± 0.61
	FHDDM	19.49 ± 0.66	19.49 ± 0.66	30.02 ± 0.73	30.03 ± 0.33	31.65 ± 0.26	30.42 ± 0.39	29.56 ± 0.40
	FTDD	19.49 ± 0.61	23.67 ± 0.65	30.95 ± 0.62	31.06 ± 0.50	**32.05 ± 0.33**	31.44 ± 0.53	31.26 ± 0.45
	HDDM$_A$	19.56 ± 0.70	23.78 ± 0.63	30.65 ± 0.39	30.74 ± 0.50	31.89 ± 0.32	30.63 ± 0.36	30.68 ± 0.44
	RDDM	19.57 ± 0.73	23.79 ± 0.60	30.82 ± 0.40	30.59 ± 0.38	31.86 ± 0.27	29.76 ± 0.32	30.53 ± 0.47
	SeqDr2	19.27 ± 0.61	19.27 ± 0.61	30.21 ± 0.69	31.08 ± 0.56	32.00 ± 0.32	31.42 ± 0.57	31.04 ± 0.46
	STEPD	19.74 ± 0.70	19.74 ± 0.70	29.92 ± 0.70	29.92 ± 0.39	31.57 ± 0.28	31.10 ± 0.54	29.70 ± 0.41
	WSTD	19.59 ± 0.92	23.80 ± 0.69	30.42 ± 0.56	31.05 ± 0.59	31.90 ± 0.33	31.32 ± 0.56	30.78 ± 0.57
Sine	ADWIN	85.67 ± 0.15	85.67 ± 0.15	85.72 ± 0.15	84.30 ± 0.19	84.38 ± 0.17	84.09 ± 0.17	84.22 ± 0.17
	DDM	86.76 ± 0.12	86.76 ± 0.12	**86.79 ± 0.11**	81.98 ± 0.45	84.96 ± 0.15	84.58 ± 0.20	84.64 ± 0.20
	ECDD	84.90 ± 0.17	84.90 ± 0.17	84.98 ± 0.17	84.55 ± 0.20	84.95 ± 0.15	84.12 ± 0.20	84.17 ± 0.15
	FHDDM	86.23 ± 0.15	86.23 ± 0.15	86.26 ± 0.14	82.24 ± 0.39	84.99 ± 0.17	76.18 ± 0.88	85.07 ± 0.16
	FTDD	86.29 ± 0.16	86.30 ± 0.16	86.35 ± 0.14	82.17 ± 0.40	84.79 ± 0.16	70.59 ± 1.06	84.74 ± 0.17
	HDDM$_A$	86.49 ± 0.14	86.50 ± 0.15	86.51 ± 0.14	82.07 ± 0.35	84.97 ± 0.16	83.64 ± 0.19	84.97 ± 0.15
	RDDM	86.57 ± 0.14	86.58 ± 0.14	86.58 ± 0.13	83.93 ± 0.46	84.93 ± 0.17	84.96 ± 0.16	84.98 ± 0.15
	SeqDr2	85.93 ± 0.13	85.93 ± 0.13	85.96 ± 0.12	83.35 ± 0.38	84.37 ± 0.18	73.84 ± 0.56	84.30 ± 0.19
	STEPD	85.27 ± 0.18	85.27 ± 0.18	85.34 ± 0.16	84.18 ± 0.23	84.46 ± 0.16	80.98 ± 0.85	84.36 ± 0.19
	WSTD	86.05 ± 0.13	86.06 ± 0.13	86.05 ± 0.12	82.75 ± 0.40	84.82 ± 0.18	73.19 ± 0.71	84.60 ± 0.16
Wavef.	ADWIN	80.01 ± 0.21	80.01 ± 0.21	79.67 ± 0.24	78.61 ± 0.26	78.95 ± 0.23	78.80 ± 0.24	78.68 ± 0.26
	DDM	79.64 ± 0.27	79.64 ± 0.27	79.37 ± 0.24	78.55 ± 0.29	79.17 ± 0.25	79.44 ± 0.18	78.46 ± 0.29
	ECDD	**81.47 ± 0.19**	**81.47 ± 0.19**	81.46 ± 0.19	79.38 ± 0.23	80.07 ± 0.21	79.89 ± 0.21	78.53 ± 0.23
	FHDDM	80.86 ± 0.20	80.86 ± 0.20	80.81 ± 0.20	78.45 ± 0.25	79.30 ± 0.24	77.82 ± 0.26	78.78 ± 0.24
	FTDD	79.18 ± 0.34	79.20 ± 0.34	78.92 ± 0.30	77.73 ± 0.35	79.14 ± 0.24	77.15 ± 0.25	78.15 ± 0.41
	HDDM$_A$	80.24 ± 0.25	80.24 ± 0.25	80.13 ± 0.26	78.63 ± 0.25	79.06 ± 0.25	78.88 ± 0.24	78.90 ± 0.30
	RDDM	80.40 ± 0.25	80.40 ± 0.25	80.27 ± 0.25	79.06 ± 0.29	79.62 ± 0.23	79.95 ± 0.22	79.22 ± 0.31
	SeqDr2	79.97 ± 0.24	79.97 ± 0.24	79.65 ± 0.25	78.50 ± 0.24	79.04 ± 0.24	77.31 ± 0.23	78.66 ± 0.27
	STEPD	80.94 ± 0.25	80.94 ± 0.25	80.86 ± 0.25	79.31 ± 0.25	79.80 ± 0.23	79.40 ± 0.24	79.31 ± 0.26
	WSTD	80.21 ± 0.23	80.21 ± 0.23	80.10 ± 0.24	78.41 ± 0.34	79.15 ± 0.21	77.73 ± 0.29	78.73 ± 0.28

Table 10
Mean accuracies of Ensembles in percentage (%) in 50K instances gradual datasets, with 95% confidence intervals, using NB.

Dataset	Ensemble	ADOB	BOLE$_4$	BOLE$_5$	DDD	FASE	LevBag	None
Agraw$_1$	ADWIN	66.93 ± 0.12	66.93 ± 0.12	67.11 ± 0.12	65.23 ± 0.12	65.62 ± 0.11	65.35 ± 0.12	65.41 ± 0.11
	DDM	65.12 ± 0.26	65.12 ± 0.26	65.40 ± 0.30	63.84 ± 0.43	65.70 ± 0.10	63.09 ± 0.18	63.92 ± 0.57
	ECDD	63.76 ± 0.11	63.76 ± 0.11	64.01 ± 0.11	62.90 ± 0.12	65.07 ± 0.10	61.71 ± 0.09	62.53 ± 0.11
	FHDDM	64.58 ± 0.11	64.58 ± 0.11	64.59 ± 0.11	64.76 ± 0.14	65.63 ± 0.11	65.12 ± 0.09	64.68 ± 0.15
	FTDD	64.79 ± 0.25	64.79 ± 0.25	64.95 ± 0.27	62.40 ± 0.41	65.62 ± 0.11	59.97 ± 0.33	62.87 ± 0.42
	HDDM$_A$	67.15 ± 0.15	67.14 ± 0.15	**67.34 ± 0.15**	64.90 ± 0.17	65.68 ± 0.09	65.42 ± 0.10	65.43 ± 0.11
	RDDM	66.92 ± 0.13	66.92 ± 0.13	67.18 ± 0.12	64.98 ± 0.17	65.71 ± 0.10	63.09 ± 0.11	65.38 ± 0.11
	SeqDr2	65.90 ± 0.17	65.90 ± 0.17	66.27 ± 0.17	64.51 ± 0.21	65.64 ± 0.11	61.09 ± 0.38	65.34 ± 0.12
	STEPD	65.27 ± 0.13	65.27 ± 0.13	65.29 ± 0.14	64.77 ± 0.14	65.60 ± 0.10	64.90 ± 0.12	64.77 ± 0.14
	WSTD	66.40 ± 0.13	66.40 ± 0.13	66.50 ± 0.14	64.78 ± 0.17	65.66 ± 0.10	63.52 ± 0.26	65.17 ± 0.14
Agraw$_2$	ADWIN	86.07 ± 0.09	86.07 ± 0.09	86.05 ± 0.09	84.20 ± 0.27	85.03 ± 0.12	84.80 ± 0.13	84.21 ± 0.30
	DDM	85.11 ± 0.13	85.11 ± 0.13	85.16 ± 0.16	82.23 ± 0.79	85.14 ± 0.11	84.43 ± 0.07	82.41 ± 0.96
	ECDD	85.35 ± 0.07	85.35 ± 0.07	85.47 ± 0.07	83.97 ± 0.23	84.83 ± 0.08	84.14 ± 0.08	83.53 ± 0.09
	FHDDM	86.27 ± 0.04	86.27 ± 0.04	86.31 ± 0.05	83.54 ± 0.16	85.14 ± 0.10	81.18 ± 0.41	84.81 ± 0.12
	FTDD	86.06 ± 0.11	86.06 ± 0.11	86.06 ± 0.12	81.89 ± 0.48	85.02 ± 0.13	78.17 ± 0.60	82.75 ± 0.51
	HDDM$_A$	86.26 ± 0.08	86.26 ± 0.08	86.31 ± 0.09	83.44 ± 0.31	85.06 ± 0.13	84.31 ± 0.22	84.57 ± 0.31
	RDDM	85.99 ± 0.12	86.00 ± 0.12	86.08 ± 0.12	84.16 ± 0.33	85.13 ± 0.10	84.63 ± 0.09	84.77 ± 0.22
	SeqDr2	86.08 ± 0.08	86.08 ± 0.08	86.07 ± 0.08	82.69 ± 0.39	85.03 ± 0.12	78.25 ± 0.66	82.98 ± 0.45
	STEPD	86.12 ± 0.07	86.12 ± 0.07	86.14 ± 0.08	84.48 ± 0.20	85.11 ± 0.09	84.12 ± 0.30	84.31 ± 0.21
	WSTD	86.47 ± 0.08	86.48 ± 0.08	**86.49 ± 0.09**	83.22 ± 0.31	85.07 ± 0.12	81.01 ± 0.63	83.90 ± 0.92
LED	ADWIN	71.05 ± 0.18	71.05 ± 0.18	71.11 ± 0.18	70.92 ± 0.24	71.82 ± 0.15	72.59 ± 0.16	67.10 ± 0.61
	DDM	72.30 ± 0.17	72.30 ± 0.17	72.30 ± 0.17	71.39 ± 0.52	72.36 ± 0.14	51.28 ± 0.19	72.30 ± 0.26
	ECDD	72.21 ± 0.16	72.21 ± 0.16	72.23 ± 0.16	70.70 ± 0.23	72.26 ± 0.17	60.09 ± 0.23	68.35 ± 0.33
	FHDDM	72.37 ± 0.16	72.37 ± 0.16	72.39 ± 0.16	72.02 ± 0.20	72.46 ± 0.14	72.47 ± 0.16	71.68 ± 0.26
	FTDD	71.85 ± 0.15	71.86 ± 0.15	71.84 ± 0.15	70.73 ± 0.19	72.08 ± 0.16	69.27 ± 0.67	71.62 ± 0.19
	HDDM$_A$	72.49 ± 0.16	72.49 ± 0.16	72.50 ± 0.16	71.27 ± 0.19	72.42 ± 0.14	72.61 ± 0.16	72.48 ± 0.15
	RDDM	72.48 ± 0.15	72.48 ± 0.15	72.49 ± 0.15	71.95 ± 0.23	72.47 ± 0.14	63.75 ± 0.26	**72.63 ± 0.15**
	SeqDr2	71.38 ± 0.16	71.38 ± 0.16	71.44 ± 0.15	71.24 ± 0.18	71.61 ± 0.14	66.81 ± 0.71	67.25 ± 0.89
	STEPD	70.11 ± 0.27	70.11 ± 0.27	70.21 ± 0.28	71.34 ± 0.32	71.63 ± 0.18	71.89 ± 0.24	68.17 ± 0.80
	WSTD	72.01 ± 0.17	72.02 ± 0.17	72.06 ± 0.17	71.55 ± 0.20	71.95 ± 0.15	70.69 ± 0.53	71.48 ± 0.28
Mixed	ADWIN	89.97 ± 0.12	89.97 ± 0.12	90.09 ± 0.11	90.29 ± 0.10	90.39 ± 0.10	90.05 ± 0.09	90.22 ± 0.11
	DDM	90.65 ± 0.13	90.65 ± 0.13	**90.74 ± 0.11**	87.41 ± 0.63	90.41 ± 0.09	90.33 ± 0.12	90.42 ± 0.11
	ECDD	88.21 ± 0.11	88.21 ± 0.11	88.24 ± 0.12	89.41 ± 0.14	90.16 ± 0.10	89.61 ± 0.17	88.78 ± 0.15
	FHDDM	89.93 ± 0.14	89.93 ± 0.14	89.97 ± 0.11	87.59 ± 0.52	90.45 ± 0.09	76.13 ± 1.35	90.52 ± 0.10
	FTDD	90.67 ± 0.10	90.67 ± 0.09	90.72 ± 0.10	88.63 ± 0.43	90.40 ± 0.09	78.68 ± 1.23	90.42 ± 0.11
	HDDM$_A$	90.31 ± 0.14	90.31 ± 0.14	90.46 ± 0.11	88.85 ± 0.47	90.44 ± 0.10	89.79 ± 0.17	90.45 ± 0.11
	RDDM	90.23 ± 0.12	90.23 ± 0.12	90.37 ± 0.10	90.11 ± 0.34	90.42 ± 0.10	90.43 ± 0.10	90.50 ± 0.09
	SeqDr2	90.27 ± 0.12	90.27 ± 0.12	90.39 ± 0.12	89.93 ± 0.16	90.39 ± 0.10	82.29 ± 0.58	90.28 ± 0.11
	STEPD	89.23 ± 0.13	89.23 ± 0.13	89.42 ± 0.10	90.26 ± 0.10	90.34 ± 0.09	87.39 ± 0.98	90.12 ± 0.11
	WSTD	90.05 ± 0.14	90.05 ± 0.14	90.17 ± 0.12	89.25 ± 0.33	90.40 ± 0.09	77.22 ± 2.03	90.40 ± 0.10
RBF	ADWIN	19.50 ± 0.64	19.50 ± 0.64	30.00 ± 0.59	30.59 ± 0.35	32.14 ± 0.25	31.32 ± 0.37	30.59 ± 0.34
	DDM	19.56 ± 0.69	19.56 ± 0.69	30.47 ± 0.61	30.96 ± 0.52	32.23 ± 0.22	29.66 ± 0.21	30.94 ± 0.49
	ECDD	19.56 ± 0.70	19.56 ± 0.70	30.29 ± 0.65	31.29 ± 0.66	**32.54 ± 0.28**	31.26 ± 0.65	31.31 ± 0.66
	FHDDM	19.27 ± 0.59	19.27 ± 0.59	29.70 ± 0.63	29.80 ± 0.36	31.67 ± 0.18	30.13 ± 0.24	29.61 ± 0.35
	FTDD	19.45 ± 0.60	23.08 ± 0.45	31.07 ± 0.45	31.25 ± 0.52	32.47 ± 0.26	31.49 ± 0.59	31.00 ± 0.49
	HDDM$_A$	19.44 ± 0.66	23.26 ± 0.54	30.80 ± 0.36	30.95 ± 0.41	32.25 ± 0.24	30.65 ± 0.25	30.92 ± 0.37
	RDDM	19.40 ± 0.66	23.15 ± 0.56	31.04 ± 0.32	30.54 ± 0.31	32.15 ± 0.20	29.74 ± 0.21	30.81 ± 0.35
	SeqDr2	19.03 ± 0.53	19.03 ± 0.53	30.05 ± 0.66	31.13 ± 0.54	32.45 ± 0.26	31.42 ± 0.61	31.12 ± 0.48
	STEPD	19.60 ± 0.58	19.60 ± 0.58	29.60 ± 0.59	29.50 ± 0.27	31.62 ± 0.20	31.00 ± 0.43	29.32 ± 0.30
	WSTD	19.08 ± 0.66	23.12 ± 0.57	30.10 ± 0.57	30.71 ± 0.51	32.22 ± 0.26	31.26 ± 0.67	30.43 ± 0.53
Sine	ADWIN	88.06 ± 0.12	88.06 ± 0.12	88.19 ± 0.11	86.24 ± 0.10	86.26 ± 0.11	86.22 ± 0.12	86.16 ± 0.10
	DDM	88.95 ± 0.11	88.95 ± 0.11	**89.04 ± 0.10**	84.58 ± 0.46	86.69 ± 0.11	86.02 ± 0.14	86.31 ± 0.26
	ECDD	86.91 ± 0.12	86.91 ± 0.12	87.00 ± 0.12	86.21 ± 0.11	86.65 ± 0.09	86.13 ± 0.10	85.62 ± 0.12
	FHDDM	88.63 ± 0.13	88.63 ± 0.13	88.68 ± 0.12	84.18 ± 0.52	86.66 ± 0.11	76.81 ± 0.92	86.76 ± 0.11
	FTDD	88.87 ± 0.09	88.87 ± 0.09	88.92 ± 0.09	84.42 ± 0.33	86.57 ± 0.11	71.88 ± 1.15	86.58 ± 0.11
	HDDM$_A$	88.81 ± 0.12	88.81 ± 0.12	88.88 ± 0.11	84.51 ± 0.54	86.69 ± 0.11	85.86 ± 0.23	86.76 ± 0.10
	RDDM	88.75 ± 0.13	88.75 ± 0.13	88.82 ± 0.11	86.30 ± 0.25	86.71 ± 0.11	86.59 ± 0.12	86.78 ± 0.11
	SeqDr2	88.39 ± 0.13	88.39 ± 0.13	88.50 ± 0.11	85.94 ± 0.11	86.28 ± 0.11	75.74 ± 0.45	86.22 ± 0.12
	STEPD	87.76 ± 0.14	87.76 ± 0.14	87.87 ± 0.13	86.26 ± 0.14	86.43 ± 0.11	83.59 ± 0.81	86.24 ± 0.12
	WSTD	88.56 ± 0.11	88.56 ± 0.11	88.64 ± 0.11	84.76 ± 0.42	86.60 ± 0.11	71.88 ± 0.84	86.63 ± 0.11
Wavef.	ADWIN	81.02 ± 0.16	81.02 ± 0.16	80.97 ± 0.17	79.85 ± 0.17	80.18 ± 0.13	79.78 ± 0.18	80.00 ± 0.13
	DDM	80.66 ± 0.16	80.66 ± 0.16	80.59 ± 0.17	79.33 ± 0.20	80.17 ± 0.13	79.85 ± 0.14	79.51 ± 0.21
	ECDD	**81.89 ± 0.13**	**81.89 ± 0.13**	**81.89 ± 0.13**	79.87 ± 0.14	80.61 ± 0.14	80.31 ± 0.14	78.96 ± 0.16
	FHDDM	81.53 ± 0.12	81.53 ± 0.12	81.50 ± 0.12	79.46 ± 0.18	80.25 ± 0.12	78.59 ± 0.19	79.75 ± 0.18
	FTDD	80.62 ± 0.12	80.62 ± 0.12	80.47 ± 0.14	79.09 ± 0.22	80.14 ± 0.14	77.84 ± 0.23	79.42 ± 0.20
	HDDM$_A$	81.15 ± 0.15	81.15 ± 0.14	81.12 ± 0.15	79.75 ± 0.18	80.20 ± 0.13	79.77 ± 0.16	79.89 ± 0.18
	RDDM	81.24 ± 0.11	81.24 ± 0.11	81.19 ± 0.12	79.82 ± 0.15	80.30 ± 0.13	80.34 ± 0.14	79.95 ± 0.13
	SeqDr2	80.90 ± 0.16	80.90 ± 0.16	80.87 ± 0.16	79.71 ± 0.21	80.19 ± 0.13	77.68 ± 0.20	79.87 ± 0.19
	STEPD	81.58 ± 0.13	81.58 ± 0.13	81.55 ± 0.12	79.97 ± 0.13	80.52 ± 0.13	80.02 ± 0.16	79.90 ± 0.12
	WSTD	81.20 ± 0.14	81.20 ± 0.14	81.17 ± 0.13	79.53 ± 0.18	80.24 ± 0.13	78.31 ± 0.22	79.80 ± 0.17

Table 11
Mean accuracies of Ensembles in percentage (%) in 100K instances gradual datasets, with 95% confidence intervals, using NB.

Dataset	Ensemble	ADOB	BOLE$_4$	BOLE$_5$	DDD	FASE	LevBag	None
Agraw$_1$	ADWIN	68.10 ± 0.09	68.10 ± 0.09	68.23 ± 0.10	65.85 ± 0.10	66.13 ± 0.08	65.90 ± 0.09	65.96 ± 0.07
	DDM	66.72 ± 0.22	66.72 ± 0.22	67.18 ± 0.25	63.47 ± 0.55	66.13 ± 0.09	63.38 ± 0.13	64.06 ± 0.63
	ECDD	64.17 ± 0.07	64.17 ± 0.07	64.37 ± 0.06	63.17 ± 0.09	65.35 ± 0.07	61.86 ± 0.06	62.77 ± 0.08
	FHDDM	64.85 ± 0.09	64.85 ± 0.09	64.82 ± 0.09	65.28 ± 0.09	66.09 ± 0.09	65.66 ± 0.08	65.07 ± 0.09
	FTDD	67.07 ± 0.20	67.07 ± 0.20	67.59 ± 0.20	63.75 ± 0.47	66.09 ± 0.08	61.42 ± 0.47	64.33 ± 0.49
	HDDM$_A$	68.31 ± 0.11	68.31 ± 0.11	**68.50 ± 0.12**	65.15 ± 0.16	66.15 ± 0.08	65.93 ± 0.08	65.93 ± 0.09
	RDDM	68.10 ± 0.09	68.10 ± 0.09	68.32 ± 0.10	65.54 ± 0.16	66.19 ± 0.07	63.40 ± 0.09	65.92 ± 0.08
	SeqDr2	67.48 ± 0.17	67.48 ± 0.17	68.10 ± 0.17	65.43 ± 0.12	66.12 ± 0.07	61.49 ± 0.37	65.92 ± 0.07
	STEPD	65.68 ± 0.10	65.68 ± 0.10	65.61 ± 0.11	65.27 ± 0.10	66.09 ± 0.08	65.38 ± 0.11	65.16 ± 0.09
	WSTD	67.18 ± 0.13	67.18 ± 0.13	67.20 ± 0.13	65.53 ± 0.14	66.14 ± 0.07	64.66 ± 0.26	65.69 ± 0.11
Agraw$_2$	ADWIN	87.34 ± 0.07	87.34 ± 0.07	87.35 ± 0.07	85.74 ± 0.08	85.94 ± 0.06	85.82 ± 0.05	85.72 ± 0.09
	DDM	86.42 ± 0.11	86.42 ± 0.11	86.49 ± 0.11	83.33 ± 0.78	85.96 ± 0.06	85.05 ± 0.07	83.83 ± 0.81
	ECDD	86.13 ± 0.07	86.13 ± 0.07	86.28 ± 0.06	84.73 ± 0.07	85.37 ± 0.07	84.76 ± 0.05	84.08 ± 0.11
	FHDDM	87.17 ± 0.04	87.17 ± 0.04	87.20 ± 0.04	84.69 ± 0.12	85.92 ± 0.05	82.25 ± 0.35	85.70 ± 0.07
	FTDD	87.35 ± 0.11	87.35 ± 0.11	87.36 ± 0.11	83.14 ± 0.48	85.92 ± 0.06	78.72 ± 0.53	84.03 ± 0.43
	HDDM$_A$	87.46 ± 0.07	87.47 ± 0.07	**87.53 ± 0.07**	84.43 ± 0.30	85.93 ± 0.06	85.05 ± 0.33	85.70 ± 0.13
	RDDM	87.24 ± 0.06	87.24 ± 0.06	87.30 ± 0.07	85.00 ± 0.36	85.96 ± 0.06	85.29 ± 0.05	85.79 ± 0.06
	SeqDr2	87.33 ± 0.08	87.33 ± 0.08	87.33 ± 0.08	83.91 ± 0.40	85.99 ± 0.05	79.03 ± 0.76	84.10 ± 0.45
	STEPD	86.96 ± 0.06	86.96 ± 0.06	86.98 ± 0.06	85.53 ± 0.13	85.87 ± 0.05	85.25 ± 0.18	85.42 ± 0.09
	WSTD	87.44 ± 0.07	87.44 ± 0.07	87.48 ± 0.07	84.43 ± 0.33	85.91 ± 0.06	82.03 ± 0.59	85.51 ± 0.26
LED	ADWIN	71.92 ± 0.17	71.92 ± 0.17	71.96 ± 0.17	72.38 ± 0.12	72.69 ± 0.11	**73.33 ± 0.11**	68.89 ± 0.56
	DDM	73.03 ± 0.13	73.03 ± 0.13	73.02 ± 0.13	71.80 ± 0.65	73.14 ± 0.11	51.73 ± 0.17	72.34 ± 0.51
	ECDD	72.88 ± 0.12	72.88 ± 0.12	72.89 ± 0.12	71.27 ± 0.21	72.97 ± 0.13	60.53 ± 0.19	68.83 ± 0.23
	FHDDM	73.12 ± 0.12	73.12 ± 0.12	73.13 ± 0.12	72.89 ± 0.12	73.22 ± 0.11	73.26 ± 0.11	72.31 ± 0.23
	FTDD	72.74 ± 0.13	72.74 ± 0.13	72.72 ± 0.12	71.48 ± 0.26	73.02 ± 0.11	69.20 ± 0.91	72.47 ± 0.17
	HDDM$_A$	73.26 ± 0.11	73.27 ± 0.11	73.27 ± 0.12	72.10 ± 0.16	73.23 ± 0.11	73.31 ± 0.12	73.22 ± 0.12
	RDDM	73.22 ± 0.12	73.22 ± 0.12	73.23 ± 0.12	72.59 ± 0.23	73.22 ± 0.11	64.39 ± 0.19	73.30 ± 0.12
	SeqDr2	72.57 ± 0.13	72.57 ± 0.13	72.57 ± 0.13	72.31 ± 0.20	72.76 ± 0.11	67.71 ± 0.81	69.98 ± 0.45
	STEPD	71.20 ± 0.21	71.20 ± 0.21	71.26 ± 0.21	72.16 ± 0.21	72.75 ± 0.13	72.55 ± 0.19	69.17 ± 0.62
	WSTD	72.98 ± 0.13	72.98 ± 0.13	73.00 ± 0.13	72.51 ± 0.17	72.99 ± 0.10	72.01 ± 0.52	72.46 ± 0.18
Mixed	ADWIN	90.27 ± 0.15	90.27 ± 0.15	90.61 ± 0.10	91.18 ± 0.06	91.22 ± 0.06	91.10 ± 0.06	91.11 ± 0.06
	DDM	90.94 ± 0.15	90.94 ± 0.15	91.14 ± 0.10	87.96 ± 0.52	91.24 ± 0.06	91.06 ± 0.10	91.22 ± 0.07
	ECDD	88.84 ± 0.07	88.84 ± 0.07	88.86 ± 0.06	90.08 ± 0.11	90.90 ± 0.07	90.71 ± 0.06	89.29 ± 0.09
	FHDDM	90.28 ± 0.15	90.28 ± 0.15	90.45 ± 0.11	89.18 ± 0.42	91.26 ± 0.06	77.03 ± 1.54	**91.30 ± 0.06**
	FTDD	91.14 ± 0.10	91.13 ± 0.10	91.26 ± 0.09	90.01 ± 0.38	91.25 ± 0.06	77.72 ± 1.93	91.23 ± 0.07
	HDDM$_A$	90.55 ± 0.14	90.55 ± 0.14	90.91 ± 0.08	88.88 ± 0.57	91.26 ± 0.06	90.59 ± 0.26	91.25 ± 0.07
	RDDM	90.37 ± 0.20	90.37 ± 0.20	90.87 ± 0.11	90.90 ± 0.39	91.24 ± 0.06	91.24 ± 0.06	91.29 ± 0.06
	SeqDr2	90.58 ± 0.14	90.58 ± 0.14	90.86 ± 0.12	91.03 ± 0.09	91.23 ± 0.06	83.23 ± 0.87	91.19 ± 0.06
	STEPD	89.76 ± 0.12	89.76 ± 0.12	90.04 ± 0.10	91.07 ± 0.07	91.18 ± 0.06	89.02 ± 0.89	90.88 ± 0.08
	WSTD	90.47 ± 0.17	90.47 ± 0.17	90.73 ± 0.13	90.07 ± 0.55	91.24 ± 0.06	78.47 ± 2.05	91.23 ± 0.07
RBF	ADWIN	19.30 ± 0.63	19.30 ± 0.63	29.93 ± 0.66	30.61 ± 0.30	32.68 ± 0.16	31.70 ± 0.32	30.56 ± 0.33
	DDM	19.45 ± 0.66	19.45 ± 0.66	30.61 ± 0.54	31.47 ± 0.50	32.71 ± 0.16	29.83 ± 0.19	31.41 ± 0.40
	ECDD	19.50 ± 0.68	19.50 ± 0.68	30.40 ± 0.51	31.42 ± 0.59	**33.12 ± 0.19**	31.48 ± 0.59	31.53 ± 0.58
	FHDDM	19.16 ± 0.50	19.16 ± 0.50	29.41 ± 0.52	29.81 ± 0.23	31.95 ± 0.13	30.14 ± 0.20	29.52 ± 0.25
	FTDD	19.28 ± 0.63	22.93 ± 0.47	30.81 ± 0.42	31.62 ± 0.44	32.88 ± 0.19	31.94 ± 0.45	31.64 ± 0.45
	HDDM$_A$	19.34 ± 0.62	22.82 ± 0.41	30.61 ± 0.37	30.98 ± 0.36	32.64 ± 0.17	30.87 ± 0.19	31.16 ± 0.33
	RDDM	19.34 ± 0.68	22.74 ± 0.53	30.84 ± 0.29	31.06 ± 0.25	32.62 ± 0.15	29.81 ± 0.13	31.23 ± 0.30
	SeqDr2	19.05 ± 0.48	19.05 ± 0.48	30.51 ± 0.49	31.44 ± 0.41	32.85 ± 0.22	31.77 ± 0.54	31.48 ± 0.43
	STEPD	19.16 ± 0.48	19.16 ± 0.48	29.29 ± 0.47	29.51 ± 0.21	31.96 ± 0.11	31.24 ± 0.38	29.14 ± 0.19
	WSTD	18.80 ± 0.49	22.88 ± 0.41	30.20 ± 0.51	30.94 ± 0.50	32.63 ± 0.17	31.65 ± 0.55	30.79 ± 0.40
Sine	ADWIN	88.65 ± 0.12	88.65 ± 0.12	88.87 ± 0.10	86.83 ± 0.08	86.85 ± 0.08	86.89 ± 0.08	86.80 ± 0.08
	DDM	89.37 ± 0.13	89.37 ± 0.13	89.51 ± 0.10	84.88 ± 0.61	87.11 ± 0.09	86.34 ± 0.16	86.58 ± 0.29
	ECDD	87.50 ± 0.07	87.50 ± 0.07	87.61 ± 0.07	86.65 ± 0.09	87.12 ± 0.07	86.81 ± 0.07	86.01 ± 0.09
	FHDDM	89.02 ± 0.15	89.02 ± 0.15	89.14 ± 0.13	85.51 ± 0.36	87.11 ± 0.08	74.87 ± 0.97	87.17 ± 0.08
	FTDD	89.41 ± 0.13	89.41 ± 0.13	**89.55 ± 0.09**	85.41 ± 0.36	87.02 ± 0.08	72.47 ± 1.05	87.04 ± 0.09
	HDDM$_A$	89.29 ± 0.12	89.29 ± 0.12	89.45 ± 0.11	85.50 ± 0.41	87.09 ± 0.09	86.05 ± 0.52	87.14 ± 0.09
	RDDM	89.36 ± 0.08	89.36 ± 0.08	89.51 ± 0.07	86.86 ± 0.18	87.15 ± 0.08	87.06 ± 0.07	87.16 ± 0.09
	SeqDr2	88.87 ± 0.13	88.87 ± 0.13	89.06 ± 0.11	86.64 ± 0.14	86.86 ± 0.08	76.11 ± 0.64	86.86 ± 0.09
	STEPD	88.51 ± 0.09	88.51 ± 0.09	88.65 ± 0.08	86.91 ± 0.08	87.00 ± 0.08	85.27 ± 0.51	86.73 ± 0.07
	WSTD	89.11 ± 0.11	89.12 ± 0.11	89.27 ± 0.09	85.84 ± 0.43	87.03 ± 0.08	73.24 ± 1.20	87.05 ± 0.09
Wavef.	ADWIN	81.46 ± 0.11	81.46 ± 0.11	81.44 ± 0.12	80.20 ± 0.13	80.46 ± 0.10	80.19 ± 0.10	80.22 ± 0.11
	DDM	80.92 ± 0.16	80.92 ± 0.16	80.90 ± 0.16	79.46 ± 0.22	80.44 ± 0.10	79.95 ± 0.09	79.52 ± 0.18
	ECDD	**81.99 ± 0.10**	**81.99 ± 0.10**	81.98 ± 0.10	79.99 ± 0.12	80.77 ± 0.11	80.43 ± 0.10	79.10 ± 0.13
	FHDDM	81.70 ± 0.09	81.70 ± 0.09	81.68 ± 0.09	79.93 ± 0.10	80.51 ± 0.09	79.33 ± 0.12	80.18 ± 0.12
	FTDD	80.93 ± 0.11	80.93 ± 0.11	80.90 ± 0.11	79.75 ± 0.16	80.46 ± 0.10	78.07 ± 0.18	79.95 ± 0.19
	HDDM$_A$	81.52 ± 0.09	81.52 ± 0.09	81.50 ± 0.09	79.94 ± 0.13	80.46 ± 0.10	80.06 ± 0.13	80.20 ± 0.11
	RDDM	81.47 ± 0.10	81.47 ± 0.10	81.46 ± 0.10	80.00 ± 0.11	80.55 ± 0.09	80.43 ± 0.10	80.13 ± 0.11
	SeqDr2	81.43 ± 0.11	81.43 ± 0.11	81.42 ± 0.12	79.92 ± 0.15	80.47 ± 0.10	77.88 ± 0.14	80.23 ± 0.10
	STEPD	81.76 ± 0.10	81.76 ± 0.10	81.73 ± 0.10	80.22 ± 0.11	80.75 ± 0.10	80.26 ± 0.10	80.13 ± 0.10
	WSTD	81.50 ± 0.09	81.50 ± 0.09	81.48 ± 0.09	79.88 ± 0.13	80.51 ± 0.09	78.83 ± 0.21	80.21 ± 0.10

Table 12
Mean accuracies of Ensembles in percentage (%) in 500K instances gradual datasets, with 95% confidence intervals, using NB.

Dataset	Ensemble	ADOB	$BOLE_4$	$BOLE_5$	DDD	FASE	LevBag	None
$Agraw_1$	ADWIN	68.87 ± 0.17	68.87 ± 0.17	68.90 ± 0.13	66.38 ± 0.03	66.48 ± 0.03	66.32 ± 0.08	66.42 ± 0.05
	DDM	67.92 ± 0.56	67.92 ± 0.56	68.33 ± 0.49	63.69 ± 0.98	66.49 ± 0.03	63.81 ± 0.24	63.93 ± 0.91
	ECDD	64.55 ± 0.05	64.55 ± 0.05	64.71 ± 0.04	63.43 ± 0.04	65.63 ± 0.05	62.04 ± 0.05	62.94 ± 0.06
	FHDDM	65.17 ± 0.08	65.17 ± 0.08	65.11 ± 0.08	65.61 ± 0.05	66.41 ± 0.07	66.07 ± 0.06	65.32 ± 0.06
	FTDD	68.88 ± 0.11	68.88 ± 0.11	69.23 ± 0.13	65.85 ± 0.15	66.48 ± 0.03	63.52 ± 0.19	66.27 ± 0.06
	$HDDM_A$	68.80 ± 0.12	68.80 ± 0.12	69.06 ± 0.16	66.15 ± 0.16	66.50 ± 0.03	66.29 ± 0.17	66.38 ± 0.05
	RDDM	68.50 ± 0.16	68.50 ± 0.16	68.80 ± 0.16	66.29 ± 0.16	66.57 ± 0.04	63.55 ± 0.07	66.36 ± 0.04
	SeqDr2	68.99 ± 0.20	68.99 ± 0.20	**69.36 ± 0.19**	65.79 ± 0.13	66.48 ± 0.03	63.42 ± 0.62	66.42 ± 0.04
	STEPD	66.15 ± 0.06	66.15 ± 0.06	66.01 ± 0.07	65.79 ± 0.07	66.46 ± 0.03	65.79 ± 0.06	65.55 ± 0.07
	WSTD	67.80 ± 0.12	67.80 ± 0.12	67.71 ± 0.16	66.24 ± 0.05	66.53 ± 0.04	65.86 ± 0.18	66.20 ± 0.05
$Agraw_2$	ADWIN	88.50 ± 0.07	88.50 ± 0.07	88.53 ± 0.06	86.78 ± 0.03	86.84 ± 0.04	86.80 ± 0.03	86.76 ± 0.04
	DDM	88.27 ± 0.10	88.27 ± 0.10	88.30 ± 0.12	84.97 ± 0.73	86.85 ± 0.04	85.58 ± 0.11	86.16 ± 0.34
	ECDD	86.91 ± 0.04	86.91 ± 0.04	86.98 ± 0.04	85.29 ± 0.03	85.89 ± 0.07	85.26 ± 0.04	84.68 ± 0.06
	FHDDM	87.99 ± 0.07	87.99 ± 0.07	87.99 ± 0.05	85.51 ± 0.32	86.65 ± 0.04	85.40 ± 0.32	86.52 ± 0.05
	FTDD	88.65 ± 0.11	88.65 ± 0.11	88.65 ± 0.10	84.64 ± 0.79	86.82 ± 0.06	80.32 ± 1.12	86.05 ± 0.62
	$HDDM_A$	88.73 ± 0.04	88.73 ± 0.04	**88.75 ± 0.04**	85.71 ± 0.33	86.84 ± 0.04	86.31 ± 0.29	86.74 ± 0.05
	RDDM	88.65 ± 0.04	88.65 ± 0.04	88.68 ± 0.04	86.70 ± 0.06	86.83 ± 0.04	85.87 ± 0.05	86.67 ± 0.07
	SeqDr2	88.66 ± 0.06	88.66 ± 0.06	88.68 ± 0.06	86.24 ± 0.62	86.84 ± 0.04	80.67 ± 0.76	86.22 ± 0.63
	STEPD	87.77 ± 0.05	87.77 ± 0.05	87.75 ± 0.05	86.38 ± 0.06	86.55 ± 0.05	86.37 ± 0.06	86.19 ± 0.07
	WSTD	88.50 ± 0.08	88.50 ± 0.08	88.49 ± 0.07	85.78 ± 0.29	86.80 ± 0.06	82.90 ± 0.59	86.61 ± 0.11
LED	ADWIN	73.13 ± 0.20	73.13 ± 0.20	73.15 ± 0.19	73.52 ± 0.12	73.62 ± 0.09	**73.79 ± 0.09**	72.60 ± 0.29
	DDM	73.40 ± 0.18	73.40 ± 0.18	73.36 ± 0.19	71.38 ± 1.61	73.71 ± 0.09	51.91 ± 0.12	72.69 ± 0.89
	ECDD	73.25 ± 0.11	73.25 ± 0.11	73.25 ± 0.11	71.57 ± 0.18	73.49 ± 0.11	60.82 ± 0.22	69.15 ± 0.15
	FHDDM	73.60 ± 0.10	73.60 ± 0.10	73.61 ± 0.10	73.69 ± 0.09	73.77 ± 0.10	73.78 ± 0.09	72.86 ± 0.13
	FTDD	73.61 ± 0.12	73.61 ± 0.12	73.60 ± 0.11	72.46 ± 0.21	73.73 ± 0.09	71.35 ± 1.24	73.35 ± 0.26
	$HDDM_A$	**73.79 ± 0.11**	**73.79 ± 0.11**	**73.79 ± 0.11**	73.00 ± 0.26	73.75 ± 0.09	73.76 ± 0.09	73.74 ± 0.11
	RDDM	**73.79 ± 0.11**	**73.79 ± 0.11**	**73.79 ± 0.11**	73.66 ± 0.10	73.77 ± 0.09	64.65 ± 0.16	73.72 ± 0.09
	SeqDr2	73.66 ± 0.11	73.66 ± 0.11	73.66 ± 0.11	73.32 ± 0.21	73.65 ± 0.09	69.94 ± 1.07	72.88 ± 0.41
	STEPD	72.55 ± 0.16	72.55 ± 0.16	72.57 ± 0.16	72.85 ± 0.21	73.59 ± 0.10	72.98 ± 0.14	69.98 ± 0.27
	WSTD	73.71 ± 0.11	73.71 ± 0.11	73.71 ± 0.11	73.61 ± 0.11	73.72 ± 0.09	73.61 ± 0.15	73.35 ± 0.06
Mixed	ADWIN	90.41 ± 0.24	90.41 ± 0.24	91.15 ± 0.14	91.91 ± 0.03	91.92 ± 0.03	91.90 ± 0.04	91.91 ± 0.04
	DDM	91.35 ± 0.13	91.35 ± 0.13	91.60 ± 0.11	89.95 ± 1.55	91.93 ± 0.03	91.70 ± 0.09	91.76 ± 0.22
	ECDD	89.36 ± 0.05	89.36 ± 0.05	89.36 ± 0.06	90.66 ± 0.05	91.53 ± 0.03	91.42 ± 0.03	89.83 ± 0.07
	FHDDM	90.43 ± 0.20	90.43 ± 0.20	90.76 ± 0.12	90.32 ± 0.55	91.93 ± 0.03	75.39 ± 2.86	**91.94 ± 0.03**
	FTDD	90.94 ± 0.33	90.94 ± 0.33	91.42 ± 0.17	90.86 ± 0.73	91.92 ± 0.03	77.32 ± 1.54	91.92 ± 0.03
	$HDDM_A$	90.43 ± 0.26	90.43 ± 0.26	91.20 ± 0.12	90.46 ± 0.74	91.93 ± 0.03	91.32 ± 0.87	**91.94 ± 0.04**
	RDDM	90.05 ± 0.26	90.05 ± 0.26	91.19 ± 0.13	91.75 ± 0.18	91.93 ± 0.03	91.91 ± 0.03	91.91 ± 0.03
	SeqDr2	89.94 ± 0.25	89.94 ± 0.25	91.05 ± 0.15	91.89 ± 0.05	91.93 ± 0.03	84.59 ± 1.43	91.93 ± 0.04
	STEPD	90.12 ± 0.08	90.12 ± 0.08	90.61 ± 0.07	91.75 ± 0.03	91.86 ± 0.03	91.05 ± 0.56	91.51 ± 0.05
	WSTD	90.26 ± 0.25	90.26 ± 0.25	91.05 ± 0.16	90.93 ± 0.77	91.93 ± 0.03	77.00 ± 3.82	91.93 ± 0.03
RBF	ADWIN	18.74 ± 1.00	18.74 ± 1.00	30.25 ± 0.91	31.00 ± 0.37	33.51 ± 0.13	33.03 ± 0.29	31.24 ± 0.29
	DDM	18.38 ± 0.60	18.38 ± 0.60	31.34 ± 0.97	33.05 ± 0.50	33.76 ± 0.15	30.93 ± 0.55	33.31 ± 0.43
	ECDD	18.38 ± 0.60	18.38 ± 0.60	31.52 ± 0.88	33.22 ± 0.63	34.39 ± 0.20	33.29 ± 0.65	33.26 ± 0.66
	FHDDM	18.06 ± 0.30	18.06 ± 0.30	29.64 ± 0.82	30.23 ± 0.16	32.61 ± 0.06	30.28 ± 0.14	29.94 ± 0.16
	FTDD	18.87 ± 1.17	21.55 ± 1.24	32.18 ± 0.71	33.35 ± 0.42	**33.86 ± 0.17**	33.59 ± 0.41	33.14 ± 0.32
	$HDDM_A$	19.09 ± 0.71	22.35 ± 0.75	31.11 ± 0.39	32.54 ± 0.37	33.72 ± 0.14	31.56 ± 0.28	32.49 ± 0.40
	RDDM	18.94 ± 0.80	21.10 ± 1.11	31.63 ± 0.41	31.90 ± 0.33	33.62 ± 0.12	29.84 ± 0.07	32.14 ± 0.26
	SeqDr2	18.38 ± 0.52	18.38 ± 0.52	31.46 ± 0.56	33.43 ± 0.39	34.09 ± 0.20	33.65 ± 0.43	33.22 ± 0.33
	STEPD	18.45 ± 0.58	18.45 ± 0.58	29.15 ± 0.64	29.50 ± 0.11	32.48 ± 0.08	31.87 ± 0.23	29.08 ± 0.13
	WSTD	18.90 ± 0.84	22.08 ± 0.91	31.00 ± 0.52	31.28 ± 0.29	33.19 ± 0.12	33.34 ± 0.56	30.99 ± 0.26
Sine	ADWIN	88.64 ± 0.13	88.64 ± 0.13	88.97 ± 0.10	87.27 ± 0.06	87.27 ± 0.05	87.37 ± 0.08	87.28 ± 0.04
	DDM	89.29 ± 0.20	89.29 ± 0.20	89.59 ± 0.13	83.43 ± 3.10	87.32 ± 0.06	86.69 ± 0.24	84.02 ± 3.07
	ECDD	87.98 ± 0.07	87.98 ± 0.07	88.09 ± 0.07	87.02 ± 0.07	87.49 ± 0.04	87.30 ± 0.05	86.36 ± 0.05
	FHDDM	89.11 ± 0.17	89.11 ± 0.17	89.32 ± 0.16	86.26 ± 0.32	87.35 ± 0.06	69.77 ± 1.61	87.36 ± 0.05
	FTDD	89.20 ± 0.16	89.20 ± 0.16	89.57 ± 0.13	85.81 ± 0.68	87.33 ± 0.05	71.73 ± 1.22	87.33 ± 0.04
	$HDDM_A$	89.14 ± 0.11	89.14 ± 0.11	89.54 ± 0.07	85.72 ± 0.66	87.31 ± 0.06	85.84 ± 1.21	87.31 ± 0.06
	RDDM	89.27 ± 0.15	89.27 ± 0.15	89.56 ± 0.13	87.35 ± 0.08	87.39 ± 0.07	87.45 ± 0.05	87.39 ± 0.05
	SeqDr2	88.75 ± 0.26	88.75 ± 0.26	89.21 ± 0.21	87.28 ± 0.05	87.29 ± 0.05	78.20 ± 0.79	87.29 ± 0.05
	STEPD	88.96 ± 0.11	88.96 ± 0.11	89.13 ± 0.09	87.35 ± 0.04	87.42 ± 0.04	87.11 ± 0.17	87.24 ± 0.05
	WSTD	89.33 ± 0.21	89.33 ± 0.21	**89.60 ± 0.14**	87.01 ± 0.21	87.33 ± 0.06	74.34 ± 1.32	87.33 ± 0.04
Wavef.	ADWIN	81.59 ± 0.10	81.59 ± 0.10	81.59 ± 0.10	80.35 ± 0.11	80.49 ± 0.12	80.39 ± 0.12	80.37 ± 0.12
	DDM	81.15 ± 0.14	81.15 ± 0.14	81.14 ± 0.15	79.89 ± 0.35	80.66 ± 0.15	80.04 ± 0.08	79.81 ± 0.29
	ECDD	**82.01 ± 0.08**	**82.01 ± 0.08**	**82.01 ± 0.08**	80.09 ± 0.10	80.89 ± 0.09	80.52 ± 0.10	79.18 ± 0.11
	FHDDM	81.80 ± 0.10	81.80 ± 0.10	81.80 ± 0.10	80.32 ± 0.13	80.60 ± 0.11	80.07 ± 0.14	80.36 ± 0.11
	FTDD	81.60 ± 0.12	81.60 ± 0.12	81.60 ± 0.12	79.99 ± 0.22	80.55 ± 0.10	78.65 ± 0.49	80.35 ± 0.11
	$HDDM_A$	81.61 ± 0.11	81.61 ± 0.11	81.61 ± 0.11	80.24 ± 0.13	80.51 ± 0.12	80.05 ± 0.25	80.35 ± 0.12
	RDDM	81.63 ± 0.11	81.63 ± 0.11	81.63 ± 0.11	80.30 ± 0.14	80.62 ± 0.12	80.51 ± 0.09	80.33 ± 0.14
	SeqDr2	81.64 ± 0.11	81.64 ± 0.11	81.64 ± 0.11	80.26 ± 0.15	80.50 ± 0.12	78.00 ± 0.37	80.37 ± 0.12
	STEPD	81.82 ± 0.11	81.82 ± 0.11	81.82 ± 0.11	80.37 ± 0.12	80.87 ± 0.10	80.42 ± 0.12	80.24 ± 0.11
	WSTD	81.68 ± 0.10	81.68 ± 0.10	81.67 ± 0.10	80.26 ± 0.21	80.58 ± 0.12	79.66 ± 0.19	80.33 ± 0.15

Table 13
Mean accuracies of Ensembles in percentage (%) in 1 Million instances gradual datasets, with 95% confidence intervals, using NB.

Dataset	Ensemble	ADOB	$BOLE_4$	$BOLE_5$	DDD	FASE	LevBag	None
$Agraw_1$	ADWIN	69.04 ± 0.13	69.04 ± 0.13	69.05 ± 0.12	66.44 ± 0.07	66.54 ± 0.04	66.36 ± 0.13	66.49 ± 0.03
	DDM	67.95 ± 0.37	67.95 ± 0.37	68.29 ± 0.41	63.43 ± 0.72	66.56 ± 0.03	63.94 ± 0.28	64.57 ± 0.85
	ECDD	64.61 ± 0.04	64.61 ± 0.04	64.77 ± 0.04	63.46 ± 0.06	65.70 ± 0.03	62.05 ± 0.04	62.96 ± 0.05
	FHDDM	65.25 ± 0.05	65.25 ± 0.05	65.17 ± 0.05	65.68 ± 0.06	66.49 ± 0.04	66.15 ± 0.05	65.36 ± 0.05
	FTDD	68.84 ± 0.28	68.84 ± 0.28	69.21 ± 0.14	66.28 ± 0.13	66.53 ± 0.03	64.03 ± 0.38	66.43 ± 0.07
	$HDDM_A$	69.03 ± 0.16	69.03 ± 0.16	69.23 ± 0.14	66.26 ± 0.14	66.55 ± 0.03	66.47 ± 0.04	66.45 ± 0.05
	RDDM	68.68 ± 0.13	68.68 ± 0.13	68.95 ± 0.15	66.41 ± 0.06	66.65 ± 0.03	63.57 ± 0.05	66.44 ± 0.06
	SeqDr2	68.93 ± 0.26	68.93 ± 0.26	**69.36 ± 0.22**	66.06 ± 0.18	66.53 ± 0.04	63.32 ± 0.59	66.49 ± 0.04
	STEPD	66.24 ± 0.05	66.24 ± 0.05	66.09 ± 0.06	65.84 ± 0.06	66.53 ± 0.02	65.87 ± 0.04	65.64 ± 0.05
	WSTD	67.99 ± 0.05	67.99 ± 0.05	67.94 ± 0.06	66.36 ± 0.04	66.58 ± 0.03	66.16 ± 0.08	66.29 ± 0.05
$Agraw_2$	ADWIN	88.72 ± 0.05	88.72 ± 0.05	88.72 ± 0.05	86.92 ± 0.02	86.96 ± 0.02	86.91 ± 0.02	86.90 ± 0.02
	DDM	88.69 ± 0.12	88.69 ± 0.12	88.71 ± 0.10	85.41 ± 0.62	86.95 ± 0.02	85.69 ± 0.07	86.55 ± 0.29
	ECDD	86.96 ± 0.03	86.96 ± 0.03	87.02 ± 0.03	85.35 ± 0.04	85.96 ± 0.08	85.33 ± 0.02	84.73 ± 0.04
	FHDDM	88.16 ± 0.03	88.16 ± 0.03	88.15 ± 0.03	85.69 ± 0.03	86.73 ± 0.02	86.04 ± 0.18	86.63 ± 0.02
	FTDD	88.86 ± 0.06	88.86 ± 0.06	88.87 ± 0.06	85.73 ± 0.16	86.94 ± 0.02	82.16 ± 0.81	86.56 ± 0.29
	$HDDM_A$	88.94 ± 0.02	88.94 ± 0.02	**88.95 ± 0.03**	86.05 ± 0.38	86.95 ± 0.03	86.08 ± 0.52	86.83 ± 0.05
	RDDM	88.82 ± 0.04	88.82 ± 0.04	88.86 ± 0.03	86.82 ± 0.04	86.93 ± 0.02	85.94 ± 0.02	86.81 ± 0.03
	SeqDr2	88.87 ± 0.05	88.87 ± 0.05	88.89 ± 0.05	86.35 ± 0.61	86.96 ± 0.02	81.79 ± 0.64	86.28 ± 0.60
	STEPD	87.84 ± 0.03	87.84 ± 0.03	87.83 ± 0.03	86.46 ± 0.04	86.69 ± 0.03	86.49 ± 0.04	86.28 ± 0.04
	WSTD	88.66 ± 0.05	88.66 ± 0.05	88.64 ± 0.05	86.26 ± 0.28	86.90 ± 0.03	83.66 ± 0.43	86.76 ± 0.10
LED	ADWIN	73.43 ± 0.13	73.43 ± 0.13	73.45 ± 0.13	73.77 ± 0.06	73.79 ± 0.06	**73.88 ± 0.05**	73.19 ± 0.21
	DDM	73.39 ± 0.11	73.39 ± 0.11	73.27 ± 0.15	70.66 ± 2.75	73.82 ± 0.04	51.97 ± 0.10	72.21 ± 1.15
	ECDD	73.35 ± 0.06	73.35 ± 0.06	73.35 ± 0.06	71.64 ± 0.13	73.58 ± 0.06	60.90 ± 0.14	69.23 ± 0.12
	FHDDM	73.69 ± 0.08	73.69 ± 0.08	73.69 ± 0.08	73.81 ± 0.06	73.87 ± 0.06	73.86 ± 0.07	72.96 ± 0.11
	FTDD	73.79 ± 0.09	73.79 ± 0.09	73.79 ± 0.09	72.85 ± 0.36	73.84 ± 0.06	71.54 ± 1.14	73.61 ± 0.24
	$HDDM_A$	73.87 ± 0.06	73.87 ± 0.06	73.87 ± 0.06	73.23 ± 0.30	73.85 ± 0.05	73.87 ± 0.06	73.84 ± 0.06
	RDDM	73.87 ± 0.07	73.87 ± 0.07	**73.88 ± 0.07**	73.79 ± 0.10	73.87 ± 0.05	64.67 ± 0.10	73.79 ± 0.05
	SeqDr2	73.82 ± 0.07	73.82 ± 0.07	73.82 ± 0.07	73.54 ± 0.16	73.81 ± 0.05	69.89 ± 1.33	73.41 ± 0.16
	STEPD	72.64 ± 0.15	72.64 ± 0.15	72.66 ± 0.15	73.00 ± 0.15	73.71 ± 0.07	73.05 ± 0.08	70.12 ± 0.30
	WSTD	73.82 ± 0.07	73.82 ± 0.07	73.82 ± 0.07	73.75 ± 0.06	73.85 ± 0.05	73.75 ± 0.09	73.46 ± 0.12
Mixed	ADWIN	90.24 ± 0.25	90.24 ± 0.25	91.36 ± 0.12	92.02 ± 0.03	92.03 ± 0.03	92.02 ± 0.03	92.02 ± 0.03
	DDM	91.47 ± 0.17	91.47 ± 0.17	91.60 ± 0.15	91.55 ± 0.47	**92.04 ± 0.03**	91.73 ± 0.13	91.89 ± 0.17
	ECDD	89.34 ± 0.04	89.34 ± 0.04	89.35 ± 0.04	90.74 ± 0.03	91.63 ± 0.03	91.51 ± 0.04	89.92 ± 0.05
	FHDDM	89.94 ± 0.24	89.94 ± 0.24	90.46 ± 0.22	90.15 ± 0.48	92.03 ± 0.03	75.04 ± 3.11	**92.04 ± 0.03**
	FTDD	91.03 ± 0.19	91.03 ± 0.19	91.56 ± 0.09	91.61 ± 0.34	92.03 ± 0.03	78.50 ± 2.52	92.03 ± 0.03
	$HDDM_A$	90.14 ± 0.19	90.14 ± 0.19	91.22 ± 0.14	90.35 ± 0.53	92.03 ± 0.03	90.83 ± 1.49	92.03 ± 0.03
	RDDM	90.18 ± 0.28	90.18 ± 0.28	91.27 ± 0.12	91.92 ± 0.08	92.03 ± 0.03	92.00 ± 0.04	91.99 ± 0.03
	SeqDr2	90.32 ± 0.26	90.32 ± 0.26	91.38 ± 0.16	92.02 ± 0.03	92.03 ± 0.03	84.11 ± 1.87	92.03 ± 0.03
	STEPD	90.20 ± 0.13	90.20 ± 0.13	90.73 ± 0.08	91.85 ± 0.03	91.96 ± 0.03	91.66 ± 0.13	91.60 ± 0.03
	WSTD	90.10 ± 0.21	90.10 ± 0.21	91.14 ± 0.16	91.63 ± 0.35	92.03 ± 0.03	75.87 ± 1.57	92.03 ± 0.03
RBF	ADWIN	18.90 ± 1.10	18.90 ± 1.10	29.95 ± 0.85	31.10 ± 0.27	33.60 ± 0.06	33.35 ± 0.14	31.56 ± 0.39
	DDM	18.10 ± 0.46	18.10 ± 0.46	31.32 ± 0.89	33.25 ± 0.33	34.02 ± 0.06	31.40 ± 0.38	33.37 ± 0.28
	ECDD	18.22 ± 0.49	18.22 ± 0.49	31.69 ± 0.72	33.18 ± 0.42	**34.47 ± 0.13**	33.19 ± 0.44	33.16 ± 0.44
	FHDDM	17.81 ± 0.16	17.81 ± 0.16	29.64 ± 0.80	30.25 ± 0.13	32.64 ± 0.05	30.28 ± 0.08	29.88 ± 0.16
	FTDD	18.93 ± 1.14	20.97 ± 1.29	32.36 ± 0.47	33.43 ± 0.25	34.14 ± 0.10	33.74 ± 0.19	33.25 ± 0.21
	$HDDM_A$	19.27 ± 0.95	22.02 ± 1.03	31.58 ± 0.49	33.26 ± 0.30	33.96 ± 0.06	31.53 ± 0.19	32.95 ± 0.25
	RDDM	18.83 ± 0.86	20.89 ± 0.70	31.94 ± 0.18	32.09 ± 0.17	33.67 ± 0.08	29.86 ± 0.05	32.10 ± 0.15
	SeqDr2	18.28 ± 0.68	18.28 ± 0.68	31.99 ± 0.73	33.48 ± 0.31	34.30 ± 0.05	33.58 ± 0.41	33.50 ± 0.30
	STEPD	18.00 ± 0.31	18.00 ± 0.31	29.13 ± 0.68	29.36 ± 0.09	32.55 ± 0.05	31.96 ± 0.14	29.03 ± 0.08
	WSTD	18.70 ± 0.64	21.36 ± 0.93	31.22 ± 0.37	31.29 ± 0.19	33.16 ± 0.07	33.65 ± 0.22	31.07 ± 0.24
Sine	ADWIN	88.65 ± 0.14	88.65 ± 0.14	88.98 ± 0.15	87.37 ± 0.05	87.37 ± 0.05	87.42 ± 0.05	87.37 ± 0.05
	DDM	89.21 ± 0.28	89.21 ± 0.28	89.73 ± 0.14	82.91 ± 3.06	87.38 ± 0.05	86.52 ± 0.46	79.93 ± 5.03
	ECDD	88.04 ± 0.05	88.04 ± 0.05	88.14 ± 0.05	87.10 ± 0.03	87.57 ± 0.03	87.37 ± 0.02	86.43 ± 0.03
	FHDDM	89.37 ± 0.18	89.37 ± 0.18	89.56 ± 0.14	86.08 ± 0.47	87.42 ± 0.06	73.30 ± 2.63	87.43 ± 0.05
	FTDD	89.41 ± 0.25	89.41 ± 0.25	89.69 ± 0.17	86.25 ± 0.51	87.41 ± 0.05	73.33 ± 1.25	87.41 ± 0.05
	$HDDM_A$	89.27 ± 0.18	89.27 ± 0.18	89.54 ± 0.14	86.34 ± 0.64	87.39 ± 0.06	85.19 ± 2.75	87.38 ± 0.07
	RDDM	89.58 ± 0.16	89.58 ± 0.16	**89.81 ± 0.11**	87.37 ± 0.11	87.47 ± 0.05	87.55 ± 0.03	87.45 ± 0.04
	SeqDr2	88.53 ± 0.21	88.53 ± 0.21	89.02 ± 0.18	87.36 ± 0.06	87.38 ± 0.05	77.68 ± 0.70	87.38 ± 0.05
	STEPD	89.04 ± 0.11	89.04 ± 0.11	89.21 ± 0.10	87.43 ± 0.04	87.48 ± 0.03	87.24 ± 0.18	87.30 ± 0.04
	WSTD	89.29 ± 0.27	89.29 ± 0.27	89.57 ± 0.23	86.73 ± 0.44	87.41 ± 0.06	74.83 ± 1.31	87.40 ± 0.05
Wavef.	ADWIN	81.62 ± 0.06	81.62 ± 0.06	81.62 ± 0.06	80.42 ± 0.07	80.52 ± 0.08	80.45 ± 0.07	80.41 ± 0.08
	DDM	81.15 ± 0.12	81.15 ± 0.12	81.14 ± 0.13	79.56 ± 0.48	80.60 ± 0.10	80.02 ± 0.06	79.89 ± 0.32
	ECDD	**82.01 ± 0.04**	**82.01 ± 0.04**	82.00 ± 0.04	80.12 ± 0.08	80.90 ± 0.06	80.52 ± 0.06	79.19 ± 0.08
	FHDDM	81.84 ± 0.06	81.84 ± 0.06	81.84 ± 0.06	80.39 ± 0.09	80.60 ± 0.07	80.32 ± 0.09	80.40 ± 0.09
	FTDD	81.66 ± 0.06	81.66 ± 0.06	81.66 ± 0.06	80.23 ± 0.12	80.53 ± 0.07	78.87 ± 0.41	80.39 ± 0.10
	$HDDM_A$	81.66 ± 0.07	81.66 ± 0.07	81.65 ± 0.07	80.34 ± 0.10	80.53 ± 0.08	80.28 ± 0.17	80.40 ± 0.09
	RDDM	81.68 ± 0.05	81.68 ± 0.05	81.68 ± 0.05	80.39 ± 0.08	80.65 ± 0.06	80.51 ± 0.06	80.40 ± 0.06
	SeqDr2	81.64 ± 0.06	81.64 ± 0.06	81.64 ± 0.06	80.31 ± 0.13	80.51 ± 0.08	78.31 ± 0.36	80.43 ± 0.07
	STEPD	81.84 ± 0.06	81.84 ± 0.06	81.84 ± 0.06	80.40 ± 0.07	80.91 ± 0.07	80.46 ± 0.08	80.27 ± 0.06
	WSTD	81.72 ± 0.06	81.72 ± 0.06	81.72 ± 0.06	80.35 ± 0.10	80.59 ± 0.06	79.97 ± 0.19	80.38 ± 0.07

Table 14
Mean accuracies of Ensembles in percentage (%) in 2 Million instances gradual datasets, with 95% confidence intervals, using NB.

Dataset	Ensemble	ADOB	BOLE$_4$	BOLE$_5$	DDD	FASE	LevBag	None
Agraw$_1$	ADWIN	69.01 ± 0.12	69.01 ± 0.12	69.08 ± 0.11	66.54 ± 0.03	66.58 ± 0.02	66.36 ± 0.15	66.55 ± 0.04
	DDM	67.97 ± 0.27	67.97 ± 0.27	68.29 ± 0.28	63.25 ± 0.96	66.60 ± 0.02	64.06 ± 0.26	64.06 ± 1.10
	ECDD	64.67 ± 0.03	64.67 ± 0.03	64.83 ± 0.03	63.46 ± 0.04	65.72 ± 0.01	62.07 ± 0.03	62.98 ± 0.03
	FHDDM	65.27 ± 0.04	65.27 ± 0.04	65.20 ± 0.04	65.69 ± 0.03	66.52 ± 0.02	66.17 ± 0.03	65.39 ± 0.03
	FTDD	69.04 ± 0.20	69.04 ± 0.20	69.35 ± 0.15	66.30 ± 0.16	66.57 ± 0.02	63.99 ± 0.42	66.52 ± 0.04
	HDDM$_A$	69.06 ± 0.16	69.06 ± 0.16	69.28 ± 0.17	66.22 ± 0.17	66.57 ± 0.03	66.46 ± 0.14	66.49 ± 0.05
	RDDM	68.78 ± 0.12	68.78 ± 0.12	69.04 ± 0.09	66.50 ± 0.04	66.68 ± 0.02	63.61 ± 0.02	66.51 ± 0.03
	SeqDr2	69.24 ± 0.33	69.24 ± 0.33	**69.51 ± 0.27**	66.09 ± 0.22	66.58 ± 0.02	63.92 ± 0.37	66.55 ± 0.04
	STEPD	66.26 ± 0.04	66.26 ± 0.04	66.12 ± 0.04	65.87 ± 0.03	66.57 ± 0.02	65.92 ± 0.03	65.65 ± 0.04
	WSTD	68.08 ± 0.04	68.08 ± 0.04	68.01 ± 0.05	66.40 ± 0.03	66.62 ± 0.02	66.35 ± 0.05	66.30 ± 0.03
Agraw$_2$	ADWIN	88.74 ± 0.06	88.74 ± 0.06	88.77 ± 0.05	86.99 ± 0.02	87.02 ± 0.02	86.98 ± 0.02	86.99 ± 0.02
	DDM	88.81 ± 0.07	88.81 ± 0.07	88.84 ± 0.06	85.64 ± 0.85	87.02 ± 0.02	85.76 ± 0.08	85.85 ± 0.87
	ECDD	87.01 ± 0.02	87.01 ± 0.02	87.07 ± 0.02	85.38 ± 0.02	85.95 ± 0.08	85.35 ± 0.02	84.76 ± 0.04
	FHDDM	88.23 ± 0.05	88.23 ± 0.05	88.21 ± 0.04	85.78 ± 0.01	86.76 ± 0.02	86.37 ± 0.11	86.68 ± 0.02
	FTDD	88.87 ± 0.12	88.87 ± 0.12	88.89 ± 0.10	86.26 ± 0.30	87.00 ± 0.02	82.74 ± 1.26	86.87 ± 0.10
	HDDM$_A$	89.03 ± 0.04	89.03 ± 0.04	**89.04 ± 0.04**	86.29 ± 0.27	87.01 ± 0.02	85.62 ± 0.89	86.95 ± 0.02
	RDDM	88.94 ± 0.03	88.94 ± 0.03	88.97 ± 0.03	86.89 ± 0.02	86.97 ± 0.02	85.97 ± 0.02	86.88 ± 0.02
	SeqDr2	88.96 ± 0.07	88.96 ± 0.07	88.99 ± 0.05	86.90 ± 0.08	87.01 ± 0.02	81.65 ± 0.61	86.96 ± 0.03
	STEPD	87.89 ± 0.03	87.89 ± 0.03	87.88 ± 0.03	86.52 ± 0.03	86.71 ± 0.03	86.57 ± 0.02	86.32 ± 0.03
	WSTD	88.74 ± 0.02	88.74 ± 0.02	88.72 ± 0.02	86.21 ± 0.27	86.95 ± 0.02	84.91 ± 0.56	86.87 ± 0.03
LED	ADWIN	73.64 ± 0.10	73.64 ± 0.10	73.65 ± 0.10	73.88 ± 0.04	73.89 ± 0.03	**73.94 ± 0.04**	73.61 ± 0.18
	DDM	73.52 ± 0.12	73.52 ± 0.12	73.41 ± 0.17	71.27 ± 1.17	73.90 ± 0.04	52.03 ± 0.08	71.68 ± 1.42
	ECDD	73.40 ± 0.03	73.40 ± 0.03	73.40 ± 0.03	71.73 ± 0.07	73.64 ± 0.03	60.97 ± 0.05	69.31 ± 0.08
	FHDDM	73.76 ± 0.03	73.76 ± 0.03	73.76 ± 0.03	73.87 ± 0.04	73.92 ± 0.04	73.93 ± 0.05	73.06 ± 0.09
	FTDD	73.91 ± 0.04	73.91 ± 0.04	73.91 ± 0.04	73.38 ± 0.34	73.92 ± 0.03	72.57 ± 0.44	73.74 ± 0.18
	HDDM$_A$	73.93 ± 0.04	73.93 ± 0.04	73.93 ± 0.04	73.69 ± 0.15	73.93 ± 0.04	73.93 ± 0.04	73.89 ± 0.05
	RDDM	73.93 ± 0.04	73.93 ± 0.04	73.93 ± 0.04	73.85 ± 0.03	73.93 ± 0.04	64.74 ± 0.06	73.86 ± 0.04
	SeqDr2	73.92 ± 0.04	73.92 ± 0.04	73.92 ± 0.04	73.77 ± 0.14	73.91 ± 0.04	70.85 ± 0.70	73.71 ± 0.09
	STEPD	72.78 ± 0.07	72.78 ± 0.07	72.79 ± 0.07	73.16 ± 0.07	73.78 ± 0.04	73.15 ± 0.04	70.23 ± 0.23
	WSTD	73.92 ± 0.04	73.92 ± 0.04	73.92 ± 0.04	73.82 ± 0.03	73.92 ± 0.04	73.89 ± 0.05	73.60 ± 0.06
Mixed	ADWIN	90.17 ± 0.12	90.17 ± 0.12	91.28 ± 0.08	**92.03 ± 0.02**	**92.03 ± 0.02**	92.02 ± 0.03	**92.03 ± 0.02**
	DDM	91.45 ± 0.15	91.45 ± 0.15	91.59 ± 0.14	89.90 ± 1.40	92.02 ± 0.02	91.75 ± 0.15	88.51 ± 3.84
	ECDD	89.36 ± 0.03	89.36 ± 0.03	89.36 ± 0.03	90.74 ± 0.03	91.61 ± 0.03	91.52 ± 0.04	89.92 ± 0.04
	FHDDM	89.97 ± 0.26	89.97 ± 0.26	90.48 ± 0.16	90.76 ± 0.22	**92.03 ± 0.02**	76.63 ± 1.89	**92.03 ± 0.02**
	FTDD	90.53 ± 0.33	90.53 ± 0.33	91.47 ± 0.12	91.43 ± 0.41	**92.03 ± 0.02**	75.90 ± 2.01	**92.03 ± 0.03**
	HDDM$_A$	90.20 ± 0.22	90.20 ± 0.22	91.26 ± 0.11	91.50 ± 0.48	92.02 ± 0.02	87.28 ± 3.01	92.02 ± 0.02
	RDDM	90.27 ± 0.16	90.27 ± 0.16	91.32 ± 0.12	92.00 ± 0.03	92.02 ± 0.03	91.99 ± 0.03	91.98 ± 0.03
	SeqDr2	89.63 ± 0.46	89.63 ± 0.46	91.22 ± 0.34	92.02 ± 0.03	**92.03 ± 0.02**	83.99 ± 2.25	**92.03 ± 0.02**
	STEPD	90.18 ± 0.10	90.18 ± 0.10	90.75 ± 0.07	91.84 ± 0.03	91.97 ± 0.02	91.89 ± 0.06	91.60 ± 0.03
	WSTD	89.95 ± 0.35	89.95 ± 0.35	91.10 ± 0.21	91.76 ± 0.18	**92.03 ± 0.02**	75.59 ± 2.36	**92.03 ± 0.03**
RBF	ADWIN	18.77 ± 0.99	18.77 ± 0.99	30.59 ± 0.86	31.08 ± 0.24	33.77 ± 0.07	33.61 ± 0.16	31.63 ± 0.25
	DDM	17.86 ± 0.23	17.86 ± 0.23	32.49 ± 0.83	33.45 ± 0.26	34.13 ± 0.06	31.87 ± 0.35	33.57 ± 0.23
	ECDD	18.10 ± 0.49	18.10 ± 0.49	32.09 ± 0.58	33.07 ± 0.18	**34.50 ± 0.17**	33.07 ± 0.19	33.07 ± 0.19
	FHDDM	17.71 ± 0.07	17.71 ± 0.07	29.78 ± 0.76	30.19 ± 0.09	32.65 ± 0.05	30.24 ± 0.06	29.85 ± 0.11
	FTDD	18.61 ± 0.76	20.24 ± 1.15	32.47 ± 0.37	33.25 ± 0.15	34.13 ± 0.07	33.83 ± 0.22	33.21 ± 0.14
	HDDM$_A$	18.97 ± 1.02	21.43 ± 1.10	32.30 ± 0.29	33.35 ± 0.18	34.08 ± 0.07	31.66 ± 0.21	33.07 ± 0.19
	RDDM	18.99 ± 0.95	20.60 ± 0.86	32.06 ± 0.09	32.01 ± 0.10	33.73 ± 0.06	29.83 ± 0.05	32.13 ± 0.10
	SeqDr2	17.97 ± 0.38	17.97 ± 0.38	31.77 ± 0.75	33.25 ± 0.21	34.34 ± 0.11	34.02 ± 0.19	33.36 ± 0.18
	STEPD	17.80 ± 0.15	17.80 ± 0.15	29.22 ± 0.64	29.30 ± 0.06	32.60 ± 0.05	32.05 ± 0.11	29.00 ± 0.09
	WSTD	18.26 ± 0.38	20.42 ± 0.84	31.25 ± 0.12	31.24 ± 0.11	33.21 ± 0.06	33.58 ± 0.16	31.15 ± 0.15
Sine	ADWIN	88.67 ± 0.16	88.67 ± 0.16	89.02 ± 0.14	87.40 ± 0.03	87.40 ± 0.02	87.44 ± 0.03	87.41 ± 0.03
	DDM	89.01 ± 0.15	89.01 ± 0.15	89.65 ± 0.16	83.03 ± 4.18	87.41 ± 0.02	86.85 ± 0.27	80.81 ± 5.41
	ECDD	88.14 ± 0.04	88.14 ± 0.04	88.23 ± 0.04	87.12 ± 0.03	87.58 ± 0.02	87.39 ± 0.02	86.45 ± 0.02
	FHDDM	89.38 ± 0.24	89.38 ± 0.24	89.59 ± 0.20	86.25 ± 0.42	87.44 ± 0.03	73.64 ± 1.90	87.45 ± 0.03
	FTDD	89.03 ± 0.26	89.03 ± 0.26	89.45 ± 0.21	86.28 ± 0.44	87.43 ± 0.03	74.96 ± 1.05	87.44 ± 0.03
	HDDM$_A$	89.06 ± 0.14	89.06 ± 0.14	89.37 ± 0.09	86.49 ± 0.94	87.41 ± 0.02	83.95 ± 1.76	87.09 ± 0.67
	RDDM	89.35 ± 0.14	89.35 ± 0.14	89.60 ± 0.10	87.45 ± 0.03	87.49 ± 0.02	87.57 ± 0.02	87.46 ± 0.03
	SeqDr2	88.62 ± 0.25	88.62 ± 0.25	89.06 ± 0.20	87.25 ± 0.26	87.40 ± 0.02	77.67 ± 0.88	87.41 ± 0.02
	STEPD	89.19 ± 0.13	89.19 ± 0.13	89.35 ± 0.11	87.46 ± 0.02	87.51 ± 0.01	87.43 ± 0.05	87.33 ± 0.03
	WSTD	89.49 ± 0.21	89.49 ± 0.21	**89.69 ± 0.16**	87.00 ± 0.36	87.43 ± 0.03	74.99 ± 1.75	87.43 ± 0.02
Wavef.	ADWIN	81.65 ± 0.02	81.65 ± 0.02	81.65 ± 0.02	80.46 ± 0.03	80.56 ± 0.02	80.50 ± 0.04	80.45 ± 0.03
	DDM	81.10 ± 0.16	81.10 ± 0.16	81.08 ± 0.16	79.99 ± 0.28	80.64 ± 0.07	80.10 ± 0.07	79.53 ± 0.30
	ECDD	**82.02 ± 0.03**	**82.02 ± 0.03**	**82.02 ± 0.03**	80.17 ± 0.05	80.94 ± 0.04	80.55 ± 0.03	79.21 ± 0.04
	FHDDM	81.86 ± 0.04	81.86 ± 0.04	81.86 ± 0.04	80.43 ± 0.04	80.62 ± 0.04	80.40 ± 0.04	80.44 ± 0.04
	FTDD	81.71 ± 0.02	81.71 ± 0.02	81.71 ± 0.02	80.34 ± 0.12	80.60 ± 0.04	79.14 ± 0.29	80.47 ± 0.04
	HDDM$_A$	81.67 ± 0.03	81.67 ± 0.03	81.67 ± 0.03	80.43 ± 0.06	80.60 ± 0.04	80.33 ± 0.13	80.46 ± 0.04
	RDDM	81.71 ± 0.03	81.71 ± 0.03	81.71 ± 0.03	80.44 ± 0.04	80.69 ± 0.04	80.54 ± 0.03	80.45 ± 0.04
	SeqDr2	81.67 ± 0.03	81.67 ± 0.03	81.67 ± 0.03	80.41 ± 0.09	80.59 ± 0.05	78.42 ± 0.30	80.48 ± 0.04
	STEPD	81.87 ± 0.04	81.87 ± 0.04	81.86 ± 0.04	80.44 ± 0.04	80.95 ± 0.04	80.50 ± 0.04	80.31 ± 0.04
	WSTD	81.75 ± 0.04	81.75 ± 0.04	81.75 ± 0.04	80.44 ± 0.04	80.61 ± 0.04	80.12 ± 0.12	80.45 ± 0.04

Table 15
Mean accuracies of Ensembles in percentage (%) in 10K instances abrupt datasets, with 95% confidence intervals, using HT.

Dataset	Ensemble	ADOB	$BOLE_4$	$BOLE_5$	DDD	FASE	LevBag	None
$Agraw_1$	ADWIN	61.90 ± 0.51	61.90 ± 0.51	62.36 ± 0.46	62.47 ± 0.25	64.11 ± 0.27	61.04 ± 0.61	62.79 ± 0.34
	DDM	61.17 ± 0.60	61.17 ± 0.60	61.66 ± 0.50	62.59 ± 0.57	**64.71 ± 0.33**	61.13 ± 0.65	63.13 ± 0.56
	ECDD	61.55 ± 0.41	61.55 ± 0.41	61.58 ± 0.40	63.29 ± 0.24	63.88 ± 0.32	60.42 ± 0.52	63.27 ± 0.33
	FHDDM	63.09 ± 0.34	63.09 ± 0.34	63.53 ± 0.33	64.31 ± 0.34	64.62 ± 0.34	62.20 ± 0.61	64.48 ± 0.39
	FTDD	60.43 ± 0.49	60.48 ± 0.48	60.75 ± 0.47	62.53 ± 0.36	64.21 ± 0.35	59.09 ± 0.44	62.64 ± 0.38
	$HDDM_A$	62.62 ± 0.58	62.60 ± 0.58	63.06 ± 0.53	63.94 ± 0.45	64.60 ± 0.38	62.10 ± 0.57	64.47 ± 0.34
	RDDM	62.48 ± 0.52	62.47 ± 0.52	62.96 ± 0.50	64.15 ± 0.34	**64.71 ± 0.36**	62.06 ± 0.62	64.69 ± 0.30
	SeqDr2	62.02 ± 0.46	62.02 ± 0.46	62.40 ± 0.43	62.78 ± 0.33	64.17 ± 0.26	58.92 ± 0.55	62.88 ± 0.38
	STEPD	62.37 ± 0.35	62.37 ± 0.35	62.79 ± 0.34	63.67 ± 0.34	63.97 ± 0.25	62.47 ± 0.72	63.78 ± 0.38
	WSTD	62.67 ± 0.47	62.67 ± 0.48	63.01 ± 0.45	63.43 ± 0.35	64.21 ± 0.27	61.94 ± 0.60	63.44 ± 0.43
$Agraw_2$	ADWIN	79.98 ± 0.28	79.98 ± 0.28	80.31 ± 0.24	78.76 ± 0.37	81.55 ± 0.30	78.92 ± 0.42	79.81 ± 0.33
	DDM	78.64 ± 0.36	78.64 ± 0.36	78.92 ± 0.34	76.79 ± 1.43	82.17 ± 0.19	81.20 ± 0.25	75.22 ± 1.80
	ECDD	81.84 ± 0.25	81.84 ± 0.25	81.99 ± 0.25	82.34 ± 0.26	**82.47 ± 0.20**	81.47 ± 0.22	81.95 ± 0.27
	FHDDM	82.15 ± 0.25	82.15 ± 0.25	82.38 ± 0.25	81.36 ± 0.29	82.12 ± 0.19	77.41 ± 0.48	82.28 ± 0.29
	FTDD	80.49 ± 0.30	80.52 ± 0.30	80.79 ± 0.27	79.05 ± 0.42	81.58 ± 0.27	73.84 ± 1.06	79.41 ± 0.66
	$HDDM_A$	81.23 ± 0.33	81.26 ± 0.32	81.56 ± 0.31	80.72 ± 0.40	81.79 ± 0.28	79.90 ± 0.33	81.56 ± 0.44
	RDDM	81.22 ± 0.35	81.26 ± 0.36	81.47 ± 0.32	81.42 ± 0.49	82.39 ± 0.19	81.49 ± 0.20	81.58 ± 0.85
	SeqDr2	79.58 ± 0.30	79.58 ± 0.30	79.93 ± 0.26	76.91 ± 0.32	81.37 ± 0.28	71.44 ± 0.65	77.81 ± 0.34
	STEPD	82.24 ± 0.23	82.24 ± 0.23	82.40 ± 0.26	81.23 ± 0.52	82.13 ± 0.22	80.94 ± 0.24	81.97 ± 0.43
	WSTD	81.67 ± 0.35	81.70 ± 0.34	81.94 ± 0.33	80.17 ± 0.53	81.86 ± 0.26	78.92 ± 0.40	81.07 ± 0.51
LED	ADWIN	67.41 ± 0.27	67.41 ± 0.27	67.63 ± 0.26	66.37 ± 0.37	68.17 ± 0.29	68.25 ± 0.32	63.48 ± 0.69
	DDM	68.24 ± 0.29	68.24 ± 0.29	68.40 ± 0.29	68.58 ± 0.37	68.48 ± 0.27	58.99 ± 0.55	69.56 ± 0.30
	ECDD	55.35 ± 4.31	55.35 ± 4.31	67.29 ± 0.74	68.63 ± 0.49	68.98 ± 0.28	63.63 ± 0.55	67.35 ± 0.43
	FHDDM	68.67 ± 0.30	68.67 ± 0.30	68.89 ± 0.28	68.94 ± 0.31	68.66 ± 0.25	68.56 ± 0.52	69.29 ± 0.37
	FTDD	65.07 ± 2.07	68.48 ± 0.31	68.05 ± 0.37	65.93 ± 0.94	68.05 ± 0.25	67.91 ± 0.51	67.01 ± 0.74
	$HDDM_A$	68.67 ± 0.29	69.01 ± 0.27	69.04 ± 0.27	68.73 ± 0.29	68.52 ± 0.26	**69.89 ± 0.31**	69.68 ± 0.30
	RDDM	68.49 ± 0.29	68.78 ± 0.26	68.82 ± 0.26	68.62 ± 0.38	68.50 ± 0.26	65.95 ± 0.47	69.78 ± 0.29
	SeqDr2	66.43 ± 0.39	66.43 ± 0.39	66.85 ± 0.28	65.21 ± 0.51	67.02 ± 0.37	64.51 ± 0.40	60.17 ± 0.59
	STEPD	65.97 ± 0.42	65.97 ± 0.42	66.35 ± 0.39	66.91 ± 0.44	66.20 ± 0.48	69.42 ± 0.37	60.78 ± 1.68
	WSTD	67.78 ± 0.36	68.12 ± 0.34	68.22 ± 0.32	68.31 ± 0.37	66.93 ± 0.39	68.27 ± 0.56	67.08 ± 1.00
Mixed	ADWIN	89.33 ± 0.24	89.33 ± 0.24	89.36 ± 0.23	87.05 ± 0.51	90.01 ± 0.18	83.46 ± 0.36	89.43 ± 0.26
	DDM	89.66 ± 0.21	89.66 ± 0.21	89.74 ± 0.20	88.07 ± 0.57	89.91 ± 0.18	88.87 ± 0.31	89.70 ± 0.29
	ECDD	88.92 ± 0.24	88.92 ± 0.24	88.98 ± 0.23	88.69 ± 0.37	90.06 ± 0.22	86.90 ± 0.66	89.01 ± 0.28
	FHDDM	89.76 ± 0.24	89.76 ± 0.24	89.93 ± 0.23	86.79 ± 0.43	89.77 ± 0.18	85.15 ± 0.66	90.14 ± 0.21
	FTDD	89.99 ± 0.23	89.99 ± 0.22	90.16 ± 0.22	88.45 ± 0.76	89.84 ± 0.18	86.48 ± 0.51	90.33 ± 0.23
	$HDDM_A$	89.93 ± 0.21	89.93 ± 0.20	90.09 ± 0.20	87.68 ± 0.65	89.96 ± 0.17	87.11 ± 0.89	90.32 ± 0.23
	RDDM	89.89 ± 0.23	89.90 ± 0.23	90.09 ± 0.23	87.51 ± 0.68	89.92 ± 0.19	88.43 ± 0.37	90.17 ± 0.24
	SeqDr2	88.51 ± 0.26	88.51 ± 0.26	88.46 ± 0.26	81.94 ± 0.30	**90.36 ± 0.18**	74.26 ± 0.52	83.77 ± 0.21
	STEPD	89.63 ± 0.22	89.63 ± 0.22	89.79 ± 0.22	87.38 ± 0.85	89.91 ± 0.19	86.13 ± 0.57	90.30 ± 0.29
	WSTD	90.04 ± 0.22	90.05 ± 0.21	90.18 ± 0.21	87.58 ± 0.90	89.83 ± 0.19	85.90 ± 0.59	**90.36 ± 0.22**
RBF	ADWIN	20.93 ± 0.90	20.93 ± 0.90	30.95 ± 0.92	31.90 ± 0.41	32.57 ± 0.40	33.47 ± 0.44	31.82 ± 0.42
	DDM	20.99 ± 0.90	20.99 ± 0.90	30.59 ± 0.92	31.72 ± 0.50	32.44 ± 0.39	33.36 ± 0.47	31.92 ± 0.44
	ECDD	21.00 ± 0.91	21.00 ± 0.91	29.80 ± 0.91	30.91 ± 0.65	32.29 ± 0.42	33.73 ± 0.38	30.83 ± 0.64
	FHDDM	20.70 ± 0.86	20.70 ± 0.86	30.63 ± 0.89	31.57 ± 0.41	32.30 ± 0.33	32.45 ± 0.45	31.34 ± 0.37
	FTDD	21.39 ± 0.95	24.90 ± 0.76	31.40 ± 0.58	31.86 ± 0.55	32.50 ± 0.43	33.85 ± 0.47	32.26 ± 0.50
	$HDDM_A$	21.08 ± 0.88	24.73 ± 0.77	31.94 ± 0.51	31.84 ± 0.39	32.49 ± 0.41	33.06 ± 0.43	32.06 ± 0.37
	RDDM	20.94 ± 0.87	24.89 ± 0.87	31.97 ± 0.52	31.90 ± 0.43	32.17 ± 0.35	32.12 ± 0.37	32.01 ± 0.39
	SeqDr2	21.03 ± 0.84	21.03 ± 0.84	30.64 ± 1.00	31.86 ± 0.51	32.50 ± 0.41	**33.86 ± 0.49**	32.34 ± 0.47
	STEPD	20.17 ± 0.84	20.17 ± 0.84	30.48 ± 0.88	31.55 ± 0.48	32.27 ± 0.37	33.30 ± 0.46	31.23 ± 0.50
	WSTD	21.61 ± 0.98	25.14 ± 0.72	30.89 ± 0.64	31.06 ± 0.65	32.41 ± 0.36	33.77 ± 0.45	30.93 ± 0.60
Sine	ADWIN	89.09 ± 0.26	89.09 ± 0.26	89.09 ± 0.26	85.94 ± 0.30	88.61 ± 0.18	85.07 ± 0.52	87.44 ± 0.17
	DDM	89.62 ± 0.24	89.62 ± 0.24	89.66 ± 0.22	87.01 ± 0.29	88.58 ± 0.17	89.35 ± 0.21	87.01 ± 0.72
	ECDD	88.50 ± 0.19	88.50 ± 0.19	88.59 ± 0.19	86.28 ± 0.27	88.04 ± 0.19	88.12 ± 0.34	86.57 ± 0.27
	FHDDM	89.33 ± 0.28	89.33 ± 0.28	89.36 ± 0.29	87.07 ± 0.27	88.43 ± 0.16	87.68 ± 0.51	88.29 ± 0.15
	FTDD	89.90 ± 0.26	89.92 ± 0.26	**89.99 ± 0.27**	87.17 ± 0.31	88.43 ± 0.18	89.07 ± 0.46	88.37 ± 0.17
	$HDDM_A$	89.64 ± 0.24	89.66 ± 0.24	89.71 ± 0.23	87.43 ± 0.31	88.58 ± 0.18	89.26 ± 0.49	88.39 ± 0.17
	RDDM	89.42 ± 0.24	89.45 ± 0.25	89.45 ± 0.25	87.34 ± 0.28	88.48 ± 0.20	89.56 ± 0.30	87.98 ± 0.20
	SeqDr2	88.71 ± 0.30	88.71 ± 0.30	88.63 ± 0.29	81.59 ± 0.31	88.48 ± 0.17	74.56 ± 0.50	83.02 ± 0.19
	STEPD	89.14 ± 0.20	89.14 ± 0.20	89.16 ± 0.20	86.99 ± 0.40	88.53 ± 0.18	88.82 ± 0.40	88.06 ± 0.19
	WSTD	89.59 ± 0.24	89.62 ± 0.25	89.64 ± 0.24	87.21 ± 0.36	88.44 ± 0.16	88.63 ± 0.54	88.38 ± 0.15
Wavef.	ADWIN	79.91 ± 0.32	79.91 ± 0.32	79.27 ± 0.37	78.14 ± 0.46	78.82 ± 0.42	80.79 ± 0.36	78.52 ± 0.42
	DDM	79.81 ± 0.31	79.81 ± 0.31	79.08 ± 0.37	78.34 ± 0.47	78.95 ± 0.41	80.16 ± 0.40	78.45 ± 0.46
	ECDD	80.60 ± 0.30	80.60 ± 0.30	80.60 ± 0.31	79.11 ± 0.43	79.69 ± 0.36	79.81 ± 0.36	78.30 ± 0.39
	FHDDM	80.61 ± 0.32	80.61 ± 0.32	80.38 ± 0.34	78.65 ± 0.48	79.14 ± 0.41	80.85 ± 0.35	78.99 ± 0.45
	FTDD	79.93 ± 0.31	80.06 ± 0.32	79.24 ± 0.37	77.90 ± 0.51	78.94 ± 0.44	80.76 ± 0.39	78.07 ± 0.58
	$HDDM_A$	80.29 ± 0.31	80.42 ± 0.30	79.82 ± 0.36	78.54 ± 0.48	78.99 ± 0.43	**81.01 ± 0.34**	78.69 ± 0.48
	RDDM	80.39 ± 0.32	80.52 ± 0.31	79.94 ± 0.36	79.03 ± 0.40	79.37 ± 0.38	80.42 ± 0.35	79.09 ± 0.47
	SeqDr2	80.04 ± 0.30	80.04 ± 0.30	79.45 ± 0.38	78.36 ± 0.43	78.84 ± 0.41	80.71 ± 0.38	78.53 ± 0.44
	STEPD	80.64 ± 0.30	80.64 ± 0.30	80.52 ± 0.30	79.01 ± 0.42	79.52 ± 0.42	81.00 ± 0.33	79.21 ± 0.42
	WSTD	80.37 ± 0.30	80.50 ± 0.30	80.02 ± 0.33	78.37 ± 0.50	79.06 ± 0.42	80.96 ± 0.38	78.77 ± 0.51

Table 16
Mean accuracies of Ensembles in percentage (%) in 20K instances abrupt datasets, with 95% confidence intervals, using HT.

Dataset	Ensemble	ADOB	$BOLE_4$	$BOLE_5$	DDD	FASE	LevBag	None
$Agraw_1$	ADWIN	62.68 ± 0.30	62.68 ± 0.30	62.99 ± 0.28	64.15 ± 0.18	65.75 ± 0.22	65.47 ± 0.55	64.56 ± 0.25
	DDM	64.42 ± 0.64	64.42 ± 0.64	64.40 ± 0.58	64.89 ± 1.09	68.46 ± 0.32	63.50 ± 0.69	64.93 ± 1.28
	ECDD	63.36 ± 0.39	63.36 ± 0.39	63.30 ± 0.38	64.50 ± 0.55	66.00 ± 0.46	61.92 ± 0.66	63.84 ± 0.44
	FHDDM	65.59 ± 0.57	65.59 ± 0.57	65.85 ± 0.55	67.40 ± 0.50	68.48 ± 0.39	65.64 ± 0.65	67.26 ± 0.59
	FTDD	60.20 ± 0.47	60.19 ± 0.47	60.46 ± 0.42	61.83 ± 0.81	66.27 ± 0.35	61.13 ± 0.60	64.04 ± 0.76
	$HDDM_A$	65.88 ± 0.55	65.87 ± 0.55	66.05 ± 0.47	67.27 ± 0.44	68.44 ± 0.36	65.91 ± 0.54	68.12 ± 0.48
	RDDM	66.00 ± 0.56	66.00 ± 0.57	66.17 ± 0.50	67.38 ± 0.48	**68.52 ± 0.32**	64.86 ± 0.62	68.19 ± 0.45
	SeqDr2	61.37 ± 0.39	61.37 ± 0.39	61.52 ± 0.35	64.36 ± 0.38	66.04 ± 0.21	61.16 ± 0.60	64.46 ± 0.48
	STEPD	63.84 ± 0.22	63.84 ± 0.22	64.16 ± 0.23	64.88 ± 0.27	65.64 ± 0.22	66.04 ± 0.53	64.97 ± 0.30
	WSTD	63.38 ± 0.39	63.38 ± 0.38	63.60 ± 0.37	65.53 ± 0.54	66.08 ± 0.28	65.23 ± 0.77	65.33 ± 0.48
$Agraw_2$	ADWIN	83.05 ± 0.16	83.05 ± 0.16	83.24 ± 0.14	82.99 ± 0.17	83.99 ± 0.21	82.50 ± 0.20	83.75 ± 0.22
	DDM	80.89 ± 0.29	80.89 ± 0.29	80.92 ± 0.28	80.61 ± 1.13	84.35 ± 0.18	83.35 ± 0.14	81.19 ± 1.60
	ECDD	83.49 ± 0.17	83.49 ± 0.17	83.49 ± 0.18	83.98 ± 0.12	84.35 ± 0.10	83.17 ± 0.10	83.49 ± 0.17
	FHDDM	84.24 ± 0.16	84.24 ± 0.16	84.35 ± 0.14	84.12 ± 0.18	84.60 ± 0.12	81.52 ± 0.33	**84.68 ± 0.17**
	FTDD	82.79 ± 0.16	82.81 ± 0.16	82.87 ± 0.17	83.31 ± 0.22	83.97 ± 0.23	77.57 ± 0.99	84.13 ± 0.24
	$HDDM_A$	83.30 ± 0.21	83.32 ± 0.21	83.46 ± 0.21	83.63 ± 0.29	84.36 ± 0.22	82.45 ± 0.35	84.44 ± 0.26
	RDDM	83.45 ± 0.28	83.46 ± 0.28	83.58 ± 0.28	83.34 ± 0.31	84.57 ± 0.13	83.52 ± 0.15	83.11 ± 1.31
	SeqDr2	82.76 ± 0.18	82.76 ± 0.18	82.97 ± 0.18	82.20 ± 0.14	83.95 ± 0.20	75.12 ± 0.54	83.04 ± 0.20
	STEPD	84.14 ± 0.20	84.14 ± 0.20	84.26 ± 0.20	84.22 ± 0.19	84.59 ± 0.13	83.18 ± 0.16	84.56 ± 0.23
	WSTD	83.72 ± 0.20	83.74 ± 0.20	83.88 ± 0.21	83.80 ± 0.21	84.32 ± 0.19	82.50 ± 0.34	84.52 ± 0.23
LED	ADWIN	69.78 ± 0.19	69.78 ± 0.19	69.95 ± 0.19	68.94 ± 0.39	70.49 ± 0.20	70.60 ± 0.28	65.49 ± 0.71
	DDM	70.76 ± 0.17	70.76 ± 0.17	70.86 ± 0.18	70.46 ± 0.26	70.98 ± 0.18	59.81 ± 0.35	71.31 ± 0.25
	ECDD	71.18 ± 0.20	71.18 ± 0.20	71.30 ± 0.19	70.10 ± 0.29	71.31 ± 0.18	64.53 ± 0.39	68.09 ± 0.42
	FHDDM	71.20 ± 0.20	71.20 ± 0.20	71.31 ± 0.21	70.82 ± 0.21	71.31 ± 0.18	70.63 ± 0.40	71.26 ± 0.32
	FTDD	70.89 ± 0.19	71.05 ± 0.19	71.00 ± 0.20	69.86 ± 0.38	70.84 ± 0.16	70.12 ± 0.42	70.51 ± 0.43
	$HDDM_A$	71.26 ± 0.17	71.41 ± 0.17	71.42 ± 0.17	70.43 ± 0.21	71.18 ± 0.16	**71.76 ± 0.20**	71.52 ± 0.18
	RDDM	71.08 ± 0.18	71.23 ± 0.18	71.26 ± 0.18	70.63 ± 0.23	71.13 ± 0.16	67.15 ± 0.38	71.73 ± 0.17
	SeqDr2	69.16 ± 0.20	69.16 ± 0.20	69.39 ± 0.20	68.57 ± 0.35	69.68 ± 0.31	66.86 ± 0.32	62.64 ± 0.77
	STEPD	68.57 ± 0.37	68.57 ± 0.37	68.78 ± 0.37	69.61 ± 0.45	69.57 ± 0.38	71.37 ± 0.25	65.16 ± 1.59
	WSTD	70.48 ± 0.29	70.64 ± 0.28	70.72 ± 0.28	70.39 ± 0.25	70.26 ± 0.24	70.36 ± 0.44	70.25 ± 0.60
Mixed	ADWIN	90.66 ± 0.13	90.66 ± 0.13	90.85 ± 0.12	89.07 ± 0.21	90.78 ± 0.12	88.69 ± 0.22	90.25 ± 0.14
	DDM	91.20 ± 0.15	91.20 ± 0.15	91.35 ± 0.15	89.06 ± 0.50	90.86 ± 0.12	**91.89 ± 0.18**	88.96 ± 0.54
	ECDD	89.70 ± 0.18	89.70 ± 0.18	89.79 ± 0.18	89.79 ± 0.24	90.83 ± 0.13	90.30 ± 0.24	89.37 ± 0.20
	FHDDM	90.84 ± 0.16	90.84 ± 0.16	91.02 ± 0.16	88.51 ± 0.36	90.77 ± 0.11	89.85 ± 0.34	90.58 ± 0.14
	FTDD	91.29 ± 0.13	91.29 ± 0.13	91.45 ± 0.12	88.95 ± 0.68	90.85 ± 0.11	90.48 ± 0.30	90.64 ± 0.15
	$HDDM_A$	90.89 ± 0.20	90.90 ± 0.19	91.10 ± 0.19	88.27 ± 0.47	90.83 ± 0.11	91.31 ± 0.38	90.29 ± 0.15
	RDDM	90.85 ± 0.15	90.85 ± 0.15	91.07 ± 0.14	89.29 ± 0.42	90.89 ± 0.11	91.78 ± 0.21	90.66 ± 0.14
	SeqDr2	90.57 ± 0.19	90.57 ± 0.19	90.72 ± 0.18	86.63 ± 0.21	90.97 ± 0.11	82.27 ± 0.30	89.03 ± 0.27
	STEPD	90.69 ± 0.14	90.69 ± 0.14	90.86 ± 0.14	89.22 ± 0.51	90.87 ± 0.11	90.39 ± 0.35	90.65 ± 0.17
	WSTD	91.01 ± 0.15	91.01 ± 0.14	91.21 ± 0.15	88.28 ± 0.66	90.85 ± 0.11	90.76 ± 0.25	90.64 ± 0.15
RBF	ADWIN	20.28 ± 0.81	20.28 ± 0.81	31.08 ± 0.95	32.10 ± 0.33	33.03 ± 0.30	35.72 ± 0.32	32.26 ± 0.38
	DDM	20.36 ± 0.81	20.36 ± 0.81	30.52 ± 0.93	31.70 ± 0.50	32.90 ± 0.31	35.53 ± 0.29	31.82 ± 0.43
	ECDD	20.37 ± 0.81	20.37 ± 0.81	29.71 ± 0.88	30.91 ± 0.62	32.92 ± 0.30	**36.24 ± 0.31**	31.23 ± 0.62
	FHDDM	19.93 ± 0.65	19.93 ± 0.65	30.98 ± 0.89	31.75 ± 0.37	32.71 ± 0.25	32.64 ± 0.34	31.39 ± 0.32
	FTDD	20.96 ± 0.84	23.48 ± 0.42	31.90 ± 0.57	32.04 ± 0.43	33.09 ± 0.30	35.99 ± 0.26	32.60 ± 0.45
	$HDDM_A$	20.31 ± 0.80	23.40 ± 0.48	32.19 ± 0.44	32.27 ± 0.38	32.99 ± 0.30	34.92 ± 0.42	32.40 ± 0.34
	RDDM	20.30 ± 0.77	23.80 ± 0.56	32.31 ± 0.40	32.24 ± 0.31	32.69 ± 0.31	33.01 ± 0.37	32.30 ± 0.37
	SeqDr2	20.84 ± 0.81	20.84 ± 0.81	30.92 ± 0.94	32.09 ± 0.43	33.07 ± 0.33	36.14 ± 0.25	32.72 ± 0.41
	STEPD	19.74 ± 0.76	19.74 ± 0.76	31.03 ± 0.74	31.59 ± 0.35	32.65 ± 0.30	34.90 ± 0.39	31.43 ± 0.40
	WSTD	20.82 ± 0.81	24.16 ± 0.64	30.70 ± 0.60	31.15 ± 0.57	32.89 ± 0.27	35.86 ± 0.37	31.12 ± 0.54
Sine	ADWIN	90.86 ± 0.18	90.86 ± 0.18	90.87 ± 0.19	88.02 ± 0.24	90.13 ± 0.11	89.53 ± 0.29	89.16 ± 0.13
	DDM	91.50 ± 0.22	91.50 ± 0.22	91.49 ± 0.21	88.70 ± 0.25	90.19 ± 0.13	**91.92 ± 0.19**	89.31 ± 0.14
	ECDD	89.48 ± 0.27	89.48 ± 0.27	89.51 ± 0.27	87.32 ± 0.19	89.05 ± 0.14	89.93 ± 0.26	86.90 ± 0.19
	FHDDM	91.14 ± 0.22	91.14 ± 0.22	91.15 ± 0.22	88.15 ± 0.34	90.13 ± 0.12	90.97 ± 0.32	89.87 ± 0.12
	FTDD	91.54 ± 0.21	91.56 ± 0.21	91.58 ± 0.21	88.35 ± 0.39	90.17 ± 0.12	91.74 ± 0.28	89.89 ± 0.13
	$HDDM_A$	91.43 ± 0.22	91.44 ± 0.22	91.46 ± 0.20	88.11 ± 0.42	90.22 ± 0.12	91.53 ± 0.45	89.89 ± 0.13
	RDDM	91.28 ± 0.23	91.29 ± 0.23	91.33 ± 0.23	88.31 ± 0.30	90.11 ± 0.13	91.87 ± 0.22	89.46 ± 0.14
	SeqDr2	90.89 ± 0.24	90.89 ± 0.24	90.84 ± 0.23	85.81 ± 0.23	90.16 ± 0.12	80.43 ± 0.35	87.04 ± 0.14
	STEPD	90.72 ± 0.25	90.72 ± 0.25	90.71 ± 0.23	88.55 ± 0.36	90.15 ± 0.12	91.30 ± 0.25	89.22 ± 0.20
	WSTD	91.52 ± 0.24	91.53 ± 0.25	91.57 ± 0.23	88.27 ± 0.32	90.14 ± 0.12	91.69 ± 0.25	89.93 ± 0.12
Wavef.	ADWIN	81.12 ± 0.22	81.12 ± 0.22	80.66 ± 0.24	79.43 ± 0.26	80.27 ± 0.23	82.56 ± 0.18	79.31 ± 0.29
	DDM	81.08 ± 0.19	81.08 ± 0.19	80.61 ± 0.20	79.40 ± 0.23	80.49 ± 0.21	81.19 ± 0.35	78.89 ± 0.22
	ECDD	81.33 ± 0.18	81.33 ± 0.18	81.24 ± 0.18	79.58 ± 0.24	80.25 ± 0.21	80.23 ± 0.21	78.65 ± 0.25
	FHDDM	81.32 ± 0.18	81.32 ± 0.18	81.02 ± 0.21	79.54 ± 0.24	80.33 ± 0.22	82.49 ± 0.26	79.52 ± 0.26
	FTDD	81.23 ± 0.19	81.30 ± 0.19	80.82 ± 0.20	79.43 ± 0.25	80.43 ± 0.22	82.41 ± 0.24	79.05 ± 0.36
	$HDDM_A$	81.14 ± 0.21	81.21 ± 0.21	80.64 ± 0.24	79.60 ± 0.26	80.40 ± 0.21	82.62 ± 0.19	79.41 ± 0.25
	RDDM	81.29 ± 0.17	81.36 ± 0.17	80.88 ± 0.19	79.83 ± 0.27	80.41 ± 0.22	81.21 ± 0.20	79.64 ± 0.26
	SeqDr2	81.16 ± 0.18	81.16 ± 0.18	80.65 ± 0.21	79.59 ± 0.24	80.11 ± 0.22	82.49 ± 0.22	79.33 ± 0.28
	STEPD	81.41 ± 0.19	81.41 ± 0.19	81.22 ± 0.21	79.72 ± 0.23	80.27 ± 0.21	81.90 ± 0.19	79.69 ± 0.22
	WSTD	81.13 ± 0.18	81.20 ± 0.18	80.75 ± 0.19	79.52 ± 0.26	80.39 ± 0.18	**82.65 ± 0.22**	79.46 ± 0.29

Table 17
Mean accuracies of Ensembles in percentage (%) in 50K instances abrupt datasets, with 95% confidence intervals, using HT.

Dataset	Ensemble	ADOB	$BOLE_4$	$BOLE_5$	DDD	FASE	LevBag	None
$Agraw_1$	ADWIN	63.28 ± 0.19	63.28 ± 0.19	63.40 ± 0.19	66.16 ± 0.21	67.12 ± 0.18	68.89 ± 0.47	65.99 ± 0.14
	DDM	67.47 ± 0.37	67.47 ± 0.37	67.36 ± 0.33	68.66 ± 1.54	72.47 ± 0.21	66.82 ± 0.47	68.03 ± 1.98
	ECDD	67.33 ± 0.21	67.33 ± 0.21	67.27 ± 0.22	66.38 ± 0.68	69.29 ± 0.34	64.38 ± 0.43	64.76 ± 0.64
	FHDDM	68.69 ± 0.30	68.69 ± 0.30	68.87 ± 0.31	70.99 ± 0.26	72.18 ± 0.22	69.22 ± 0.32	71.50 ± 0.28
	FTDD	62.73 ± 0.52	62.73 ± 0.52	62.73 ± 0.48	65.98 ± 1.11	68.73 ± 0.45	63.87 ± 0.67	67.23 ± 0.89
	$HDDM_A$	68.32 ± 0.32	68.32 ± 0.32	68.36 ± 0.25	71.31 ± 0.30	72.38 ± 0.23	69.16 ± 0.43	**72.57 ± 0.33**
	RDDM	68.68 ± 0.23	68.68 ± 0.23	68.86 ± 0.21	71.47 ± 0.28	72.31 ± 0.22	67.61 ± 0.29	72.43 ± 0.31
	SeqDr2	62.73 ± 0.31	62.73 ± 0.31	62.71 ± 0.27	66.12 ± 0.32	67.83 ± 0.24	65.62 ± 0.48	67.29 ± 0.36
	STEPD	64.03 ± 0.14	64.03 ± 0.14	64.21 ± 0.14	66.40 ± 0.29	67.33 ± 0.30	68.88 ± 0.19	66.18 ± 0.29
	WSTD	63.98 ± 0.33	63.98 ± 0.33	64.08 ± 0.30	70.43 ± 0.50	69.25 ± 0.61	69.09 ± 0.45	69.16 ± 0.72
$Agraw_2$	ADWIN	84.67 ± 0.19	84.67 ± 0.19	84.71 ± 0.18	85.27 ± 0.16	86.16 ± 0.09	84.77 ± 0.16	85.39 ± 0.15
	DDM	83.46 ± 0.34	83.46 ± 0.34	83.48 ± 0.31	83.20 ± 0.80	86.41 ± 0.08	85.00 ± 0.10	83.60 ± 1.14
	ECDD	84.82 ± 0.10	84.82 ± 0.10	84.83 ± 0.11	84.90 ± 0.08	85.41 ± 0.07	84.25 ± 0.07	84.40 ± 0.08
	FHDDM	85.77 ± 0.09	85.77 ± 0.09	85.83 ± 0.09	86.09 ± 0.12	86.48 ± 0.08	84.54 ± 0.14	**86.54 ± 0.13**
	FTDD	84.01 ± 0.29	84.02 ± 0.29	84.02 ± 0.26	83.69 ± 0.53	86.41 ± 0.08	82.65 ± 0.22	84.46 ± 0.44
	$HDDM_A$	84.97 ± 0.18	84.97 ± 0.18	84.99 ± 0.18	85.74 ± 0.33	86.42 ± 0.08	84.56 ± 0.20	85.76 ± 0.32
	RDDM	85.17 ± 0.13	85.18 ± 0.13	85.25 ± 0.13	85.62 ± 0.29	86.50 ± 0.08	85.11 ± 0.12	86.09 ± 0.19
	SeqDr2	83.70 ± 0.18	83.70 ± 0.18	83.77 ± 0.15	83.57 ± 0.47	86.35 ± 0.08	77.65 ± 0.55	84.32 ± 0.42
	STEPD	85.50 ± 0.10	85.50 ± 0.10	85.55 ± 0.10	86.12 ± 0.17	86.37 ± 0.07	85.14 ± 0.10	85.95 ± 0.16
	WSTD	85.51 ± 0.09	85.51 ± 0.09	85.54 ± 0.10	85.50 ± 0.44	86.40 ± 0.08	84.66 ± 0.18	85.86 ± 0.42
LED	ADWIN	71.58 ± 0.19	71.58 ± 0.19	71.67 ± 0.19	71.40 ± 0.22	72.11 ± 0.17	72.39 ± 0.16	67.13 ± 0.58
	DDM	72.38 ± 0.16	72.38 ± 0.16	72.42 ± 0.16	71.72 ± 0.37	72.58 ± 0.15	60.10 ± 0.28	71.93 ± 0.48
	ECDD	72.49 ± 0.16	72.49 ± 0.16	72.55 ± 0.16	70.95 ± 0.25	72.63 ± 0.17	64.66 ± 0.29	68.69 ± 0.32
	FHDDM	72.62 ± 0.16	72.62 ± 0.16	72.67 ± 0.16	72.28 ± 0.20	72.79 ± 0.15	72.71 ± 0.19	72.10 ± 0.29
	FTDD	72.41 ± 0.17	72.47 ± 0.17	72.46 ± 0.17	71.41 ± 0.27	72.45 ± 0.15	72.35 ± 0.22	72.20 ± 0.21
	$HDDM_A$	72.75 ± 0.16	72.81 ± 0.16	72.82 ± 0.16	71.68 ± 0.22	72.74 ± 0.15	72.83 ± 0.16	72.81 ± 0.16
	RDDM	72.61 ± 0.16	72.67 ± 0.16	72.68 ± 0.16	71.98 ± 0.23	72.70 ± 0.14	67.80 ± 0.28	**72.88 ± 0.15**
	SeqDr2	71.57 ± 0.17	71.57 ± 0.17	71.67 ± 0.17	71.24 ± 0.23	71.89 ± 0.16	70.00 ± 0.22	66.53 ± 1.24
	STEPD	70.55 ± 0.26	70.55 ± 0.26	70.65 ± 0.25	71.64 ± 0.33	71.97 ± 0.19	72.47 ± 0.21	67.85 ± 1.10
	WSTD	72.41 ± 0.18	72.48 ± 0.18	72.52 ± 0.18	72.05 ± 0.22	72.34 ± 0.17	72.62 ± 0.17	71.99 ± 0.31
Mixed	ADWIN	92.28 ± 0.10	92.28 ± 0.10	92.41 ± 0.10	91.35 ± 0.19	92.23 ± 0.09	91.31 ± 0.25	91.69 ± 0.12
	DDM	93.28 ± 0.15	93.28 ± 0.15	93.33 ± 0.14	90.96 ± 0.54	92.40 ± 0.09	**94.23 ± 0.15**	91.28 ± 0.37
	ECDD	90.33 ± 0.12	90.33 ± 0.12	90.43 ± 0.13	90.31 ± 0.16	91.40 ± 0.10	91.89 ± 0.13	89.78 ± 0.14
	FHDDM	92.61 ± 0.13	92.61 ± 0.13	92.64 ± 0.12	89.77 ± 0.36	92.39 ± 0.09	92.61 ± 0.30	92.13 ± 0.08
	FTDD	93.15 ± 0.12	93.15 ± 0.12	93.21 ± 0.12	90.33 ± 0.75	92.42 ± 0.09	93.16 ± 0.25	92.05 ± 0.09
	$HDDM_A$	92.85 ± 0.14	92.85 ± 0.14	92.91 ± 0.13	89.82 ± 0.56	92.42 ± 0.09	93.26 ± 0.27	92.11 ± 0.07
	RDDM	92.66 ± 0.12	92.66 ± 0.12	92.73 ± 0.12	90.62 ± 0.61	92.01 ± 0.12	94.16 ± 0.16	91.60 ± 0.14
	SeqDr2	92.58 ± 0.13	92.58 ± 0.13	92.64 ± 0.12	90.00 ± 0.27	92.50 ± 0.08	86.68 ± 0.18	90.70 ± 0.11
	STEPD	92.12 ± 0.13	92.12 ± 0.13	92.16 ± 0.13	90.46 ± 0.36	91.79 ± 0.12	92.90 ± 0.27	91.14 ± 0.10
	WSTD	92.90 ± 0.12	92.90 ± 0.12	92.95 ± 0.11	89.39 ± 0.74	92.40 ± 0.09	92.66 ± 0.29	92.03 ± 0.11
RBF	ADWIN	19.88 ± 0.67	19.88 ± 0.67	31.69 ± 0.89	32.42 ± 0.33	33.26 ± 0.22	37.84 ± 0.21	32.35 ± 0.33
	DDM	20.06 ± 0.64	20.06 ± 0.64	31.74 ± 0.77	32.73 ± 0.31	33.24 ± 0.23	38.19 ± 0.20	32.54 ± 0.37
	ECDD	20.13 ± 0.71	20.13 ± 0.71	31.66 ± 0.60	33.14 ± 0.32	33.91 ± 0.27	**39.03 ± 0.16**	33.17 ± 0.32
	FHDDM	19.64 ± 0.55	19.64 ± 0.55	31.23 ± 0.73	31.70 ± 0.22	32.76 ± 0.20	32.87 ± 0.26	31.35 ± 0.25
	FTDD	20.63 ± 0.73	23.39 ± 0.64	32.63 ± 0.31	32.98 ± 0.39	33.67 ± 0.24	38.47 ± 0.20	32.70 ± 0.42
	$HDDM_A$	19.99 ± 0.64	22.92 ± 0.52	32.58 ± 0.39	32.78 ± 0.33	33.25 ± 0.23	37.10 ± 0.29	32.57 ± 0.30
	RDDM	19.96 ± 0.76	23.46 ± 0.42	32.68 ± 0.31	32.62 ± 0.22	32.99 ± 0.20	35.48 ± 0.40	32.40 ± 0.28
	SeqDr2	20.47 ± 0.72	20.47 ± 0.72	31.80 ± 0.74	32.99 ± 0.34	33.57 ± 0.26	38.43 ± 0.18	32.62 ± 0.36
	STEPD	19.34 ± 0.59	19.34 ± 0.59	31.30 ± 0.76	31.60 ± 0.30	32.62 ± 0.19	36.31 ± 0.30	31.10 ± 0.23
	WSTD	20.29 ± 0.72	23.06 ± 0.48	31.80 ± 0.57	32.24 ± 0.49	33.40 ± 0.24	38.18 ± 0.22	31.81 ± 0.38
Sine	ADWIN	92.27 ± 0.13	92.27 ± 0.13	92.33 ± 0.13	90.23 ± 0.14	91.51 ± 0.09	92.49 ± 0.27	90.24 ± 0.10
	DDM	94.19 ± 0.17	94.19 ± 0.17	94.18 ± 0.17	91.35 ± 0.18	91.95 ± 0.12	**95.12 ± 0.11**	91.06 ± 0.15
	ECDD	90.71 ± 0.26	90.71 ± 0.26	90.72 ± 0.26	87.64 ± 0.15	89.50 ± 0.09	91.46 ± 0.16	87.12 ± 0.13
	FHDDM	93.72 ± 0.16	93.72 ± 0.16	93.70 ± 0.16	90.48 ± 0.33	91.93 ± 0.11	94.10 ± 0.16	91.52 ± 0.13
	FTDD	94.32 ± 0.19	94.33 ± 0.19	94.33 ± 0.18	91.07 ± 0.31	91.95 ± 0.11	94.49 ± 0.17	91.55 ± 0.15
	$HDDM_A$	93.97 ± 0.18	93.97 ± 0.18	93.99 ± 0.18	90.77 ± 0.37	91.96 ± 0.11	94.78 ± 0.19	91.52 ± 0.14
	RDDM	93.86 ± 0.20	93.87 ± 0.20	93.88 ± 0.20	91.23 ± 0.25	91.90 ± 0.11	94.86 ± 0.11	91.19 ± 0.14
	SeqDr2	92.95 ± 0.18	92.95 ± 0.18	92.94 ± 0.17	90.02 ± 0.21	91.92 ± 0.11	86.94 ± 0.32	90.37 ± 0.11
	STEPD	92.58 ± 0.21	92.58 ± 0.21	92.58 ± 0.20	90.62 ± 0.23	91.65 ± 0.11	94.26 ± 0.19	90.37 ± 0.21
	WSTD	93.91 ± 0.17	93.92 ± 0.17	93.92 ± 0.17	90.78 ± 0.27	91.96 ± 0.11	94.00 ± 0.17	91.52 ± 0.13
Wavef.	ADWIN	82.27 ± 0.11	82.27 ± 0.11	82.01 ± 0.11	80.69 ± 0.13	81.75 ± 0.17	83.89 ± 0.11	79.51 ± 0.18
	DDM	82.64 ± 0.14	82.64 ± 0.14	82.39 ± 0.16	80.56 ± 0.20	82.05 ± 0.11	82.50 ± 0.30	79.28 ± 0.20
	ECDD	81.87 ± 0.13	81.87 ± 0.13	81.71 ± 0.14	79.93 ± 0.14	80.66 ± 0.14	80.58 ± 0.13	79.00 ± 0.17
	FHDDM	81.82 ± 0.13	81.82 ± 0.13	81.48 ± 0.14	80.46 ± 0.18	81.65 ± 0.15	83.84 ± 0.14	79.81 ± 0.14
	FTDD	82.66 ± 0.14	82.69 ± 0.13	82.46 ± 0.13	80.68 ± 0.20	81.83 ± 0.18	83.81 ± 0.14	79.37 ± 0.21
	$HDDM_A$	82.40 ± 0.12	82.42 ± 0.12	82.08 ± 0.13	80.67 ± 0.16	81.95 ± 0.14	83.95 ± 0.12	79.58 ± 0.16
	RDDM	82.44 ± 0.12	82.47 ± 0.12	81.94 ± 0.10	80.76 ± 0.16	81.58 ± 0.13	81.99 ± 0.20	79.94 ± 0.16
	SeqDr2	82.62 ± 0.13	82.62 ± 0.13	82.45 ± 0.12	80.85 ± 0.24	81.86 ± 0.17	83.79 ± 0.14	79.48 ± 0.20
	STEPD	81.94 ± 0.12	81.94 ± 0.12	81.70 ± 0.13	80.30 ± 0.13	80.76 ± 0.12	82.45 ± 0.12	80.00 ± 0.15
	WSTD	82.22 ± 0.14	82.25 ± 0.14	81.79 ± 0.16	80.56 ± 0.19	81.66 ± 0.13	**83.96 ± 0.12**	79.63 ± 0.18

Table 18
Mean accuracies of Ensembles in percentage (%) in 100K instances abrupt datasets, with 95% confidence intervals, using HT.

Dataset	Ensemble	ADOB	$BOLE_4$	$BOLE_5$	DDD	FASE	LevBag	None
$Agraw_1$	ADWIN	67.84 ± 0.13	63.98 ± 0.18	63.98 ± 0.18	64.01 ± 0.17	66.97 ± 0.17	63.72 ± 0.11	66.44 ± 0.12
	DDM	74.47 ± 0.21	69.67 ± 0.33	69.67 ± 0.33	69.45 ± 0.31	72.68 ± 1.35	70.97 ± 0.26	71.01 ± 2.08
	ECDD	70.77 ± 0.16	68.79 ± 0.18	68.79 ± 0.18	68.79 ± 0.19	67.80 ± 0.49	65.37 ± 0.25	66.25 ± 0.71
	FHDDM	73.93 ± 0.18	69.86 ± 0.21	69.86 ± 0.21	70.02 ± 0.21	73.20 ± 0.20	69.52 ± 0.18	72.91 ± 0.25
	FTDD	65.45 ± 0.53	65.45 ± 0.53	65.20 ± 0.44	71.56 ± 0.82	71.09 ± 0.54	67.08 ± 0.66	70.38 ± 1.01
	$HDDM_A$	69.96 ± 0.20	69.96 ± 0.20	70.07 ± 0.20	73.97 ± 0.29	74.38 ± 0.22	71.15 ± 0.17	74.74 ± 0.34
	RDDM	70.17 ± 0.26	70.17 ± 0.26	70.26 ± 0.25	73.94 ± 0.26	74.57 ± 0.21	69.04 ± 0.16	**74.85 ± 0.28**
	SeqDr2	69.07 ± 0.18	64.53 ± 0.28	64.53 ± 0.28	64.38 ± 0.24	67.94 ± 0.31	65.15 ± 0.15	68.73 ± 0.32
	STEPD	68.53 ± 0.24	64.25 ± 0.14	64.25 ± 0.14	64.34 ± 0.13	67.85 ± 0.52	63.18 ± 0.32	66.89 ± 0.27
	WSTD	65.02 ± 0.42	65.02 ± 0.42	64.90 ± 0.35	73.12 ± 0.41	71.73 ± 0.61	70.98 ± 0.30	71.62 ± 0.74
$Agraw_2$	ADWIN	86.86 ± 0.08	85.95 ± 0.09	85.95 ± 0.09	85.96 ± 0.09	86.45 ± 0.07	85.85 ± 0.15	86.25 ± 0.09
	DDM	84.66 ± 0.29	84.66 ± 0.29	84.63 ± 0.26	84.70 ± 0.81	87.58 ± 0.07	85.82 ± 0.11	84.81 ± 0.83
	ECDD	85.13 ± 0.08	85.13 ± 0.08	85.16 ± 0.07	85.13 ± 0.06	85.76 ± 0.05	84.55 ± 0.04	84.65 ± 0.07
	FHDDM	86.40 ± 0.07	86.40 ± 0.07	86.42 ± 0.07	87.45 ± 0.07	87.62 ± 0.05	84.55 ± 0.14	**87.69 ± 0.05**
	FTDD	85.32 ± 0.31	85.32 ± 0.31	85.31 ± 0.28	85.25 ± 0.55	87.55 ± 0.05	83.32 ± 0.41	85.86 ± 0.51
	$HDDM_A$	86.19 ± 0.11	86.20 ± 0.11	86.21 ± 0.10	87.20 ± 0.11	87.61 ± 0.05	85.69 ± 0.15	87.44 ± 0.14
	RDDM	85.82 ± 0.20	85.82 ± 0.20	85.84 ± 0.19	87.14 ± 0.26	87.61 ± 0.06	85.91 ± 0.10	87.17 ± 0.19
	SeqDr2	85.25 ± 0.27	85.25 ± 0.27	85.24 ± 0.25	85.20 ± 0.60	87.62 ± 0.06	79.86 ± 0.45	85.83 ± 0.56
	STEPD	86.07 ± 0.06	86.07 ± 0.06	86.08 ± 0.06	87.10 ± 0.16	87.28 ± 0.07	85.92 ± 0.09	86.70 ± 0.10
	WSTD	86.44 ± 0.08	86.44 ± 0.08	86.45 ± 0.07	86.72 ± 0.45	87.64 ± 0.05	85.76 ± 0.15	87.40 ± 0.15
LED	ADWIN	72.29 ± 0.14	72.29 ± 0.14	72.34 ± 0.14	72.72 ± 0.16	72.84 ± 0.11	73.14 ± 0.10	68.87 ± 0.73
	DDM	73.06 ± 0.14	73.06 ± 0.14	73.08 ± 0.14	72.58 ± 0.21	73.23 ± 0.11	60.47 ± 0.25	72.65 ± 0.30
	ECDD	72.99 ± 0.12	72.99 ± 0.12	73.02 ± 0.11	71.35 ± 0.19	73.17 ± 0.13	65.03 ± 0.23	68.97 ± 0.23
	FHDDM	73.24 ± 0.12	73.24 ± 0.12	73.26 ± 0.12	73.01 ± 0.14	73.39 ± 0.11	73.35 ± 0.12	72.51 ± 0.23
	FTDD	73.02 ± 0.14	73.05 ± 0.14	73.04 ± 0.13	72.10 ± 0.19	73.18 ± 0.12	72.94 ± 0.15	72.93 ± 0.18
	$HDDM_A$	73.38 ± 0.12	**73.41 ± 0.12**	**73.41 ± 0.12**	72.38 ± 0.16	73.34 ± 0.11	73.39 ± 0.11	73.37 ± 0.11
	RDDM	73.24 ± 0.12	73.27 ± 0.13	73.28 ± 0.13	72.74 ± 0.19	73.31 ± 0.12	68.27 ± 0.22	73.39 ± 0.12
	SeqDr2	72.72 ± 0.14	72.72 ± 0.14	72.75 ± 0.14	72.27 ± 0.22	72.86 ± 0.11	70.41 ± 0.32	69.56 ± 1.06
	STEPD	71.62 ± 0.14	71.62 ± 0.14	71.69 ± 0.14	72.45 ± 0.18	72.90 ± 0.15	73.00 ± 0.15	69.04 ± 0.73
	WSTD	73.22 ± 0.12	73.25 ± 0.12	73.27 ± 0.12	72.76 ± 0.20	73.19 ± 0.11	73.23 ± 0.12	72.81 ± 0.20
Mixed	ADWIN	93.06 ± 0.08	93.06 ± 0.08	93.16 ± 0.08	92.12 ± 0.09	92.63 ± 0.08	92.73 ± 0.21	92.00 ± 0.11
	DDM	95.26 ± 0.12	95.26 ± 0.12	95.29 ± 0.12	92.71 ± 0.21	93.31 ± 0.07	**95.71 ± 0.09**	92.79 ± 0.12
	ECDD	90.66 ± 0.11	90.66 ± 0.11	90.75 ± 0.11	90.44 ± 0.09	91.50 ± 0.06	92.44 ± 0.08	89.75 ± 0.10
	FHDDM	94.28 ± 0.15	94.28 ± 0.15	94.34 ± 0.15	91.87 ± 0.16	93.31 ± 0.06	94.29 ± 0.13	93.15 ± 0.06
	FTDD	95.26 ± 0.11	95.26 ± 0.11	95.31 ± 0.11	92.11 ± 0.34	93.33 ± 0.06	94.74 ± 0.24	93.13 ± 0.06
	$HDDM_A$	95.00 ± 0.11	95.00 ± 0.11	95.05 ± 0.11	91.85 ± 0.28	93.34 ± 0.06	94.73 ± 0.25	93.12 ± 0.07
	RDDM	94.61 ± 0.16	94.61 ± 0.16	94.66 ± 0.16	92.44 ± 0.31	93.00 ± 0.09	95.68 ± 0.10	92.46 ± 0.21
	SeqDr2	94.23 ± 0.12	94.23 ± 0.12	94.29 ± 0.12	92.15 ± 0.12	93.36 ± 0.06	90.14 ± 0.17	92.42 ± 0.08
	STEPD	93.27 ± 0.14	93.27 ± 0.14	93.30 ± 0.14	91.56 ± 0.20	92.28 ± 0.09	94.78 ± 0.20	91.23 ± 0.09
	WSTD	94.79 ± 0.12	94.79 ± 0.12	94.83 ± 0.12	92.06 ± 0.43	93.31 ± 0.06	94.04 ± 0.19	93.09 ± 0.06
RBF	ADWIN	19.49 ± 0.62	19.49 ± 0.62	32.00 ± 0.79	32.91 ± 0.27	33.45 ± 0.19	39.02 ± 0.15	32.67 ± 0.25
	DDM	19.58 ± 0.57	19.58 ± 0.57	32.64 ± 0.71	34.10 ± 0.35	33.69 ± 0.17	39.18 ± 0.16	33.64 ± 0.26
	ECDD	19.58 ± 0.57	19.58 ± 0.57	34.05 ± 0.57	35.29 ± 0.21	34.87 ± 0.22	**39.87 ± 0.14**	34.86 ± 0.23
	FHDDM	19.31 ± 0.55	19.31 ± 0.55	31.46 ± 0.64	31.93 ± 0.13	32.71 ± 0.13	33.11 ± 0.19	31.30 ± 0.13
	FTDD	20.14 ± 0.70	23.19 ± 0.37	34.31 ± 0.28	34.02 ± 0.29	34.23 ± 0.25	39.49 ± 0.17	33.32 ± 0.32
	$HDDM_A$	19.67 ± 0.62	22.62 ± 0.50	32.66 ± 0.23	33.46 ± 0.34	33.50 ± 0.16	38.22 ± 0.20	32.87 ± 0.27
	RDDM	19.71 ± 0.78	23.00 ± 0.40	32.57 ± 0.28	33.01 ± 0.20	33.26 ± 0.15	36.96 ± 0.30	32.80 ± 0.20
	SeqDr2	19.82 ± 0.66	19.82 ± 0.66	33.00 ± 0.59	34.22 ± 0.29	34.41 ± 0.16	39.31 ± 0.16	33.61 ± 0.20
	STEPD	19.26 ± 0.54	19.26 ± 0.54	31.35 ± 0.71	31.64 ± 0.18	32.55 ± 0.13	37.01 ± 0.22	31.17 ± 0.16
	WSTD	19.84 ± 0.63	22.83 ± 0.37	32.23 ± 0.30	32.80 ± 0.34	33.54 ± 0.18	39.01 ± 0.20	32.45 ± 0.30
Sine	ADWIN	93.32 ± 0.12	93.32 ± 0.12	93.38 ± 0.12	91.05 ± 0.10	92.16 ± 0.07	94.51 ± 0.20	90.54 ± 0.08
	DDM	96.23 ± 0.08	96.23 ± 0.08	96.23 ± 0.08	92.57 ± 0.25	93.11 ± 0.08	**96.80 ± 0.06**	92.31 ± 0.09
	ECDD	91.94 ± 0.25	91.94 ± 0.25	91.94 ± 0.24	87.78 ± 0.10	89.67 ± 0.07	92.06 ± 0.11	87.15 ± 0.10
	FHDDM	96.06 ± 0.10	96.06 ± 0.10	96.05 ± 0.11	91.79 ± 0.32	93.10 ± 0.08	96.07 ± 0.11	92.59 ± 0.10
	FTDD	96.27 ± 0.07	96.28 ± 0.07	96.28 ± 0.06	92.16 ± 0.28	93.11 ± 0.07	95.99 ± 0.16	92.61 ± 0.10
	$HDDM_A$	96.29 ± 0.10	96.29 ± 0.10	96.30 ± 0.10	92.11 ± 0.35	93.11 ± 0.07	96.39 ± 0.18	92.57 ± 0.11
	RDDM	96.00 ± 0.13	96.00 ± 0.13	96.01 ± 0.13	92.49 ± 0.25	93.10 ± 0.07	96.56 ± 0.08	92.41 ± 0.11
	SeqDr2	94.87 ± 0.16	94.87 ± 0.16	94.89 ± 0.16	92.06 ± 0.23	93.11 ± 0.07	90.53 ± 0.29	92.01 ± 0.09
	STEPD	94.50 ± 0.21	94.50 ± 0.21	94.48 ± 0.21	91.51 ± 0.22	92.64 ± 0.11	96.16 ± 0.11	90.96 ± 0.16
	WSTD	96.19 ± 0.13	96.19 ± 0.13	96.19 ± 0.13	92.14 ± 0.26	93.11 ± 0.08	95.57 ± 0.14	92.59 ± 0.10
Wavef.	ADWIN	82.63 ± 0.08	82.63 ± 0.08	82.38 ± 0.09	80.93 ± 0.13	81.94 ± 0.20	84.54 ± 0.07	79.66 ± 0.15
	DDM	83.46 ± 0.10	83.46 ± 0.10	83.33 ± 0.10	81.35 ± 0.17	82.57 ± 0.12	83.27 ± 0.27	79.35 ± 0.24
	ECDD	82.10 ± 0.11	82.10 ± 0.11	81.95 ± 0.11	80.03 ± 0.12	80.79 ± 0.11	80.67 ± 0.10	79.12 ± 0.14
	FHDDM	81.98 ± 0.11	81.98 ± 0.11	81.66 ± 0.11	81.17 ± 0.16	82.15 ± 0.14	84.52 ± 0.10	79.85 ± 0.13
	FTDD	83.42 ± 0.10	83.44 ± 0.10	83.33 ± 0.11	81.27 ± 0.17	82.22 ± 0.16	84.50 ± 0.09	79.59 ± 0.19
	$HDDM_A$	82.77 ± 0.10	82.78 ± 0.10	82.44 ± 0.10	81.08 ± 0.15	82.29 ± 0.15	84.60 ± 0.08	79.52 ± 0.17
	RDDM	82.85 ± 0.11	82.87 ± 0.11	82.41 ± 0.12	81.01 ± 0.12	82.05 ± 0.11	82.37 ± 0.17	79.97 ± 0.14
	SeqDr2	83.39 ± 0.09	83.39 ± 0.09	83.26 ± 0.11	81.20 ± 0.15	81.98 ± 0.14	84.53 ± 0.09	79.59 ± 0.16
	STEPD	82.08 ± 0.10	82.08 ± 0.10	81.84 ± 0.11	80.48 ± 0.11	80.94 ± 0.10	82.82 ± 0.10	80.16 ± 0.11
	WSTD	82.62 ± 0.08	82.63 ± 0.08	82.22 ± 0.10	81.22 ± 0.13	82.29 ± 0.12	**84.64 ± 0.08**	79.83 ± 0.15

Table 19
Mean accuracies of Ensembles in percentage (%) in 10K instances gradual datasets, with 95% confidence intervals, using HT.

Dataset	Ensemble	ADOB	$BOLE_4$	$BOLE_5$	DDD	FASE	LevBag	None
$Agraw_1$	ADWIN	60.35 ± 0.48	60.35 ± 0.48	60.90 ± 0.47	61.63 ± 0.19	62.13 ± 0.22	60.67 ± 0.59	61.60 ± 0.26
	DDM	59.47 ± 0.54	59.47 ± 0.54	60.04 ± 0.48	61.26 ± 0.51	62.63 ± 0.25	59.80 ± 0.52	61.57 ± 0.47
	ECDD	60.00 ± 0.45	60.00 ± 0.45	60.07 ± 0.44	61.95 ± 0.24	62.38 ± 0.24	58.63 ± 0.50	61.98 ± 0.20
	FHDDM	61.32 ± 0.37	61.32 ± 0.37	61.75 ± 0.36	62.56 ± 0.32	62.64 ± 0.24	60.55 ± 0.59	62.56 ± 0.24
	FTDD	58.42 ± 0.39	58.51 ± 0.40	58.82 ± 0.38	60.73 ± 0.48	62.28 ± 0.20	57.92 ± 0.37	61.33 ± 0.29
	$HDDM_A$	60.70 ± 0.46	60.68 ± 0.47	61.25 ± 0.41	62.01 ± 0.33	62.33 ± 0.28	60.61 ± 0.45	62.27 ± 0.36
	RDDM	60.72 ± 0.42	60.72 ± 0.43	61.21 ± 0.39	62.39 ± 0.32	62.59 ± 0.22	60.29 ± 0.63	**62.92 ± 0.27**
	SeqDr2	59.96 ± 0.36	59.96 ± 0.36	60.49 ± 0.34	61.25 ± 0.19	62.18 ± 0.27	58.59 ± 0.43	61.31 ± 0.26
	STEPD	60.63 ± 0.39	60.63 ± 0.39	61.09 ± 0.37	62.03 ± 0.32	62.31 ± 0.27	60.80 ± 0.57	62.15 ± 0.25
	WSTD	60.54 ± 0.39	60.54 ± 0.39	60.99 ± 0.37	61.57 ± 0.33	62.12 ± 0.23	60.53 ± 0.49	61.77 ± 0.38
$Agraw_2$	ADWIN	77.47 ± 0.33	77.47 ± 0.33	77.84 ± 0.28	76.67 ± 0.37	78.79 ± 0.33	77.03 ± 0.35	76.69 ± 0.40
	DDM	75.89 ± 0.33	75.89 ± 0.33	76.16 ± 0.33	75.35 ± 1.19	79.50 ± 0.15	78.02 ± 0.22	73.62 ± 1.59
	ECDD	78.43 ± 0.23	78.43 ± 0.23	78.57 ± 0.23	79.57 ± 0.24	79.54 ± 0.20	78.14 ± 0.20	79.16 ± 0.25
	FHDDM	79.17 ± 0.20	79.17 ± 0.20	79.38 ± 0.20	77.61 ± 0.47	79.43 ± 0.19	73.64 ± 0.64	78.81 ± 0.37
	FTDD	77.36 ± 0.38	77.39 ± 0.37	77.70 ± 0.32	74.28 ± 0.66	78.86 ± 0.28	68.18 ± 0.87	74.56 ± 0.76
	$HDDM_A$	77.68 ± 0.33	77.71 ± 0.33	78.04 ± 0.30	77.55 ± 0.54	79.16 ± 0.27	76.31 ± 0.41	78.27 ± 0.54
	RDDM	78.19 ± 0.33	78.22 ± 0.34	78.43 ± 0.32	78.50 ± 0.77	**79.66 ± 0.18**	78.14 ± 0.24	78.65 ± 0.98
	SeqDr2	77.69 ± 0.33	77.69 ± 0.33	78.01 ± 0.28	75.96 ± 0.30	78.75 ± 0.27	70.15 ± 0.67	76.33 ± 0.37
	STEPD	78.82 ± 0.24	78.82 ± 0.24	79.05 ± 0.25	78.20 ± 0.51	79.30 ± 0.25	77.52 ± 0.29	78.59 ± 0.50
	WSTD	78.68 ± 0.26	78.71 ± 0.27	78.99 ± 0.28	76.19 ± 0.76	79.15 ± 0.25	74.82 ± 0.60	77.35 ± 0.66
LED	ADWIN	65.59 ± 0.45	65.59 ± 0.45	65.99 ± 0.34	64.50 ± 0.49	66.45 ± 0.29	68.22 ± 0.29	62.21 ± 0.43
	DDM	66.74 ± 0.38	66.74 ± 0.38	67.15 ± 0.32	66.47 ± 0.60	67.02 ± 0.25	57.37 ± 0.53	67.76 ± 0.42
	ECDD	52.64 ± 1.73	52.64 ± 1.73	61.15 ± 0.71	66.57 ± 0.45	67.24 ± 0.30	61.43 ± 0.56	65.10 ± 0.40
	FHDDM	67.14 ± 0.57	67.14 ± 0.57	67.57 ± 0.28	66.99 ± 0.35	67.12 ± 0.26	67.92 ± 0.29	66.86 ± 0.35
	FTDD	57.46 ± 2.54	65.75 ± 0.45	64.94 ± 0.53	62.59 ± 0.97	66.01 ± 0.33	65.87 ± 0.48	62.88 ± 0.89
	$HDDM_A$	67.18 ± 0.32	67.49 ± 0.28	67.40 ± 0.28	67.02 ± 0.29	66.96 ± 0.27	**68.37 ± 0.29**	67.58 ± 0.31
	RDDM	67.20 ± 0.32	67.53 ± 0.28	67.54 ± 0.28	67.22 ± 0.29	67.08 ± 0.28	64.33 ± 0.51	67.81 ± 0.29
	SeqDr2	65.02 ± 0.43	65.02 ± 0.43	65.70 ± 0.34	64.35 ± 0.42	65.75 ± 0.27	65.19 ± 0.41	59.70 ± 0.70
	STEPD	64.59 ± 0.35	64.59 ± 0.35	65.11 ± 0.33	64.97 ± 0.49	64.78 ± 0.39	68.06 ± 0.35	59.39 ± 1.39
	WSTD	64.47 ± 1.06	66.30 ± 0.33	66.34 ± 0.36	65.19 ± 0.53	65.30 ± 0.37	67.44 ± 0.40	63.99 ± 0.81
Mixed	ADWIN	83.63 ± 0.24	83.63 ± 0.24	83.71 ± 0.22	83.39 ± 0.21	83.60 ± 0.24	81.92 ± 0.16	82.97 ± 0.27
	DDM	84.12 ± 0.22	84.12 ± 0.22	**84.23 ± 0.22**	81.68 ± 0.46	83.65 ± 0.25	83.90 ± 0.25	83.49 ± 0.28
	ECDD	82.26 ± 0.23	82.26 ± 0.23	82.31 ± 0.22	82.40 ± 0.42	83.44 ± 0.25	83.20 ± 0.23	82.84 ± 0.31
	FHDDM	83.30 ± 0.23	83.30 ± 0.23	83.39 ± 0.23	81.49 ± 0.54	83.83 ± 0.27	79.84 ± 0.45	83.98 ± 0.28
	FTDD	83.86 ± 0.21	83.86 ± 0.21	83.97 ± 0.22	80.64 ± 0.46	83.58 ± 0.24	80.20 ± 0.36	83.50 ± 0.23
	$HDDM_A$	83.52 ± 0.19	83.53 ± 0.19	83.61 ± 0.18	81.26 ± 0.57	83.57 ± 0.26	82.78 ± 0.25	83.39 ± 0.27
	RDDM	83.61 ± 0.22	83.61 ± 0.21	83.71 ± 0.21	83.15 ± 0.40	83.52 ± 0.28	83.79 ± 0.26	83.70 ± 0.31
	SeqDr2	83.41 ± 0.21	83.41 ± 0.21	83.46 ± 0.22	82.06 ± 0.41	83.53 ± 0.21	75.49 ± 0.33	83.28 ± 0.27
	STEPD	82.36 ± 0.21	82.36 ± 0.21	82.43 ± 0.22	82.66 ± 0.44	82.97 ± 0.25	82.61 ± 0.36	83.04 ± 0.31
	WSTD	83.26 ± 0.21	83.27 ± 0.20	83.38 ± 0.20	81.79 ± 0.54	83.41 ± 0.23	80.69 ± 0.41	83.26 ± 0.27
RBF	ADWIN	20.97 ± 0.87	20.97 ± 0.87	31.09 ± 1.09	31.71 ± 0.39	32.66 ± 0.37	33.48 ± 0.48	31.87 ± 0.38
	DDM	21.04 ± 0.88	21.04 ± 0.88	30.56 ± 1.07	31.65 ± 0.58	32.48 ± 0.38	32.98 ± 0.49	32.00 ± 0.46
	ECDD	21.03 ± 0.87	21.03 ± 0.87	29.68 ± 0.98	30.79 ± 0.64	32.44 ± 0.37	33.74 ± 0.44	30.87 ± 0.65
	FHDDM	20.47 ± 0.74	20.47 ± 0.74	30.65 ± 0.97	31.48 ± 0.52	32.39 ± 0.35	32.29 ± 0.37	31.21 ± 0.43
	FTDD	21.38 ± 0.89	24.87 ± 0.82	31.42 ± 0.65	31.70 ± 0.50	32.58 ± 0.40	**33.80 ± 0.47**	32.10 ± 0.52
	$HDDM_A$	21.12 ± 0.88	24.82 ± 0.89	31.93 ± 0.59	31.94 ± 0.43	32.55 ± 0.39	33.03 ± 0.44	32.02 ± 0.39
	RDDM	21.01 ± 0.86	24.72 ± 0.81	32.12 ± 0.49	31.86 ± 0.40	32.18 ± 0.35	31.80 ± 0.34	31.93 ± 0.38
	SeqDr2	21.16 ± 0.97	21.16 ± 0.97	30.60 ± 1.07	31.68 ± 0.51	32.52 ± 0.38	33.72 ± 0.43	32.20 ± 0.49
	STEPD	20.16 ± 0.86	20.16 ± 0.86	30.75 ± 0.82	31.33 ± 0.60	32.23 ± 0.42	33.33 ± 0.41	30.99 ± 0.51
	WSTD	21.81 ± 1.08	25.46 ± 0.83	30.95 ± 0.64	30.97 ± 0.61	32.49 ± 0.40	33.70 ± 0.49	31.12 ± 0.64
Sine	ADWIN	82.42 ± 0.22	82.42 ± 0.22	82.48 ± 0.22	81.92 ± 0.21	82.65 ± 0.19	82.91 ± 0.21	81.71 ± 0.21
	DDM	83.22 ± 0.24	83.22 ± 0.24	83.27 ± 0.21	81.02 ± 0.43	83.09 ± 0.18	**84.05 ± 0.20**	82.43 ± 0.28
	ECDD	81.39 ± 0.28	81.39 ± 0.28	81.53 ± 0.27	81.41 ± 0.40	82.52 ± 0.17	83.07 ± 0.19	81.50 ± 0.22
	FHDDM	82.47 ± 0.24	82.47 ± 0.24	82.50 ± 0.22	80.97 ± 0.42	83.26 ± 0.19	82.36 ± 0.34	82.95 ± 0.21
	FTDD	83.36 ± 0.24	83.38 ± 0.24	83.48 ± 0.24	80.59 ± 0.34	83.07 ± 0.18	82.30 ± 0.36	82.28 ± 0.24
	$HDDM_A$	82.70 ± 0.27	82.72 ± 0.27	82.79 ± 0.25	80.70 ± 0.41	83.05 ± 0.17	82.76 ± 0.22	82.41 ± 0.27
	RDDM	82.81 ± 0.23	82.84 ± 0.23	82.91 ± 0.23	82.25 ± 0.30	82.96 ± 0.18	83.95 ± 0.18	82.66 ± 0.19
	SeqDr2	82.42 ± 0.25	82.42 ± 0.25	82.52 ± 0.24	81.18 ± 0.31	82.39 ± 0.19	77.41 ± 0.35	82.05 ± 0.24
	STEPD	81.67 ± 0.22	81.67 ± 0.22	81.76 ± 0.20	81.43 ± 0.36	82.31 ± 0.21	83.56 ± 0.23	81.48 ± 0.24
	WSTD	82.53 ± 0.21	82.55 ± 0.22	82.52 ± 0.21	81.19 ± 0.34	82.95 ± 0.19	82.72 ± 0.27	82.14 ± 0.22
Wavef.	ADWIN	79.68 ± 0.30	79.68 ± 0.30	78.98 ± 0.33	77.69 ± 0.38	78.15 ± 0.38	80.62 ± 0.35	77.85 ± 0.39
	DDM	79.40 ± 0.30	79.40 ± 0.30	78.70 ± 0.32	77.96 ± 0.46	78.26 ± 0.37	80.11 ± 0.44	77.97 ± 0.43
	ECDD	80.33 ± 0.28	80.33 ± 0.28	80.26 ± 0.28	78.84 ± 0.37	79.34 ± 0.34	79.53 ± 0.34	78.05 ± 0.37
	FHDDM	80.20 ± 0.30	80.20 ± 0.30	79.88 ± 0.29	77.61 ± 0.38	78.30 ± 0.38	80.57 ± 0.32	77.77 ± 0.39
	FTDD	79.45 ± 0.35	79.58 ± 0.35	78.68 ± 0.36	76.90 ± 0.39	78.07 ± 0.38	80.52 ± 0.35	76.68 ± 0.43
	$HDDM_A$	79.79 ± 0.32	79.92 ± 0.32	79.25 ± 0.35	77.84 ± 0.43	78.20 ± 0.40	80.80 ± 0.32	77.82 ± 0.47
	RDDM	80.07 ± 0.32	80.20 ± 0.32	79.64 ± 0.35	78.31 ± 0.43	78.57 ± 0.35	80.37 ± 0.34	78.42 ± 0.37
	SeqDr2	79.74 ± 0.31	79.74 ± 0.31	79.16 ± 0.36	77.83 ± 0.41	78.15 ± 0.41	80.49 ± 0.37	77.73 ± 0.40
	STEPD	80.29 ± 0.34	80.29 ± 0.34	80.07 ± 0.36	78.28 ± 0.38	78.69 ± 0.41	**80.99 ± 0.27**	78.29 ± 0.39
	WSTD	79.82 ± 0.33	79.95 ± 0.34	79.39 ± 0.39	77.52 ± 0.41	78.18 ± 0.38	80.78 ± 0.38	77.57 ± 0.51

Table 20
Mean accuracies of ensembles in percentage (%) in 20K instances gradual datasets, with 95% confidence intervals, using HT.

Dataset	Ensemble	ADOB	$BOLE_4$	$BOLE_5$	DDD	FASE	LevBag	None
$Agraw_1$	ADWIN	62.40 ± 0.27	62.40 ± 0.27	62.66 ± 0.26	63.43 ± 0.28	64.66 ± 0.18	63.88 ± 0.58	63.65 ± 0.24
	DDM	62.81 ± 0.60	62.81 ± 0.60	62.94 ± 0.54	63.79 ± 0.91	66.98 ± 0.35	62.58 ± 0.56	64.10 ± 1.17
	ECDD	62.38 ± 0.47	62.38 ± 0.47	62.33 ± 0.47	63.73 ± 0.56	65.04 ± 0.49	60.69 ± 0.68	62.94 ± 0.32
	FHDDM	64.42 ± 0.45	64.42 ± 0.45	64.61 ± 0.40	66.18 ± 0.43	67.01 ± 0.38	64.67 ± 0.67	65.73 ± 0.50
	FTDD	59.95 ± 0.50	59.94 ± 0.50	60.13 ± 0.46	60.42 ± 0.69	65.31 ± 0.36	60.60 ± 0.50	61.47 ± 0.81
	$HDDM_A$	65.04 ± 0.52	65.03 ± 0.52	65.09 ± 0.47	65.59 ± 0.48	66.67 ± 0.31	64.34 ± 0.58	66.30 ± 0.43
	RDDM	64.25 ± 0.42	64.25 ± 0.43	64.51 ± 0.39	66.55 ± 0.36	**67.13 ± 0.37**	63.92 ± 0.50	66.84 ± 0.40
	SeqDr2	60.81 ± 0.50	60.81 ± 0.50	60.96 ± 0.47	62.05 ± 0.52	64.76 ± 0.17	61.80 ± 0.69	63.07 ± 0.65
	STEPD	63.02 ± 0.19	63.02 ± 0.19	63.36 ± 0.19	63.88 ± 0.21	64.68 ± 0.24	64.53 ± 0.52	64.05 ± 0.21
	WSTD	62.77 ± 0.33	62.77 ± 0.33	62.94 ± 0.29	63.87 ± 0.63	64.84 ± 0.30	64.22 ± 0.60	64.61 ± 0.37
$Agraw_2$	ADWIN	81.67 ± 0.19	81.67 ± 0.19	81.90 ± 0.19	81.94 ± 0.17	82.42 ± 0.20	81.44 ± 0.19	82.47 ± 0.24
	DDM	79.65 ± 0.28	79.65 ± 0.28	79.64 ± 0.24	78.68 ± 1.24	82.81 ± 0.14	81.77 ± 0.14	79.00 ± 1.91
	ECDD	81.91 ± 0.15	81.91 ± 0.15	81.95 ± 0.16	82.66 ± 0.13	82.94 ± 0.11	81.52 ± 0.10	82.27 ± 0.14
	FHDDM	82.71 ± 0.18	82.71 ± 0.18	82.88 ± 0.17	81.97 ± 0.24	83.06 ± 0.11	79.96 ± 0.28	82.98 ± 0.17
	FTDD	81.27 ± 0.23	81.28 ± 0.24	81.33 ± 0.23	81.14 ± 0.23	82.29 ± 0.21	74.46 ± 0.91	82.21 ± 0.35
	$HDDM_A$	81.67 ± 0.17	81.69 ± 0.18	81.86 ± 0.17	81.95 ± 0.29	82.86 ± 0.20	80.99 ± 0.20	82.98 ± 0.28
	RDDM	82.11 ± 0.25	82.13 ± 0.25	82.23 ± 0.25	82.00 ± 0.59	**83.19 ± 0.12**	81.82 ± 0.13	82.64 ± 0.96
	SeqDr2	81.67 ± 0.19	81.67 ± 0.19	81.88 ± 0.18	81.76 ± 0.23	82.47 ± 0.19	77.21 ± 0.50	82.54 ± 0.24
	STEPD	82.35 ± 0.17	82.35 ± 0.17	82.48 ± 0.17	82.31 ± 0.24	83.07 ± 0.12	81.48 ± 0.14	83.08 ± 0.24
	WSTD	82.27 ± 0.16	82.29 ± 0.16	82.46 ± 0.16	81.89 ± 0.30	82.83 ± 0.16	80.52 ± 0.33	82.94 ± 0.22
LED	ADWIN	69.05 ± 0.19	69.05 ± 0.19	69.23 ± 0.18	67.85 ± 0.33	69.66 ± 0.22	**70.95 ± 0.18**	64.38 ± 0.60
	DDM	70.25 ± 0.16	70.25 ± 0.16	70.33 ± 0.17	69.58 ± 0.24	70.20 ± 0.17	59.13 ± 0.46	70.54 ± 0.19
	ECDD	70.20 ± 0.37	70.20 ± 0.37	70.47 ± 0.19	69.12 ± 0.28	70.32 ± 0.18	63.13 ± 0.42	67.16 ± 0.43
	FHDDM	70.36 ± 0.19	70.36 ± 0.19	70.48 ± 0.19	70.02 ± 0.23	70.37 ± 0.15	70.67 ± 0.24	69.79 ± 0.34
	FTDD	69.27 ± 0.32	69.60 ± 0.22	69.35 ± 0.21	67.02 ± 0.75	69.66 ± 0.14	69.44 ± 0.39	67.67 ± 0.84
	$HDDM_A$	70.39 ± 0.17	70.53 ± 0.16	70.55 ± 0.16	69.59 ± 0.20	70.21 ± 0.17	70.92 ± 0.20	70.42 ± 0.19
	RDDM	70.37 ± 0.17	70.52 ± 0.16	70.54 ± 0.16	70.19 ± 0.22	70.24 ± 0.17	66.36 ± 0.42	70.66 ± 0.19
	SeqDr2	68.67 ± 0.25	68.67 ± 0.25	68.94 ± 0.23	68.84 ± 0.27	68.94 ± 0.25	68.26 ± 0.30	61.89 ± 0.63
	STEPD	67.74 ± 0.32	67.74 ± 0.32	68.01 ± 0.32	68.82 ± 0.37	68.46 ± 0.31	70.58 ± 0.22	64.25 ± 1.39
	WSTD	69.45 ± 0.23	69.64 ± 0.23	69.70 ± 0.24	69.35 ± 0.27	69.07 ± 0.18	70.38 ± 0.31	68.57 ± 0.50
Mixed	ADWIN	87.38 ± 0.14	87.38 ± 0.14	87.49 ± 0.13	87.48 ± 0.15	87.76 ± 0.14	88.11 ± 0.16	86.99 ± 0.15
	DDM	87.77 ± 0.13	87.77 ± 0.13	87.90 ± 0.12	86.11 ± 0.38	87.73 ± 0.16	**89.10 ± 0.17**	87.29 ± 0.19
	ECDD	86.13 ± 0.17	86.13 ± 0.17	86.21 ± 0.17	87.01 ± 0.22	87.66 ± 0.15	87.96 ± 0.19	86.69 ± 0.18
	FHDDM	87.11 ± 0.15	87.11 ± 0.15	87.22 ± 0.14	85.51 ± 0.38	87.80 ± 0.17	87.01 ± 0.24	87.52 ± 0.17
	FTDD	87.97 ± 0.14	87.97 ± 0.13	88.09 ± 0.13	85.48 ± 0.43	87.72 ± 0.15	87.21 ± 0.28	87.11 ± 0.16
	$HDDM_A$	87.32 ± 0.15	87.32 ± 0.14	87.41 ± 0.14	85.97 ± 0.45	87.71 ± 0.16	88.11 ± 0.21	87.23 ± 0.18
	RDDM	87.40 ± 0.12	87.40 ± 0.12	87.51 ± 0.12	87.59 ± 0.28	87.76 ± 0.16	89.01 ± 0.18	87.53 ± 0.18
	SeqDr2	87.49 ± 0.15	87.49 ± 0.15	87.62 ± 0.14	86.62 ± 0.26	87.75 ± 0.14	83.39 ± 0.28	87.36 ± 0.18
	STEPD	86.70 ± 0.15	86.70 ± 0.15	86.76 ± 0.14	87.36 ± 0.17	87.60 ± 0.15	88.39 ± 0.21	87.30 ± 0.15
	WSTD	87.40 ± 0.14	87.41 ± 0.14	87.49 ± 0.13	85.92 ± 0.36	87.67 ± 0.14	87.24 ± 0.23	87.17 ± 0.17
RBF	ADWIN	20.12 ± 0.78	20.12 ± 0.78	31.28 ± 0.88	32.00 ± 0.34	33.05 ± 0.28	35.65 ± 0.33	32.34 ± 0.41
	DDM	20.30 ± 0.83	20.30 ± 0.83	30.42 ± 0.80	31.92 ± 0.50	32.93 ± 0.28	35.46 ± 0.30	31.80 ± 0.54
	ECDD	20.27 ± 0.81	20.27 ± 0.81	29.49 ± 0.76	31.12 ± 0.62	32.81 ± 0.33	36.19 ± 0.33	31.25 ± 0.62
	FHDDM	19.99 ± 0.66	19.99 ± 0.66	31.36 ± 0.79	31.57 ± 0.30	32.72 ± 0.27	32.86 ± 0.35	31.26 ± 0.35
	FTDD	20.91 ± 0.82	23.80 ± 0.60	31.75 ± 0.51	32.20 ± 0.45	33.02 ± 0.32	35.90 ± 0.29	32.69 ± 0.44
	$HDDM_A$	20.27 ± 0.81	23.80 ± 0.56	32.31 ± 0.39	32.32 ± 0.38	32.92 ± 0.32	35.08 ± 0.38	32.32 ± 0.37
	RDDM	20.28 ± 0.78	23.83 ± 0.58	32.34 ± 0.36	32.18 ± 0.31	32.72 ± 0.31	32.85 ± 0.43	32.19 ± 0.35
	SeqDr2	20.98 ± 0.94	20.98 ± 0.94	31.14 ± 0.91	32.29 ± 0.42	33.10 ± 0.29	35.89 ± 0.27	32.76 ± 0.37
	STEPD	19.75 ± 0.74	19.75 ± 0.74	31.06 ± 0.57	31.52 ± 0.40	32.66 ± 0.31	35.01 ± 0.35	31.21 ± 0.36
	WSTD	21.01 ± 0.88	24.31 ± 0.61	30.42 ± 0.52	31.13 ± 0.57	32.87 ± 0.30	**35.93 ± 0.30**	31.06 ± 0.53
Sine	ADWIN	86.49 ± 0.15	86.49 ± 0.15	86.54 ± 0.16	85.91 ± 0.14	86.88 ± 0.11	88.70 ± 0.16	86.10 ± 0.13
	DDM	87.41 ± 0.20	87.41 ± 0.20	87.51 ± 0.19	84.29 ± 0.34	87.30 ± 0.11	89.33 ± 0.17	86.68 ± 0.14
	ECDD	85.52 ± 0.14	85.52 ± 0.14	85.61 ± 0.13	85.26 ± 0.22	86.56 ± 0.15	87.48 ± 0.19	85.04 ± 0.18
	FHDDM	86.62 ± 0.21	86.62 ± 0.21	86.63 ± 0.22	84.84 ± 0.46	87.41 ± 0.10	88.26 ± 0.20	87.03 ± 0.13
	FTDD	87.98 ± 0.17	87.99 ± 0.17	88.03 ± 0.18	84.71 ± 0.34	87.27 ± 0.12	88.15 ± 0.26	86.76 ± 0.11
	$HDDM_A$	86.89 ± 0.15	86.90 ± 0.15	86.93 ± 0.15	84.39 ± 0.32	87.33 ± 0.12	88.56 ± 0.20	86.79 ± 0.13
	RDDM	86.83 ± 0.19	86.84 ± 0.19	86.87 ± 0.20	86.27 ± 0.32	87.25 ± 0.13	**89.47 ± 0.16**	86.83 ± 0.13
	SeqDr2	86.85 ± 0.15	86.85 ± 0.15	86.85 ± 0.15	85.81 ± 0.26	86.88 ± 0.10	84.36 ± 0.29	86.52 ± 0.15
	STEPD	86.12 ± 0.18	86.12 ± 0.18	86.13 ± 0.17	86.05 ± 0.23	86.87 ± 0.12	89.05 ± 0.17	85.89 ± 0.12
	WSTD	86.84 ± 0.20	86.85 ± 0.19	86.86 ± 0.18	85.13 ± 0.30	87.27 ± 0.12	88.43 ± 0.24	86.67 ± 0.10
Wavef.	ADWIN	80.92 ± 0.22	80.92 ± 0.22	80.54 ± 0.25	79.06 ± 0.25	79.85 ± 0.28	82.56 ± 0.20	78.63 ± 0.24
	DDM	81.04 ± 0.20	81.04 ± 0.20	80.50 ± 0.22	79.27 ± 0.25	80.16 ± 0.26	81.36 ± 0.24	78.52 ± 0.25
	ECDD	81.19 ± 0.18	81.19 ± 0.18	81.07 ± 0.18	79.40 ± 0.23	80.06 ± 0.22	80.09 ± 0.21	78.50 ± 0.23
	FHDDM	81.09 ± 0.20	81.09 ± 0.20	80.78 ± 0.24	79.12 ± 0.22	79.90 ± 0.23	82.41 ± 0.21	78.84 ± 0.26
	FTDD	81.02 ± 0.20	81.09 ± 0.21	80.40 ± 0.25	79.23 ± 0.25	80.11 ± 0.24	82.34 ± 0.21	78.36 ± 0.30
	$HDDM_A$	81.04 ± 0.20	81.10 ± 0.20	80.57 ± 0.22	79.29 ± 0.23	79.96 ± 0.27	82.55 ± 0.20	78.78 ± 0.28
	RDDM	81.10 ± 0.19	81.16 ± 0.18	80.67 ± 0.21	79.46 ± 0.25	80.13 ± 0.20	81.19 ± 0.20	79.10 ± 0.28
	SeqDr2	81.07 ± 0.19	81.07 ± 0.19	80.65 ± 0.20	79.46 ± 0.20	79.90 ± 0.27	82.38 ± 0.21	78.69 ± 0.24
	STEPD	81.14 ± 0.20	81.14 ± 0.20	80.94 ± 0.21	79.41 ± 0.24	79.83 ± 0.22	81.84 ± 0.17	79.29 ± 0.23
	WSTD	81.01 ± 0.20	81.07 ± 0.19	80.65 ± 0.23	79.20 ± 0.24	80.07 ± 0.20	**82.57 ± 0.21**	78.78 ± 0.24

Table 21
Mean accuracies of ensembles in percentage (%) in 50K instances gradual datasets, with 95% confidence intervals, using HT.

Dataset	Ensemble	ADOB	$BOLE_4$	$BOLE_5$	DDD	FASE	LevBag	None
$Agraw_1$	ADWIN	63.10 ± 0.23	63.10 ± 0.23	63.24 ± 0.22	65.83 ± 0.22	66.60 ± 0.14	68.19 ± 0.48	65.59 ± 0.15
	DDM	66.91 ± 0.45	66.91 ± 0.45	66.74 ± 0.44	68.06 ± 1.38	71.44 ± 0.21	66.70 ± 0.43	68.46 ± 1.74
	ECDD	67.19 ± 0.24	67.19 ± 0.24	67.12 ± 0.26	65.91 ± 0.76	69.14 ± 0.32	63.68 ± 0.35	64.95 ± 0.74
	FHDDM	68.34 ± 0.21	68.34 ± 0.21	68.54 ± 0.20	70.26 ± 0.24	**71.56 ± 0.21**	68.73 ± 0.30	70.35 ± 0.31
	FTDD	62.68 ± 0.58	62.67 ± 0.58	62.69 ± 0.53	65.93 ± 1.23	68.18 ± 0.40	63.18 ± 0.62	65.95 ± 0.82
	$HDDM_A$	67.70 ± 0.20	67.70 ± 0.20	67.74 ± 0.22	70.29 ± 0.25	71.19 ± 0.21	68.54 ± 0.39	71.39 ± 0.26
	RDDM	68.24 ± 0.21	68.24 ± 0.21	68.39 ± 0.21	70.90 ± 0.31	71.48 ± 0.21	67.06 ± 0.43	71.43 ± 0.31
	SeqDr2	62.62 ± 0.33	62.62 ± 0.33	62.59 ± 0.29	65.59 ± 0.29	67.33 ± 0.19	65.91 ± 0.43	66.59 ± 0.38
	STEPD	63.78 ± 0.19	63.78 ± 0.19	63.98 ± 0.19	65.82 ± 0.21	66.85 ± 0.21	68.23 ± 0.25	65.77 ± 0.25
	WSTD	64.25 ± 0.50	64.25 ± 0.50	64.24 ± 0.42	68.79 ± 0.54	68.07 ± 0.42	68.56 ± 0.43	67.93 ± 0.65
$Agraw_2$	ADWIN	84.08 ± 0.18	84.08 ± 0.18	84.12 ± 0.16	84.75 ± 0.21	85.53 ± 0.09	84.21 ± 0.11	84.85 ± 0.21
	DDM	82.71 ± 0.30	82.71 ± 0.30	82.82 ± 0.27	82.53 ± 0.84	85.82 ± 0.11	84.33 ± 0.11	83.17 ± 1.16
	ECDD	84.14 ± 0.11	84.14 ± 0.11	84.16 ± 0.12	84.44 ± 0.11	84.86 ± 0.06	83.62 ± 0.06	83.81 ± 0.10
	FHDDM	85.09 ± 0.11	85.09 ± 0.11	85.15 ± 0.11	85.22 ± 0.16	85.87 ± 0.09	83.57 ± 0.16	85.87 ± 0.13
	FTDD	83.20 ± 0.29	83.21 ± 0.29	83.23 ± 0.25	82.76 ± 0.42	85.80 ± 0.09	81.67 ± 0.25	83.88 ± 0.45
	$HDDM_A$	84.34 ± 0.19	84.35 ± 0.19	84.35 ± 0.18	84.48 ± 0.37	85.84 ± 0.09	83.67 ± 0.20	85.19 ± 0.32
	RDDM	84.63 ± 0.14	84.64 ± 0.14	84.71 ± 0.13	85.47 ± 0.30	**85.89 ± 0.09**	84.36 ± 0.10	85.69 ± 0.22
	SeqDr2	83.51 ± 0.17	83.51 ± 0.17	83.54 ± 0.15	83.03 ± 0.41	85.77 ± 0.10	79.59 ± 0.43	84.13 ± 0.43
	STEPD	84.91 ± 0.08	84.91 ± 0.08	84.94 ± 0.08	85.36 ± 0.15	85.80 ± 0.09	84.29 ± 0.10	85.45 ± 0.15
	WSTD	84.74 ± 0.11	84.74 ± 0.11	84.78 ± 0.10	83.99 ± 0.44	85.82 ± 0.10	83.64 ± 0.20	85.29 ± 0.42
LED	ADWIN	71.20 ± 0.17	71.20 ± 0.17	71.29 ± 0.17	70.80 ± 0.21	71.75 ± 0.15	72.31 ± 0.15	66.81 ± 0.59
	DDM	72.25 ± 0.16	72.25 ± 0.16	72.29 ± 0.16	71.53 ± 0.27	72.35 ± 0.15	59.74 ± 0.20	72.33 ± 0.23
	ECDD	72.18 ± 0.16	72.18 ± 0.16	72.22 ± 0.16	70.68 ± 0.25	72.26 ± 0.17	64.30 ± 0.30	68.32 ± 0.33
	FHDDM	72.33 ± 0.17	72.33 ± 0.17	72.37 ± 0.17	72.05 ± 0.19	72.46 ± 0.14	72.50 ± 0.17	71.66 ± 0.26
	FTDD	71.94 ± 0.15	72.00 ± 0.16	71.98 ± 0.16	70.94 ± 0.18	72.00 ± 0.16	71.99 ± 0.16	71.61 ± 0.17
	$HDDM_A$	72.44 ± 0.16	72.50 ± 0.16	72.51 ± 0.16	71.43 ± 0.19	72.42 ± 0.14	72.53 ± 0.15	72.47 ± 0.14
	RDDM	72.43 ± 0.15	72.49 ± 0.15	72.50 ± 0.15	71.98 ± 0.18	72.46 ± 0.14	67.53 ± 0.30	**72.62 ± 0.15**
	SeqDr2	71.42 ± 0.16	71.42 ± 0.16	71.52 ± 0.16	71.22 ± 0.19	71.61 ± 0.15	70.64 ± 0.26	66.85 ± 0.93
	STEPD	70.23 ± 0.24	70.23 ± 0.24	70.37 ± 0.23	71.19 ± 0.34	71.55 ± 0.20	72.21 ± 0.21	67.97 ± 0.93
	WSTD	72.00 ± 0.18	72.06 ± 0.18	72.11 ± 0.19	71.54 ± 0.21	71.79 ± 0.16	72.35 ± 0.16	71.36 ± 0.32
Mixed	ADWIN	90.56 ± 0.10	90.56 ± 0.10	90.64 ± 0.10	90.52 ± 0.09	90.88 ± 0.08	92.78 ± 0.12	90.37 ± 0.12
	DDM	91.79 ± 0.10	91.79 ± 0.10	91.85 ± 0.10	88.08 ± 0.57	91.06 ± 0.08	93.13 ± 0.15	90.84 ± 0.10
	ECDD	89.21 ± 0.14	89.21 ± 0.14	89.27 ± 0.13	89.27 ± 0.13	90.12 ± 0.10	91.09 ± 0.11	88.69 ± 0.14
	FHDDM	91.08 ± 0.14	91.08 ± 0.14	91.08 ± 0.14	88.09 ± 0.43	91.12 ± 0.08	92.32 ± 0.13	90.93 ± 0.09
	FTDD	91.72 ± 0.11	91.72 ± 0.11	91.75 ± 0.10	89.43 ± 0.31	91.08 ± 0.08	92.26 ± 0.17	90.75 ± 0.08
	$HDDM_A$	91.23 ± 0.11	91.24 ± 0.11	91.26 ± 0.11	88.45 ± 0.59	91.09 ± 0.09	92.61 ± 0.14	90.78 ± 0.09
	RDDM	91.33 ± 0.09	91.33 ± 0.09	91.37 ± 0.09	90.25 ± 0.47	90.98 ± 0.09	**93.17 ± 0.13**	90.74 ± 0.09
	SeqDr2	90.95 ± 0.11	90.95 ± 0.11	90.98 ± 0.11	90.47 ± 0.13	91.04 ± 0.08	89.41 ± 0.19	90.56 ± 0.09
	STEPD	90.28 ± 0.13	90.28 ± 0.13	90.30 ± 0.11	90.29 ± 0.13	90.44 ± 0.10	92.73 ± 0.14	89.81 ± 0.10
	WSTD	91.04 ± 0.10	91.04 ± 0.10	91.04 ± 0.10	89.98 ± 0.35	91.05 ± 0.09	92.59 ± 0.11	90.68 ± 0.09
RBF	ADWIN	19.91 ± 0.71	19.91 ± 0.71	31.61 ± 0.87	32.43 ± 0.25	33.24 ± 0.24	37.88 ± 0.19	32.40 ± 0.31
	DDM	20.00 ± 0.63	20.00 ± 0.63	31.82 ± 0.78	32.66 ± 0.36	33.16 ± 0.21	38.03 ± 0.24	32.53 ± 0.34
	ECDD	20.06 ± 0.68	20.06 ± 0.68	31.57 ± 0.64	33.10 ± 0.31	33.95 ± 0.30	**39.00 ± 0.17**	33.17 ± 0.32
	FHDDM	19.65 ± 0.56	19.65 ± 0.56	31.25 ± 0.67	31.77 ± 0.22	32.68 ± 0.19	32.96 ± 0.25	31.34 ± 0.23
	FTDD	20.61 ± 0.74	23.21 ± 0.55	32.79 ± 0.31	32.97 ± 0.39	33.68 ± 0.27	38.33 ± 0.21	32.69 ± 0.43
	$HDDM_A$	19.93 ± 0.60	22.92 ± 0.51	32.56 ± 0.33	32.71 ± 0.31	33.29 ± 0.22	37.36 ± 0.29	32.58 ± 0.29
	RDDM	19.87 ± 0.72	23.39 ± 0.39	32.60 ± 0.29	32.50 ± 0.27	32.99 ± 0.22	35.59 ± 0.38	32.38 ± 0.28
	SeqDr2	20.49 ± 0.74	20.49 ± 0.74	31.86 ± 0.70	32.95 ± 0.37	33.55 ± 0.28	38.25 ± 0.16	32.73 ± 0.35
	STEPD	19.34 ± 0.63	19.34 ± 0.63	31.32 ± 0.70	31.63 ± 0.26	32.64 ± 0.22	36.22 ± 0.35	31.08 ± 0.28
	WSTD	20.23 ± 0.67	22.99 ± 0.44	31.80 ± 0.48	32.39 ± 0.48	33.47 ± 0.28	38.15 ± 0.19	31.91 ± 0.38
Sine	ADWIN	90.30 ± 0.12	90.30 ± 0.12	90.34 ± 0.12	89.18 ± 0.11	90.27 ± 0.10	93.89 ± 0.12	89.06 ± 0.09
	DDM	92.24 ± 0.20	92.24 ± 0.20	92.24 ± 0.19	89.09 ± 0.38	90.86 ± 0.10	**94.08 ± 0.12**	90.27 ± 0.10
	ECDD	89.91 ± 0.25	89.91 ± 0.25	89.94 ± 0.24	87.23 ± 0.19	88.84 ± 0.10	90.50 ± 0.16	86.33 ± 0.14
	FHDDM	91.52 ± 0.18	91.52 ± 0.18	91.51 ± 0.18	88.98 ± 0.33	90.89 ± 0.12	93.78 ± 0.10	90.42 ± 0.10
	FTDD	91.93 ± 0.18	91.94 ± 0.18	91.96 ± 0.17	89.03 ± 0.27	90.86 ± 0.11	93.57 ± 0.16	90.27 ± 0.11
	$HDDM_A$	91.77 ± 0.21	91.77 ± 0.21	91.80 ± 0.19	88.87 ± 0.28	90.87 ± 0.12	93.60 ± 0.13	90.33 ± 0.11
	RDDM	91.65 ± 0.22	91.65 ± 0.22	91.69 ± 0.22	90.25 ± 0.22	90.85 ± 0.10	**94.08 ± 0.11**	90.34 ± 0.09
	SeqDr2	90.72 ± 0.12	90.72 ± 0.12	90.73 ± 0.11	90.32 ± 0.14	90.64 ± 0.11	90.57 ± 0.19	90.03 ± 0.11
	STEPD	90.17 ± 0.19	90.17 ± 0.19	90.18 ± 0.19	90.02 ± 0.16	90.46 ± 0.10	93.76 ± 0.13	89.16 ± 0.20
	WSTD	91.37 ± 0.19	91.37 ± 0.19	91.36 ± 0.19	89.76 ± 0.31	90.84 ± 0.10	93.74 ± 0.14	90.24 ± 0.12
Wavef.	ADWIN	82.23 ± 0.11	82.23 ± 0.11	82.00 ± 0.12	80.52 ± 0.16	81.63 ± 0.24	83.85 ± 0.12	79.42 ± 0.17
	DDM	82.67 ± 0.14	82.67 ± 0.14	82.45 ± 0.15	80.44 ± 0.19	82.12 ± 0.16	82.60 ± 0.34	79.18 ± 0.23
	ECDD	81.80 ± 0.13	81.80 ± 0.13	81.63 ± 0.14	79.87 ± 0.14	80.59 ± 0.15	80.52 ± 0.13	78.95 ± 0.16
	FHDDM	81.77 ± 0.13	81.77 ± 0.13	81.43 ± 0.14	80.50 ± 0.21	81.68 ± 0.13	83.71 ± 0.13	79.49 ± 0.16
	FTDD	82.58 ± 0.12	82.61 ± 0.12	82.35 ± 0.14	80.39 ± 0.18	81.90 ± 0.19	83.68 ± 0.14	79.01 ± 0.21
	$HDDM_A$	82.37 ± 0.11	82.39 ± 0.11	82.01 ± 0.12	80.66 ± 0.18	81.89 ± 0.19	**83.92 ± 0.13**	79.44 ± 0.18
	RDDM	82.48 ± 0.13	82.50 ± 0.13	81.99 ± 0.13	80.81 ± 0.14	81.63 ± 0.15	81.98 ± 0.19	79.71 ± 0.14
	SeqDr2	82.56 ± 0.12	82.56 ± 0.12	82.40 ± 0.12	80.46 ± 0.20	81.63 ± 0.26	83.74 ± 0.14	79.15 ± 0.21
	STEPD	81.84 ± 0.13	81.84 ± 0.13	81.61 ± 0.13	80.13 ± 0.13	80.60 ± 0.13	82.52 ± 0.12	79.85 ± 0.13
	WSTD	82.10 ± 0.14	82.13 ± 0.14	81.73 ± 0.15	80.55 ± 0.20	81.85 ± 0.16	83.85 ± 0.12	79.44 ± 0.19

Table 22
Mean accuracies of ensembles in percentage (%) in 100K instances gradual datasets, with 95% confidence intervals, using HT.

Dataset	Ensemble	ADOB	$BOLE_4$	$BOLE_5$	DDD	FASE	LevBag	None
$Agraw_1$	ADWIN	63.96 ± 0.19	63.96 ± 0.19	63.97 ± 0.18	66.85 ± 0.15	67.57 ± 0.12	70.59 ± 0.28	66.25 ± 0.14
	DDM	69.50 ± 0.26	69.50 ± 0.26	69.31 ± 0.25	71.01 ± 1.66	73.82 ± 0.29	68.27 ± 0.32	71.72 ± 1.76
	ECDD	68.72 ± 0.22	68.72 ± 0.22	68.74 ± 0.23	67.52 ± 0.59	70.54 ± 0.13	64.63 ± 0.42	65.79 ± 0.64
	FHDDM	69.72 ± 0.23	69.72 ± 0.23	69.87 ± 0.23	72.54 ± 0.23	73.67 ± 0.25	70.54 ± 0.20	72.61 ± 0.31
	FTDD	64.69 ± 0.35	64.68 ± 0.35	64.62 ± 0.32	70.32 ± 0.98	70.64 ± 0.61	66.92 ± 0.64	69.51 ± 1.08
	$HDDM_A$	69.89 ± 0.18	69.89 ± 0.19	70.01 ± 0.18	73.23 ± 0.25	73.74 ± 0.22	70.77 ± 0.21	74.25 ± 0.29
	RDDM	70.05 ± 0.21	70.04 ± 0.21	70.13 ± 0.22	73.43 ± 0.33	**73.95 ± 0.22**	68.85 ± 0.22	74.43 ± 0.33
	SeqDr2	64.56 ± 0.26	64.56 ± 0.26	64.46 ± 0.25	67.48 ± 0.31	68.68 ± 0.21	69.55 ± 0.30	68.54 ± 0.38
	STEPD	64.05 ± 0.12	64.05 ± 0.12	64.14 ± 0.12	67.52 ± 0.52	68.04 ± 0.20	69.73 ± 0.14	66.63 ± 0.27
	WSTD	64.94 ± 0.54	64.94 ± 0.54	64.87 ± 0.48	72.13 ± 0.54	71.13 ± 0.48	70.60 ± 0.31	70.90 ± 0.80
$Agraw_2$	ADWIN	85.62 ± 0.10	85.62 ± 0.10	85.64 ± 0.10	86.18 ± 0.07	86.53 ± 0.07	85.60 ± 0.10	85.94 ± 0.09
	DDM	84.19 ± 0.29	84.19 ± 0.29	84.23 ± 0.27	84.23 ± 0.88	87.27 ± 0.05	85.40 ± 0.07	84.47 ± 0.82
	ECDD	84.90 ± 0.08	84.90 ± 0.08	84.93 ± 0.08	84.90 ± 0.08	85.41 ± 0.05	84.24 ± 0.05	84.40 ± 0.06
	FHDDM	86.12 ± 0.09	86.12 ± 0.09	86.14 ± 0.08	86.82 ± 0.18	**87.35 ± 0.06**	84.42 ± 0.15	87.31 ± 0.05
	FTDD	84.87 ± 0.28	84.87 ± 0.28	84.84 ± 0.26	84.17 ± 0.57	87.26 ± 0.05	82.77 ± 0.36	85.59 ± 0.51
	$HDDM_A$	85.87 ± 0.15	85.88 ± 0.15	85.87 ± 0.14	86.49 ± 0.20	87.33 ± 0.06	85.03 ± 0.11	87.14 ± 0.15
	RDDM	85.76 ± 0.13	85.77 ± 0.13	85.76 ± 0.13	86.90 ± 0.26	87.33 ± 0.06	85.48 ± 0.08	86.97 ± 0.18
	SeqDr2	85.00 ± 0.26	85.00 ± 0.26	84.99 ± 0.24	85.57 ± 0.55	87.31 ± 0.06	82.03 ± 0.40	85.60 ± 0.56
	STEPD	85.77 ± 0.05	85.77 ± 0.05	85.80 ± 0.05	86.75 ± 0.14	87.06 ± 0.06	85.50 ± 0.10	86.53 ± 0.12
	WSTD	86.05 ± 0.07	86.06 ± 0.07	86.07 ± 0.06	86.11 ± 0.35	87.33 ± 0.06	85.29 ± 0.13	86.98 ± 0.33
LED	ADWIN	72.18 ± 0.14	72.18 ± 0.14	72.24 ± 0.15	72.44 ± 0.15	72.65 ± 0.11	73.12 ± 0.11	68.74 ± 0.60
	DDM	73.05 ± 0.12	73.05 ± 0.12	73.06 ± 0.12	72.44 ± 0.22	73.13 ± 0.12	60.29 ± 0.26	72.54 ± 0.34
	ECDD	72.86 ± 0.12	72.86 ± 0.12	72.88 ± 0.12	71.19 ± 0.20	72.98 ± 0.13	64.87 ± 0.22	68.79 ± 0.23
	FHDDM	73.10 ± 0.12	73.10 ± 0.12	73.13 ± 0.12	72.90 ± 0.12	73.20 ± 0.11	73.26 ± 0.12	72.30 ± 0.23
	FTDD	72.88 ± 0.13	72.91 ± 0.13	72.90 ± 0.13	71.80 ± 0.19	72.99 ± 0.11	72.80 ± 0.14	72.53 ± 0.15
	$HDDM_A$	73.24 ± 0.11	73.27 ± 0.11	73.27 ± 0.11	72.31 ± 0.16	73.22 ± 0.11	73.23 ± 0.11	73.21 ± 0.12
	RDDM	73.20 ± 0.12	73.22 ± 0.12	73.23 ± 0.12	72.66 ± 0.20	73.21 ± 0.11	68.11 ± 0.22	**73.30 ± 0.12**
	SeqDr2	72.58 ± 0.13	72.58 ± 0.13	72.62 ± 0.13	72.26 ± 0.21	72.74 ± 0.11	71.54 ± 0.20	69.43 ± 0.54
	STEPD	71.50 ± 0.16	71.50 ± 0.16	71.58 ± 0.16	72.23 ± 0.19	72.68 ± 0.13	72.88 ± 0.15	68.75 ± 0.71
	WSTD	73.00 ± 0.12	73.04 ± 0.12	73.05 ± 0.12	72.54 ± 0.18	72.90 ± 0.11	73.12 ± 0.11	72.40 ± 0.18
Mixed	ADWIN	92.10 ± 0.07	92.10 ± 0.07	92.16 ± 0.07	91.69 ± 0.08	92.01 ± 0.08	94.93 ± 0.09	91.24 ± 0.09
	DDM	94.23 ± 0.11	94.23 ± 0.11	94.21 ± 0.11	91.25 ± 0.33	92.65 ± 0.06	95.18 ± 0.08	92.42 ± 0.08
	ECDD	90.38 ± 0.17	90.38 ± 0.17	90.47 ± 0.16	89.93 ± 0.09	90.88 ± 0.06	92.04 ± 0.08	89.20 ± 0.10
	FHDDM	93.39 ± 0.13	93.39 ± 0.13	93.42 ± 0.12	91.28 ± 0.33	92.70 ± 0.06	94.71 ± 0.11	92.52 ± 0.07
	FTDD	93.91 ± 0.13	93.91 ± 0.13	93.93 ± 0.13	91.67 ± 0.24	92.67 ± 0.06	94.56 ± 0.13	92.43 ± 0.06
	$HDDM_A$	93.91 ± 0.14	93.91 ± 0.14	93.93 ± 0.14	91.58 ± 0.29	92.66 ± 0.06	94.85 ± 0.12	92.43 ± 0.08
	RDDM	93.91 ± 0.13	93.91 ± 0.13	93.94 ± 0.13	92.35 ± 0.17	92.58 ± 0.07	**95.21 ± 0.08**	92.37 ± 0.08
	SeqDr2	93.00 ± 0.09	93.00 ± 0.09	93.03 ± 0.08	92.36 ± 0.07	92.63 ± 0.06	92.22 ± 0.20	92.29 ± 0.05
	STEPD	92.05 ± 0.09	92.05 ± 0.09	92.06 ± 0.09	91.39 ± 0.13	91.71 ± 0.11	94.79 ± 0.10	90.64 ± 0.09
	WSTD	93.37 ± 0.07	93.37 ± 0.07	93.37 ± 0.08	91.80 ± 0.27	92.66 ± 0.07	94.78 ± 0.11	92.38 ± 0.06
RBF	ADWIN	19.45 ± 0.60	19.45 ± 0.60	31.98 ± 0.76	32.96 ± 0.24	33.44 ± 0.17	39.01 ± 0.13	32.58 ± 0.23
	DDM	19.58 ± 0.57	19.58 ± 0.57	32.63 ± 0.84	34.18 ± 0.28	33.70 ± 0.18	39.20 ± 0.19	33.67 ± 0.24
	ECDD	19.68 ± 0.62	19.68 ± 0.62	34.19 ± 0.61	35.33 ± 0.19	34.95 ± 0.17	**39.91 ± 0.14**	34.85 ± 0.20
	FHDDM	19.32 ± 0.55	19.32 ± 0.55	31.35 ± 0.67	31.85 ± 0.14	32.71 ± 0.12	33.31 ± 0.21	31.30 ± 0.13
	FTDD	20.28 ± 0.86	23.46 ± 0.77	34.31 ± 0.29	34.13 ± 0.30	34.26 ± 0.22	39.45 ± 0.15	33.27 ± 0.32
	$HDDM_A$	19.67 ± 0.62	22.62 ± 0.50	32.54 ± 0.24	33.49 ± 0.30	33.56 ± 0.20	38.40 ± 0.17	32.92 ± 0.26
	RDDM	19.72 ± 0.78	23.02 ± 0.41	32.57 ± 0.20	32.89 ± 0.22	33.24 ± 0.15	37.04 ± 0.29	32.84 ± 0.19
	SeqDr2	19.81 ± 0.65	19.81 ± 0.65	32.92 ± 0.93	34.49 ± 0.29	34.26 ± 0.18	39.22 ± 0.15	33.61 ± 0.21
	STEPD	19.27 ± 0.55	19.27 ± 0.55	31.22 ± 0.79	31.61 ± 0.15	32.56 ± 0.13	36.82 ± 0.26	31.16 ± 0.15
	WSTD	19.84 ± 0.63	22.80 ± 0.35	32.20 ± 0.23	32.69 ± 0.25	33.75 ± 0.20	39.05 ± 0.16	32.26 ± 0.25
Sine	ADWIN	92.25 ± 0.12	92.25 ± 0.12	92.29 ± 0.12	90.50 ± 0.09	91.53 ± 0.07	96.15 ± 0.08	89.96 ± 0.09
	DDM	95.28 ± 0.15	95.28 ± 0.15	95.28 ± 0.15	90.88 ± 0.34	92.56 ± 0.08	**96.25 ± 0.09**	92.00 ± 0.09
	ECDD	91.54 ± 0.24	91.54 ± 0.24	91.56 ± 0.24	87.62 ± 0.10	89.37 ± 0.07	91.57 ± 0.12	86.80 ± 0.12
	FHDDM	94.81 ± 0.20	94.81 ± 0.20	94.78 ± 0.19	90.95 ± 0.33	92.56 ± 0.07	96.07 ± 0.10	92.04 ± 0.08
	FTDD	95.00 ± 0.15	95.00 ± 0.15	95.00 ± 0.14	91.53 ± 0.22	92.54 ± 0.07	95.76 ± 0.12	91.92 ± 0.09
	$HDDM_A$	95.11 ± 0.14	95.11 ± 0.14	95.11 ± 0.13	90.98 ± 0.27	92.56 ± 0.07	95.97 ± 0.10	91.98 ± 0.09
	RDDM	94.93 ± 0.15	94.93 ± 0.15	94.96 ± 0.14	92.14 ± 0.24	92.54 ± 0.07	96.20 ± 0.08	91.99 ± 0.09
	SeqDr2	93.44 ± 0.13	93.44 ± 0.13	93.46 ± 0.13	92.48 ± 0.09	92.45 ± 0.09	93.94 ± 0.17	91.88 ± 0.08
	STEPD	92.78 ± 0.22	92.78 ± 0.22	92.79 ± 0.22	91.40 ± 0.18	92.02 ± 0.10	95.98 ± 0.09	90.38 ± 0.15
	WSTD	94.71 ± 0.16	94.71 ± 0.16	94.71 ± 0.16	91.60 ± 0.28	92.53 ± 0.08	95.93 ± 0.10	91.93 ± 0.10
Wavef.	ADWIN	82.60 ± 0.08	82.60 ± 0.08	82.37 ± 0.08	80.82 ± 0.12	81.96 ± 0.16	84.54 ± 0.08	79.57 ± 0.15
	DDM	83.46 ± 0.09	83.46 ± 0.09	83.32 ± 0.09	81.29 ± 0.13	82.51 ± 0.15	83.28 ± 0.20	79.37 ± 0.25
	ECDD	82.06 ± 0.10	82.06 ± 0.10	81.90 ± 0.11	80.01 ± 0.12	80.75 ± 0.11	80.65 ± 0.10	79.09 ± 0.13
	FHDDM	81.96 ± 0.11	81.96 ± 0.11	81.63 ± 0.11	81.15 ± 0.23	82.16 ± 0.15	84.47 ± 0.08	79.67 ± 0.14
	FTDD	83.39 ± 0.08	83.40 ± 0.08	83.30 ± 0.09	81.47 ± 0.27	82.33 ± 0.15	84.43 ± 0.09	79.28 ± 0.21
	$HDDM_A$	82.78 ± 0.11	82.79 ± 0.11	82.48 ± 0.11	81.07 ± 0.17	82.33 ± 0.13	**84.62 ± 0.08**	79.47 ± 0.12
	RDDM	82.85 ± 0.11	82.87 ± 0.12	82.40 ± 0.12	81.00 ± 0.16	82.10 ± 0.13	82.36 ± 0.16	79.81 ± 0.13
	SeqDr2	83.36 ± 0.07	83.36 ± 0.07	83.24 ± 0.08	81.22 ± 0.19	82.03 ± 0.16	84.55 ± 0.10	79.41 ± 0.17
	STEPD	82.01 ± 0.09	82.01 ± 0.09	81.77 ± 0.11	80.41 ± 0.11	80.85 ± 0.09	82.84 ± 0.11	80.10 ± 0.10
	WSTD	82.63 ± 0.09	82.64 ± 0.09	82.23 ± 0.11	81.18 ± 0.20	82.24 ± 0.13	84.59 ± 0.08	79.47 ± 0.16

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